T2K 3.23×10^{19}-POT Muon-Neutrino Disappearance Analysis

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Abstract

We report the results of the Rutherford-Oxford ν_μ-disappearance analysis on the dataset collected during the first physics run which spanned the period from January to June 2010. This first physics dataset corresponds to an integrated JPARC neutrino beam exposure of 3.23×10^{19} POT [1]. The oscillation analysis was performed using both GENIE [2] and NEUT [3].

Folding in the ND280 measurement, GENIE predicts 22.42 $\pm$ 4.74 (stat) $^{+3.58}_{-3.48}$ (syst) 1-ring μ-like events without oscillations but only 8 were observed. The statistical significance of the deficit is 2.5 standard deviations. NEUT predicts 22.81 $\pm$ 4.78 (stat) $^{+3.58}_{-3.49}$ (syst), which implies that the statistical significance of the deficit is 2.5 standard deviations.

The observed reconstructed energy spectrum of 1-ring μ-like events is consistent with 2-flavour neutrino oscillations. A 2-flavour neutrino oscillation fit of the GENIE prediction to the 3.23×10^{19} POT dataset, with $sin^2 2\theta_{23}$ constrained in the physical region and $|\Delta m^2_{23}|$ constrained in a $\sim 15\sigma$ window around the MINOS best-fit value, yields $|\Delta m^2_{23}|$ = 2.8$\times10^{-3}$ eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.81 as the best-fit values. For NEUT, the corresponding values are $|\Delta m^2_{23}|$ = 2.9$\times10^{-3}$ eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.83.

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13 Effect of SuperK CCnonQE efficiency variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 54

14 Effect of SuperK NC efficiency variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. . . . 55

15 Effect of SuperK $\nu_e CC$ efficiency variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. . . . 56

16 Effect of ND280 efficiency variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. . . . 57

17 Effect of flux normalization variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. . . . 58

18 Effect of CCQE cross section variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. . . . 59

19 Effect of CC1π/CCQE cross section ratio variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 60

20 Effect of CCother/CCQE cross section ratio variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 61

21 Effect of NC/CCQE cross section ratio variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 62

22 Effect of FSI variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 63
23 Effect of SuperK CCQE efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 64
24 Effect of SuperK CCnonQE efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 65
25 Effect of SuperK NC efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 66
26 Effect of SuperK ν_e CC efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 67
27 Effect of ND280 efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 68
28 Effect of beam normalization variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 69
29 Effect of CCQE cross section variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 70
30 Effect of CC1π/CCQE cross section variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 71
31 Effect of CCother/CCQE cross section ratio variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 72
32 Effect of NC/CCQE cross section ratio variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 73
33 Effect of FSI variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied. 74

1. Introduction

In this technical note, a description is given of the methods used by the Rutherford-Oxford group to measure the ν_μ-disappearance oscillation parameters $\sin^2(2\theta_{23})$ and Δm^2_{23} from data acquired during the first T2K physics run (the 2010a dataset). The best-fit values of these oscillation parameters, and the 68%, 90% and 95% confidence contours are also reported. The T2K 2010a dataset was acquired from January - June 2010 and corresponds to an integrated JPARC neutrino beam exposure of 3.23×10^{19} POT.

For this analysis, the ND280 group supplied the total number of ND280 CC-like events, but no energy spectrum was available. This information was used to make a prediction of the SuperK reconstructed energy spectrum of 1 μ-like ring events; this is described in sections 2.

The oscillation parameters were measured by comparing this predicted SuperK spectrum with the spectrum reconstructed from the 2010a data. This was done using a likelihood-ratio method as described in section 3. Confidence contours were calculated using both the standard PDG method of "constant $\Delta\chi^2$" [9] and the method of Feldman-Cousins [10]; these methods are described in section 3.4. Systematic uncertainties in the oscillation parameters are evaluated and incorporated into the confidence contour as described in section 4.

The results of the measurements of $\sin^2 2\theta_{23}$ and Δm^2_{23} are given in section 5. Planned improvements to the present analysis are discussed in section 6. A summary is given in section 7.

2. SuperK reconstructed energy spectrum prediction from ND280 measurement

2.1. Formal derivation of predicted SuperK reconstructed energy spectrum formula from measured ND280 CC inclusive reconstructed energy spectrum

A schematic outline of the method used for predicting the SuperK reconstructed energy spectrum of 1 μ-like ring events from the ND280[1] measurement of CC inclusive reconstructed energy spectrum is shown in Fig. 1.

The predicted reconstructed energy (E_r) spectrum ($\frac{dN^{exp}_{SK}}{dE_r}$) of selected 1 μ-like ring events at SuperK, is the sum of the corresponding spectra for all event types m^2 whose contribution is tracked separately.

$$\frac{dN^{exp}_{SK}(E_r)}{dE_r} = \sum_m \frac{dN^{exp}_{SK;m}(E_r)}{dE_r} \tag{1}$$

Let $\epsilon^{reco}_{SK;m}(E_t)$ be the SuperK efficiency for reconstructing a neutrino event of type m and true energy E_t, and $\epsilon^{sel}_{SK;m}(E_r)$ be the SuperK efficiency for selecting an event of type m and reconstructed energy E_r. If $R_{SK;m}(E_t, E_r)$ is the SuperK energy transfer matrix[3] for events of type m, the expected reconstructed energy spectrum can be expressed in terms of the total true neutrino energy spectrum ($\frac{dN^{tot}_{SK;m}(E_t)}{dE_t}$):

$$\frac{dN^{exp}_{SK}(E_r)}{dE_r} = \sum_m \int dE_t \frac{dN^{tot}_{SK;m}(E_t)}{dE_t} R_{SK;m}(E_t, E_r)\epsilon^{sel}_{SK;m}(E_r)\epsilon^{reco}_{SK;m}(E_t) \tag{4}$$

The above expression can be rewritten to explicitly include the survival probability $P^{surv}_m(E_t)$ for neutrinos of

[1] Throughout this note we use 'ND280' to refer to the 'ND280 off-axis tracker'.

[2] In this work, m = ν_μ CCQE, ν_μ CC1π, ν_μ CC other, $\nu_\mu(\nu_\tau)$ NC, $\bar{\nu_\mu}$ CCQE, $\bar{\nu_\mu}$ CC1π, $\bar{\nu_\mu}$ CC other $\bar{\nu_\mu}(\bar{\nu_\tau})$ NC, ν_e CCQE, ν_e CC1π, ν_e CC other and ν_e NC.

[3] The quantity $R_{SK;m}(E_t, E_r)$ is defined by

$$\frac{dN_{SK;m}}{dE_r} = \int dE_t \frac{d^2N_{SK;m}}{dE_r dE_t} = \int dE_t \frac{dN_{SK;m}}{dE_t} R_{SK;m} \tag{2}$$

This implies that

$$R_{SK;m} = \frac{d^2N_{SK;m}}{dE_r dE_t} \Big/ \frac{dN_{SK;m}}{dE_t} \tag{3}$$

This also defines the normalization properties all the $R_{SK;m}(E_t, E_r)$ matrix: the elements for each E_r row (or column, depending on how the matrix is represented) should add up to 1.

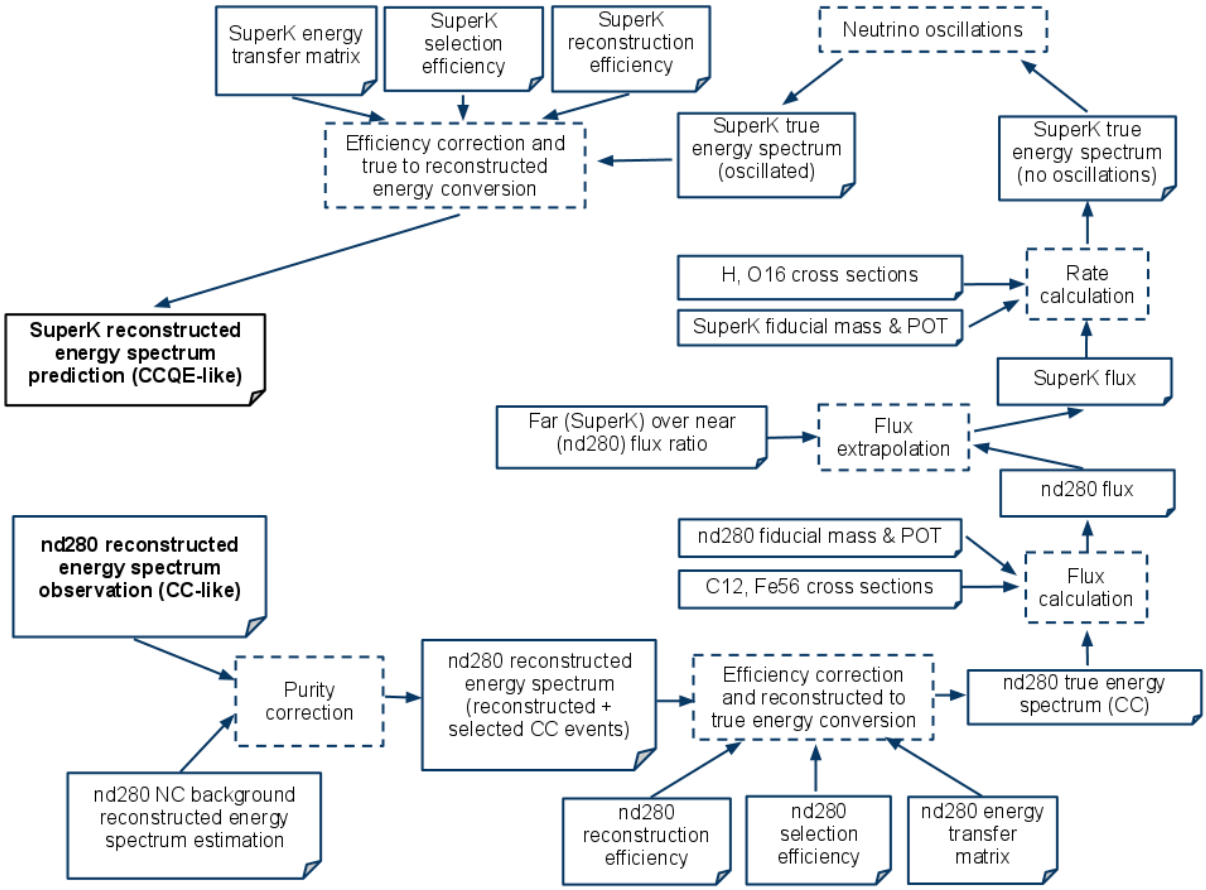


Figure 1: Outline of the procedure for predicting the SuperK reconstructed energy spectrum of ν_μ CCQE-like events from the ν_μ CC inclusive reconstructed energy spectrum measured in ND280.

energy E_t, and the total non-oscillated true neutrino energy spectrum $\frac{dN^{tot;unosc}_{SK;m}}{dE_t}$:

$$\frac{dN^{exp}_{SK}(E_r)}{dE_r} = \sum_m \int dE_t \frac{dN^{tot;unosc}_{SK;m}(E_t)}{dE_t} P^{surv}_m(E_t) R_{SK;m}(E_t, E_r) \epsilon^{sel}_{SK;m}(E_r) \epsilon^{reco}_{SK;m}(E_t) \tag{5}$$

The total non-oscillated true neutrino energy spectrum is related to the estimated neutrino flux at SuperK ($\frac{d\Phi_{SK}(E_t)}{dE_t}$) in the following way:

$$\begin{aligned} \frac{dN^{exp}_{SK}(E_r)}{dE_r} = N^{POT}_{SK} M^{fid}_{SK} \frac{N_A}{A_{H20}} \sum_m \int dE_t \Bigg(& \frac{d\Phi_{SK}(E_t)}{dE_t} \times \\ & \sigma^{tot}_{H20}(E_t) f_m(E_t) P^{surv}_m(E_t) R_{SK;m}(E_t, E_r) \epsilon^{sel}_{SK;m}(E_r) \epsilon^{reco}_{SK;m}(E_t) \Bigg) \end{aligned} \tag{6}$$

where N^{POT}_{SK} is the proton-on-target exposure for the analyzed SuperK data sample, M^{fid}_{SK} is the SuperK fiducial volume mass, $\sigma^{tot}_{H20}(E_t)$ is the total neutrino interaction cross section in water at energy E_t, $f_m(E_t$ is the fraction of events of type m, N_A is Avogadro's number and A_{H20} is the water mass number.

The absolute neutrino flux $\frac{d\Phi_{SK}(E_t)}{dE_t}$ in Eq. 6, for which significant uncertainties exist, can be constrained if it is expressed in terms of the observed reconstructed neutrino energy spectrum from an ND280 inclusive charged-current sample. From the definition of the far-over-near ratio

$$\frac{d\Phi_{SK}(E_t)}{dE_t} = R^{F/N}(E_t)\frac{d\Phi_{ND280}(E_t)}{dE_t} \tag{7}$$

where $\frac{d\Phi_{ND280}(E_t)}{dE_t}$ is the neutrino flux at ND280. This latter quantity can be related to the total CC true neutrino energy spectrum $\frac{dN^{tot}_{ND280;CC}(E_t)}{dE_t}$ as follows:

$$\frac{d\Phi_{SK}(E_t)}{dE_t} = \frac{R^{F/N}(E_t)A_{ND280}}{N^{POT}_{ND280}M^{fid}_{ND280}N_A\sigma^{CC}_{ND280}(E_t)}\frac{dN^{tot}_{ND280;CC}(E_t)}{dE_t} \tag{8}$$

where N^{POT}_{ND280} is the proton-on-target exposure for the analyzed ND280 data sample, M^{fid}_{ND280} is the fiducial mass for the ND280 'target' region, $\sigma^{CC}_{ND280}(E_t)$ is the total charged-current cross section for the ND280 'target' region mixture at energy E_t and A_{ND280} is the mass number for the ND280 'target' mixture.

The total CC true neutrino energy spectrum can be expressed in terms of the observed reconstructed neutrino energy spectrum (which has the NC background subtracted):

$$\begin{aligned}\frac{d\Phi_{SK}(E_t)}{dE_t} = \frac{R^{F/N}(E_t)A_{ND280}}{N^{POT}_{ND280}M^{fid}_{ND280}\sigma^{CC}_{ND280}(E_t)}\int dE_{r\prime}\Bigg(\\ \left(\frac{dN^{obs}_{ND280;CC-like}(E_{r\prime})}{dE_{r\prime}} - \frac{dN^{exp}_{ND280;NC-bkg}(E_{r\prime})}{dE_{r\prime}}\right)\frac{R^T_{ND280;CC}(E_t,E_{r\prime})}{\epsilon^{reco}_{ND280;CC}(E_t)\epsilon^{sel}_{ND280;CC}(E_{r\prime})}\Bigg)\end{aligned} \tag{9}$$

where $\epsilon^{reco}_{ND280;CC}(E_t)$ is the ND280 efficiency for reconstructing a charged-current neutrino event with true energy E_t, $\epsilon^{sel}_{ND280;CC}(E_{r\prime})$ is the ND280 selection efficiency for a charged-current event reconstructed with energy $E_{r\prime}$, and $R^T_{ND280;CC}(E_t, E_{r\prime})$ is the transpose of the ND280 energy transfer matrix for charged-current events.

Combining Eq. 6 and 9 we obtain

$$\begin{aligned}\frac{dN^{exp}_{SK}(E_r)}{dE_r} = \frac{N^{POT}_{SK}M^{fid}_{SK}A_{ND280}}{N^{POT}_{ND280}M^{fid}_{ND280}A_{H20}}\sum_m\int dE_t\int dE_{r\prime}\Bigg(R^{F/N}(E_t)\frac{\sigma^{tot}_{H20}(E_t)f_m(E_t)}{\sigma^{CC}_{ND280}(E_t)}P^{surv}_m(E_t)\times \\ \left(\frac{dN^{obs}_{ND280;CC-like}(E_{r\prime})}{dE_{r\prime}} - \frac{dN^{exp}_{ND280;NC-bkg}(E_{r\prime})}{dE_{r\prime}}\right)\times \\ R_{SK;m}(E_t,E_r)R^T_{ND280;CC}(E_t,E_{r\prime})\frac{\epsilon^{sel}_{SK;m}(E_r)\epsilon^{reco}_{SK;m}(E_t)}{\epsilon^{reco}_{ND280;CC}(E_t)\epsilon^{sel}_{ND280;CC}(E_{r\prime})}\Bigg)\end{aligned} \tag{10}$$

2.2. 3.23×10^{19}-POT analysis specifics and discussion of ND280 input

In the present analysis, there is no spectral information in the inclusive CC ND280 measurement. In this case, Eq. 10 simplifies to

$$\begin{aligned}\frac{dN^{exp}_{SK}(E_r)}{dE_r} = N^{POT}_{SK}M^{fid}_{SK}\frac{N_A}{A_{H20}}C_{ND280}\sum_m\int dE_t\Bigg(\\ \frac{d\Phi_{SK}(E_t)}{dE_t}\sigma^{tot}_{H20}(E_t)f_m(E_t)P^{surv}_m(E_t)R_{SK;m}(E_t,E_r)\epsilon^{sel}_{SK;m}(E_r)\epsilon^{reco}_{SK;m}(E_t)\Bigg)\end{aligned} \tag{11}$$

where C_{ND280} is the ND280 data-driven correction factor on the flux times cross section product. We note that the far over near ratio $R^{F/N}(E_t)$, discussed previously, does not appear in the above formula. This follows from the fact that the ND280 measurement has no energy dependence.

In Eq. 11, C_{ND280} is given by

$$C_{ND280} = \frac{N^{obs}_{ND280;CC-like}}{N^{exp}_{ND280;CC} + N^{exp}_{ND280;NC-bkg}} \tag{12}$$

where $N^{obs}_{ND280;CC-like}$ is the observed number of CC-like events, $N^{exp}_{ND280;NC-bkg}$ is the expected NC contamination

of the observation, and $N^{exp}_{ND280;CC}$ is the expected number of CC events.

In the present analysis C_{ND280} is 1.1549 for GENIE and 1.06127 for NEUT[14, 15]. The difference in the required correction factor for GENIE and NEUT reflects generator-level tuning choices; mainly, it reflects the dramatic difference in the choice of the input CCQE axial mass. The authors of GENIE use a value of M_A = 0.99 GeV/c^2 which is better-motivated theoretically. One the other hand, the authors of NEUT use a higher value (M_A = 1.2 GeV/c^2) motivated by K2K and MiniBooNE measurements of an increased event rate (in C^{12}, O^{16} targets) compared with expectations informed by previous measurements in deuterium targets (although the observed excess is only about 1σ). It is plausible, although not satisfying, that M_A should have a larger values for nuclear targets. However, it is also likely that the discrepancy is caused, amongst other things, by underestimating the K2K and MiniBooNE neutrino flux, or by the existence of unmodelled CCQE-like contributions (2p2h). So, increasing M_A may or may not be appropriate. The issue is far from being resolved and reflects a genuine physics uncertainty: the absolute neutrino cross section at around 1 GeV is known only to about 20-30%.

In any case, this (or any) T2K oscillation analysis is credible only if it is shown explicitly that it does not depend on semi-arbitrary generator choices, and that measurements from the near detector suite effectively constrain uncertainties in the product of flux and cross-section. It also needs to be shown that any difference in the oscillation results is well-understood in terms of the quoted cross-section systematics. This is the main motivation behind pursuing this analysis using both GENIE and NEUT.

2.3. Implementation details

So far, the SuperK spectrum prediction method has been presented in a formal manner. In practice, the procedure for calculating the spectrum can be simplified. By using appropriate Monte Carlo 'templates', representing the simulated number of events in N-dimensional bins of kinematical quantities, one can avoid performing integrations such as the one appearing in Eq. 6[4]

In this analysis, oscillation parameters are determined by fitting the reconstructed energy (E_r) spectrum of 1-ring μ-like events observed at SuperK, oscillation weights are functions of the true energy (E_t), and all considered systematics affecting the SuperK spectrum prediction either have no kinematical dependence or depend on only one of true or reconstructed energy. Therefore, in the context of this analysis, it is sufficient to use 2-dimensional templates representing the number of SuperK events, as estimated by MC, as functions of E_r and E_t. Let $N^{MC}_{SK;m;r;t}$ be such a template for events of mode m in the 1 μ-like ring sample, where indices r and t denote, respectively, reconstructed and true energy bin numbers. The construction of these templates is discussed in detail in Sec. 2.4. The predicted number $N^{exp}_{SK;r}$ of SuperK events in the rth reconstructed energy bin can be written in terms of the $N^{MC}_{SK;m;r;t}$ templates as

$$N^{exp}_{SK;r} = C_{ND280} \sum_m \sum_t P^{surv}_{m;t} N^{MC}_{SK;m;r;t} \tag{14}$$

where C_{ND280} is the ND280 correction factor defined in Eq. 12 and $P^{surv}_{m;t}$ is oscillation survival probability applied in the tth true energy bin of the SuperK MC template which corresponds to mode m. Application of term $P^{surv}_{m;t}$ is discussed in Sec. 2.6.

Only the value of C_{ND280} is needed for the prediction of the nominal SuperK spectrum. However, for systematic studies, one needs to be able to calculate how the value of C_{ND280} is modified as a result of systematic parameter variations. Therefore it is necessary to express C_{ND280} in terms of appropriate ND280 MC templates. In this analysis, all considered systematics affecting C_{ND280} either have no kinematical dependence or depend on true energy only.

[4]On the SuperK side of the calculation, avoiding integrations such as the one in Eq. 6 using MC templates is just a matter of convenience and computational efficiency. However, on the ND280 side, avoiding such integrations and, instead, using MC templates is a necessity. Indeed, for ND280, event rate calculations folding in the flux and neutrino cross section to should be generalized to take into account the effects arising from the convolution of a non-uniform flux with non-uniform detector mass distribution. In this case, the event rate can be written as

$$N^{exp} = N^{POT} N_A \int d^3x \int d^3p \frac{d^6\Phi(\mathbf{x},\mathbf{p})}{d^3x d^3p} \epsilon(E) \sum_a \frac{\sigma_a(E)}{A_a} \int dl \rho_a \tag{13}$$

where the vector $\mathbf{x}$ is the flux neutrino point of origin and the vector $\mathbf{p}$ is the flux neutrino momentum. In the above equation, the sum over a is the sum over nuclear targets in the detectors. The mass number of each nuclear target is A_a and the total interaction cross section is $\sigma_a(E)$. The final integral in Eq. 13 calculates the density-weighted path length for nuclear target a with density ρ_a along the given flux netrino trajectory. This complicated integration is already performed by GENIE and NEUT using Monte Carlo techniques and the result is built into the generated templates. It is practically impossible and unnecessary to reproduce this time-consuming Monte Carlo integration in the context of the oscillation analysis.

Therefore, in the context of this analysis, it is sufficient to use 1-dimensional templates representing the number of ND280 events, as estimated by MC, as functions of E_t. Let $N^{MC}_{ND280;m\prime;t}$ be such a template for events of mode $m\prime$ in the ND280 CC inclusive sample, where index t denotes true energy bin number. The construction of these templates is discussed in Sec. 2.5. Using the ND280 MC templates, C_{ND280} can be written as

$$C_{ND280} = N^{obs}_{ND280;CC-like} / \sum_{m\prime} N^{MC}_{ND280;m\prime} \tag{15}$$

Using Eq. 15, Eq. 14 can be written as

$$N^{exp}_{SK;r} = (N^{obs}_{ND280;CC-like} / \sum_{m\prime} N^{MC}_{ND280;m\prime}) \cdot \sum_{m} \sum_{t} P^{surv}_{m;t} N^{MC}_{SK;m;r;t} \tag{16}$$

This is the actual formula used in this analysis for predicting the SuperK reconstructed energy spectrum.

2.4. Construction of SuperK MC templates

The $N^{MC}_{SK;m;r;t}$ templates are shown in Fig. 2 (GENIE) and Fig. 3 (NEUT). Separate templates are used for ν_μ CCQE[5], ν_μ CC1π, ν_μ CC other, $\nu_\mu(\nu_\tau)$ NC, $\bar{\nu_\mu}$ CCQE, $\bar{\nu_\mu}$ CC1π, $\bar{\nu_\mu}$ CC other, $\bar{\nu_\mu}(\bar{\nu_\tau})$ NC, ν_e CCQE, ν_e CC1π, ν_e CC other and ν_e NC. The events contributing to the construction of these 2-D templates are the ones that pass the '1-ring μ-like (tight)' cuts. The templates span the 0-10 GeV range in both E_r and E_t variables using 50 MeV bins.

The templates were generated using the 2010a GENIE and NEUT SuperK samples and then they were reweighted to the 2010d(v2) flux [21]. Details of the 2010a GENIE and NEUT SuperK samples and of the applied selection cuts are given below.

SuperK MC samples

- GENIE SuperK sample:
  The 2010a GENIE SuperK Monte Carlo sample was generated at the Rutherford Appleton Laboratory PPD Tier2 cluster using GENIE v2.6.0 and the 2010a JNUBEAM flux [4]. The GENIE decay flags for D^+, D^-, D^0, $\bar{D^0}$, D_s^+, D_s^-, Λ_c^+, Σ_c^0, Σ_c^+ and Σ_c^{++} were set to 'true' (non-default GENIE settings) as the SuperK detector simulation would not decay charmed hadrons. Default GENIE v2.6.0 values were used for all other physics options.

  The GENIE vector files were proccessed by P. de Perio at the SciNet cluster in Toronto. The SK software used is the same as that used for the official 10a NEUT production (SKDETSIM-v11p79_for_atmnu_apr10, ATMPD-10a, SKOFL-10a, ROOT-v5.26) [11]. The 'T2K PMT flasher reduction' cut was not included when filling the output DSTs. According to experts 'this is a small effect ($< 1\%$)' [13].

  Using the GENIE v2.6.0 cross section calculations for water and the 2010a JNUBEAM flux, we calculate 1325.4200 ν_μ events / 22.5 kt fiducial mass / 10^{21} POT, 65.1023 $\bar{\nu_\mu}$ events / 22.5 kt fiducial mass / 10^{21} POT and 25.9600 beam ν_e events / 22.5 kt fiducial mass / 10^{21} POT. After applying fiducial volume cuts to the generated samples, we find that 604403 ν_μ, 303747 $\bar{\nu_\mu}$ and 621525 beam ν_e events have survived. From the above, we calculate the following absolute normalizations for the 2010a samples: 4.560000×10^{23} POT for the ν_μ sample, 4.665680×10^{24} POT for the $\bar{\nu_\mu}$ sample and 2.394164×10^{25} POT for the beam ν_e sample. Therefore, the MC scaling factors to the data beam exposure (3.23×10^{19} POT) are 7.08333×10^{-5} (ν_μ), 6.92289×10^{-6} ($\bar{\nu_\mu}$), 1.34911×10^{-6} (beam ν_e).

- NEUT SuperK sample:
  The 2010a NEUT SuperK Monte Carlo sample was generated and processed at the Kamioka cluster. NEUT v5.0.7 and the 2010a JNUBEAM flux were used. The corresponding SuperK DSTs were kindly provided by P. de Perio. The MC scaling factors to the data POT exposure are the ones provided by Y.Hayato and J.Kameda [12]: 7.35361×10^{-5} (ν_μ), 7.17816×10^{-6} ($\bar{\nu_\mu}$), 1.48428×10^{-6} (beam ν_e).

[5] A commonly asked question is why the ν_μ-CCQE template is not just a 'line' along the diagonal. Conservation of energy and momentum is the anwer to this question. Recall that the ν_μ-CCQE template is generated from a sample of ν_μ interactions in water and that, in oxygen, the Fermi level k_F is 225 MeV. On the other hand, recall that the hit bound-nucleon momentum is 'esoteric' and that the reconstructed neutrino energy formula naturally assumes that the hit nucleon was at rest.

Fully contained fiducial volume	1 μ-like ring	1 μ-like ring (tight)	>1 μ-like ring
Number of PMT hits in highest charge Outer Detector cluster $<=15$	Number of PMT hits in highest charge Outer Detector cluster $<=15$	Number of PMT hits in highest charge Outer Detector cluster $<=15$	Number of PMT hits in highest charge Outer Detector cluster $<=15$
Vertex $>$ 2m from Inner Detector wall	Vertex $>$ 2m from Inner Detector wall	Vertex $>$ 2m from Inner Detector wall	Vertex $>$ 2m from Inner Detector wall
$E_{visible} > 30$ MeV	$E_{visible} > 30$ MeV	$E_{visible} > 30$ MeV	$E_{visible} > 30$ MeV
	Exactly 1 ring	Exactly 1 ring	>1 ring
	ip[0] = 3 (μ-like ring)	ip[0] = 3 (μ-like ring)	Ring with largest amome is μ-like
		amomm[0] $>$ 200 MeV	
		$<=1$ decay e^-	

Table 1: Standard SuperK ν_μ event selection cuts. In this analysis, the 1-ring μ-like (tight) event selection is used. Potential benefits from including the multi-ring μ-like events will be explored in future revision of this work.

SuperK selection cuts

The standard SuperK cuts for obtaining a) 'fully contained fiducial volume (FCFV)', b) '1 μ-like ring', c) '1 μ-like ring (tight)' and d) '>1 μ-like ring' samples are shown in Tab. 1. In this analysis, the 1 μ-like ring (tight) event selection is used.

2.5. ND280 MC templates

The ND280 MC templates were provided directly by the ND280 group [14]. In total 21 histograms were provided representing the number of events as a function of true energy for 1) ν_μ CC QE, 2) ν_μ CC resonance, 3) ν_μ CC coherent, 4) ν_μ CC other, 5) ν_μ NC resonance, 6) ν_μ NC coherent, 7) ν_μ NC other, 8) $\bar{\nu_\mu}$ CC QE, 9) $\bar{\nu_\mu}$ CC resonance, 10) $\bar{\nu_\mu}$ CC coherent, 11) $\bar{\nu_\mu}$ CC other, 12) $\bar{\nu_\mu}$ NC resonance, 13) $\bar{\nu_\mu}$ NC coherent, 14) $\bar{\nu_\mu}$ NC other, 15) ν_e CC QE, 16) ν_e CC resonance, 17) ν_e CC coherent, 18) ν_e CC other, 19) ν_e NC resonance, 20) ν_e NC coherent and 21) ν_e NC other. These templates were generated by applying the Saclay CC inclusive selection cuts to the ND280 MCP1 GENIE and NEUT samples. The GENIE and NEUT versions used in the ND280 MCP1 production are the same as those used at the SuperK 2010a production. They were reweighted to the 2010d(v2) flux [21]. More details are given in [14].

2.6. Applying neutrino oscillations

Oscillations are applied to the ν_μ-CCQE, ν_μ-CC1π, ν_μ-CCother, $\bar{\nu_\mu}$-CCQE, $\bar{\nu_\mu}$-CC1π and $\bar{\nu_\mu}$-CCother MC templates (as function of E_t). Oscillations are not applied to the ν_μ-NC and $\bar{\nu_\mu}$-NC MC templates since they serve as proxies for the $\nu_\mu(+\nu_\tau)$-NC and $\bar{\nu_\mu}(+\bar{\nu_\tau})$-NC MC templates which are unchanged under $\nu_\mu \rightarrow \nu_\tau$ and $\bar{\nu_\mu} \rightarrow \bar{\nu_\tau}$ oscillations. We neglect contributions from ν_τ-CC and $\bar{\nu_\tau}$-CC as their energy threshold is around 3.5 GeV and their effect is negligible. The effect of sub-dominant (and unknown) ν_e oscillation is also neglected. The following oscillation survival probability is applied:

$$P_{surv} = 1 - sin^2 2\theta_{23} \cdot sin^2\Big(1.267 \cdot \frac{\Delta m^2_{23}(eV^2/c^4) \cdot 295.335(km)}{E_t(GeV)}\Big) \tag{17}$$

The same $sin^2 2\theta_{23}$ and Δm^2_{23} parameters are used for both ν_μ and $\bar{\nu_\mu}$ oscillations.

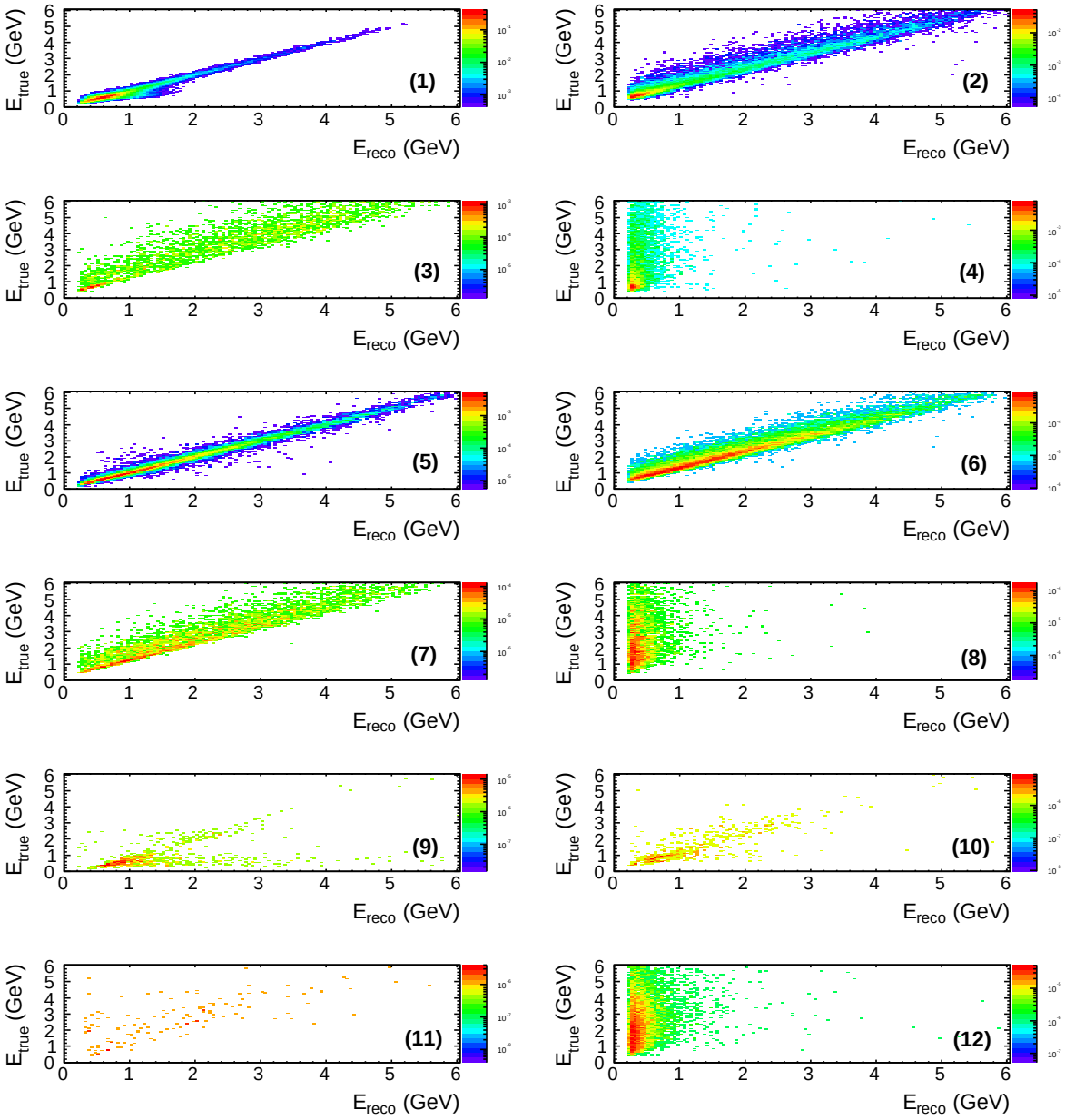


Figure 2: GENIE (E_t, E_r) 1-ring μ-like MC templates. The templates are shown for ν_μ CCQE (1), ν_μ CC1π (2), ν_μ CC other (3), $\nu_\mu(\nu_\tau)$ NC (4), $\bar{\nu_\mu}$ CCQE (5), $\bar{\nu_\mu}$ CC1π (6), $\bar{\nu_\mu}$ CC other (7), $\bar{\nu_\mu}(\bar{\nu_\tau})$ NC (8), ν_e CCQE (9), ν_e CC1π (10), ν_e CC other (11) and ν_e NC (12). The templates are zoomed-in for illustration purposes. The range 0-10 GeV is used in the analysis.

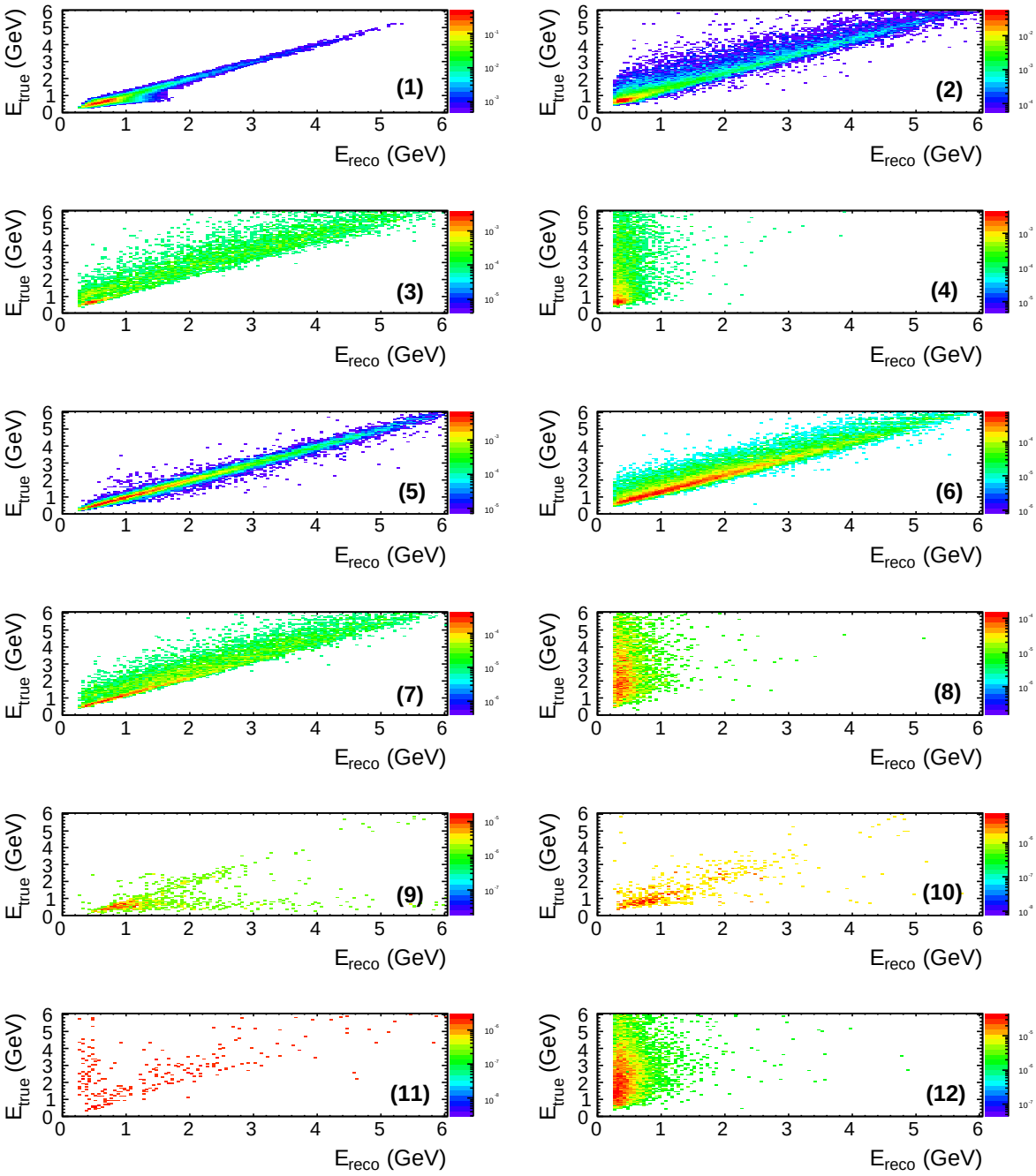


Figure 3: NEUT (E_t, E_r) 1-ring μ-like MC templates. The order of templates is the same as in Fig.2.

2.7. Nominal GENIE and NEUT expectations under various oscillation hypotheses

In this section we present plots and tables comparing the GENIE and NEUT SuperK predictions (after extrapolation) under various oscillation hypotheses. Figs. 4 and 5 show, respectively, the GENIE and NEUT predicted (after extrapolation) non-oscillated reconstructed-energy spectrum of 1-ring μ-like events passing the tight cuts (see Tab. 1). The contributions from various true neutrino reaction modes are also shown. The corresponding oscillated predictions ($sin^2 2\theta_{23}$=1.0 and Δm^2_{23}=2.4$\times 10^{-3}$ eV^2/c^4) are shown in Fig. 6 for GENIE and Fig. 7 for NEUT. In Tab. 2 we summarize Figs. 4 - 7.

In Fig. 8 we show the GENIE predicted 1-ring μ-like event reconstructed energy spectrum (after extrapolation) for a fixed Δm^2_{23} and various $sin^2 2\theta_{23}$ values. The corresponding plot for NEUT is shown in Fig. 9. Tab. 3 summarizes Figs. 8 and 9. In Fig. 10 we show the GENIE predicted 1-ring μ-like event reconstructed energy spectrum (after extrapolation) for a fixed $sin^2 2\theta_{23}$ and various Δm^2_{23} values. The corresponding plot for NEUT is shown in Fig. 11. Tab. 4 summarizes Figs. 10 and 11.

In Fig. 12 we show the GENIE predicted integrated number of 1-ring μ-like events (after extrapolation) for a grid of $sin^2 2\theta_{23}$ and Δm^2_{23} values. The corresponding plot for NEUT is shown in Fig. 13. Finally, in Fig. 14 we show (for the same grid of $sin^2 2\theta_{23}$ and Δm^2_{23} values) the percentage difference of the NEUT and GENIE predictions of integrated number of 1-ring μ-like events (after extrapolation).

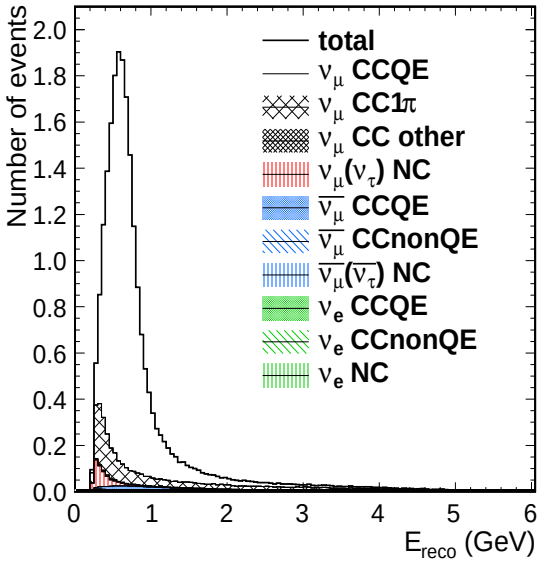


Figure 4: Predicted non-oscillated reconstructed-energy spectrum of 1-ring μ-like events (after extrapolation) and contributions from various true neutrino reaction modes (GENIE).

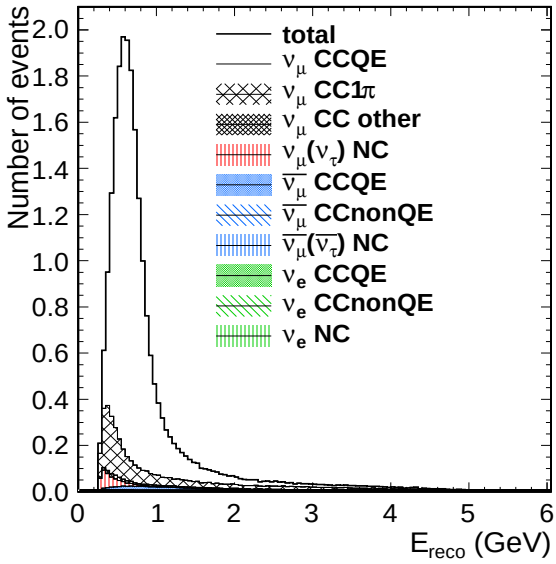


Figure 5: Predicted non-oscillated reconstructed-energy spectrum of 1-ring μ-like events (after extrapolation) and contributions from various true neutrino reaction modes (NEUT).

	(a)		(b)	
	GENIE	NEUT	GENIE	NEUT
Total	22.4193	22.8136	6.2043	6.3395
ν_μ CCQE	18.0269 (80.408%)	18.0707 (79.210%)	3.5785 (57.678%)	3.5944 (56.699%)
ν_μ CC1π	2.8527 (12.724%)	3.0883 (13.537%)	1.4488 (23.352%)	1.5163 (23.918%)
ν_μ CCother	0.2995 (1.336%)	0.5219 (2.288%)	0.2514 (4.052%)	0.4056 (6.398%)
$\nu_\mu(\nu_\tau)$ NC	0.4720 (2.106%)	0.3767 (1.651%)	0.4720 (7.608%)	0.3767 (5.942%)
$\bar{\nu_\mu}$ CCQE	0.4854 (2.165%)	0.5027 (2.203%)	0.2375 (3.828%)	0.2526 (3.985%)
$\bar{\nu_\mu}$ CCnonQE	0.2383 (1.063%)	0.2160 (0.947%)	0.1716 (2.765%)	0.1564 (2.468%)
$\bar{\nu_\mu}(\bar{\nu_\tau})$ NC	0.0273 (0.122%)	0.0219 (0.096%)	0.0273 (0.440%)	0.0219 (0.345%)
ν_e CCQE	0.0016 (0.007%)	0.0017 (0.008%)	0.0016 (0.026%)	0.0017 (0.027%)
ν_e CCnonQE	0.0012 (0.005%)	0.0015 (0.006%)	0.0012 (0.020%)	0.0015 (0.023%)
ν_e NC	0.0144 (0.064%)	0.0124 (0.054%)	0.0144 (0.232%)	0.0124 (0.195%)

Table 2: Expected numbers of SuperK 1-ring μ-like events from GENIE and NEUT (after extrapolation) for (a) no oscillation and (b) oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4.

E_{reco}	(a)		(b)		(c)		(d)		(e)	
(GeV)	GENIE	NEUT	GENIE	NEUT	GENIE	NEUT	GENIE	NEUT	GENIE	NEUT
< 0.50	7.442	6.526	1.691	1.379	2.841	2.408	3.991	3.438	5.142	4.467
< 0.75	15.528	15.114	2.490	2.239	5.098	4.814	7.705	7.389	10.313	9.964
< 1.00	18.392	18.315	3.252	3.065	6.280	6.115	9.308	9.165	12.336	12.215
< 1.50	20.101	20.245	4.180	4.100	7.364	7.329	10.548	10.558	13.732	13.787
< 2.00	20.839	21.063	4.751	4.729	7.969	7.996	11.186	11.263	14.404	14.592
< 4.00	22.068	22.417	5.864	5.956	9.105	9.248	12.346	12.540	15.586	15.833
< 10.00	22.419	22.814	6.204	6.340	9.447	9.634	12.690	12.929	15.933	16.224

Table 3: Expected numbers of SuperK 1-ring μ-like events from GENIE and NEUT (after extrapolation) for various reconstructed energy slices and oscillation hypotheses. The non-oscillated expectations are shown in (a), while (b)-(e) show the oscillated expectation for Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 and various neutrino mixing angles: (b) $sin^2 2\theta_{23}$ = 1.0, (c) $sin^2 2\theta_{23}$ = 0.8, (d) $sin^2 2\theta_{23}$ = 0.6 and (e) $sin^2 2\theta_{23}$ = 0.4.

E_{reco}	(a)		(b)		(c)		(d)		(e)	
(GeV)	GENIE	NEUT	GENIE	NEUT	GENIE	NEUT	GENIE	NEUT	GENIE	NEUT
< 0.50	7.442	6.526	1.416	1.170	1.691	1.379	3.752	3.252	3.919	3.245
< 0.75	15.528	15.114	3.128	2.977	2.490	2.239	5.070	4.718	6.501	6.189
< 1.00	18.392	18.315	4.303	4.264	3.252	3.065	5.357	5.038	8.336	8.203
< 1.50	20.101	20.245	5.421	5.516	4.180	4.100	5.909	5.647	9.136	9.139
< 2.00	20.839	21.063	6.039	6.198	4.751	4.729	6.364	6.145	9.213	9.231
< 4.00	22.068	22.417	7.186	7.462	5.864	5.956	7.385	7.269	9.623	9.672
< 10.00	22.419	22.814	7.529	7.850	6.204	6.340	7.716	7.641	9.868	9.947

Table 4: Expected numbers of SuperK 1-ring μ-like events from GENIE and NEUT (after extrapolation) for various reconstructed energy slices and oscillation hypotheses. The non-oscillated expectations are shown in (a), while (b)-(e) show the oscillated expectation for $sin^2 2\theta_{23}$ = 1.0 and squared neutrino mass differences: (b) Δm^2_{23} = 2.0×10^{-3} eV^2/c^4, (c) Δm^2_{23} = 2.4×10^{-3} eV^2/c^4, (d) Δm^2_{23} = 3.3×10^{-3} eV^2/c^4 and (e) Δm^2_{23} = 8.0×10^{-3} eV^2/c^4

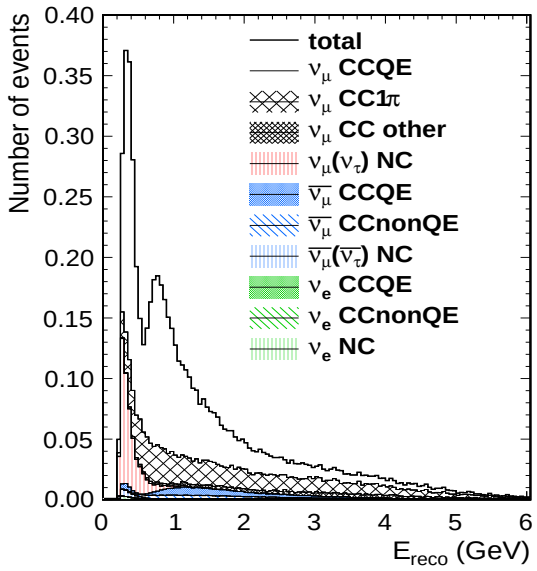


Figure 6: Predicted oscillated ($sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4) reconstructed-energy spectrum of 1-ring μ-like events (after extrapolation) and contributions from various true neutrino reaction modes (GENIE).

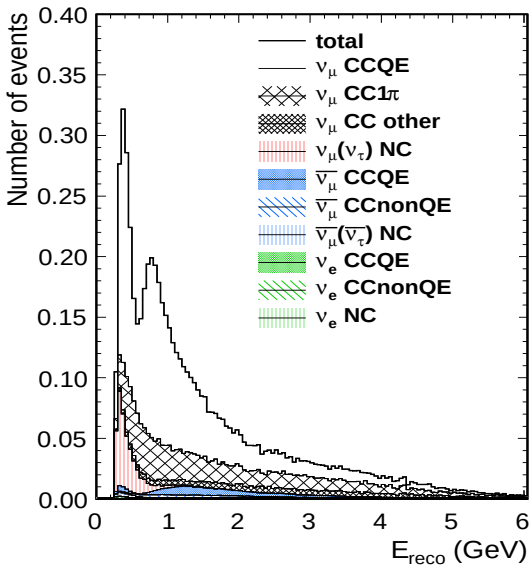


Figure 7: Predicted oscillated ($sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4) reconstructed-energy spectrum of 1-ring μ-like events (after extrapolation) and contributions from various true neutrino reaction modes (NEUT).

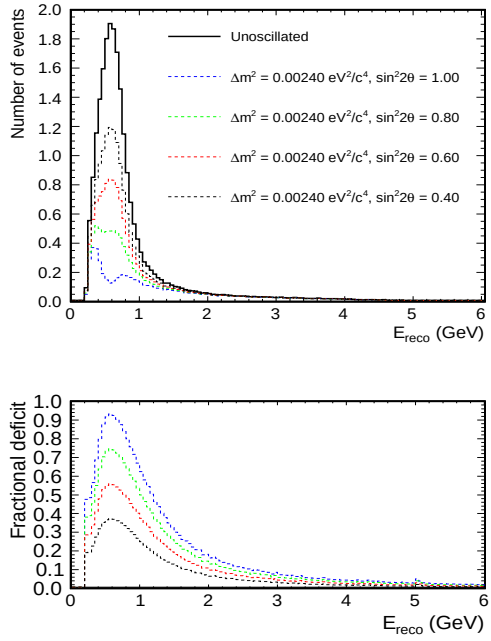


Figure 8: Predicted reconstructed-energy spectrum of 1-ring μ-like events (after extrapolation) for Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 1.0, 0.8, 0.6, 0.4. The non-oscillated spectrum is also shown (GENIE).

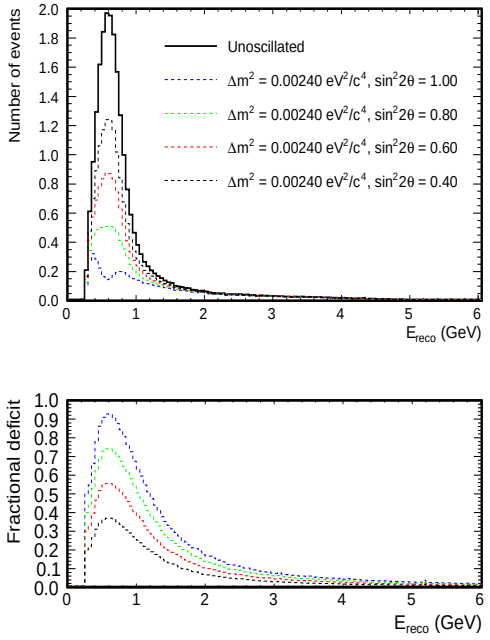


Figure 9: Predicted reconstructed-energy spectrum of 1-ring μ-like events (after extrapolation) for Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 1.0, 0.8, 0.6, 0.4. The non-oscillated spectrum is also shown (NEUT).

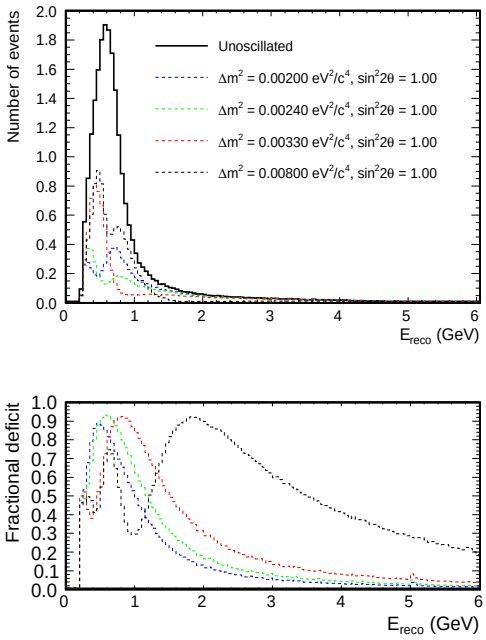


Figure 10: Predicted reconstructed-energy spectrum of 1-ring μ-like events (after extrapolation) for $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.0×10^{-3} eV^2/c^4, 2.4×10^{-3} eV^2/c^4, 3.3×10^{-3} eV^2/c^4 and 8.0×10^{-3} eV^2/c^4. The non-oscillated spectrum is also shown (GENIE).

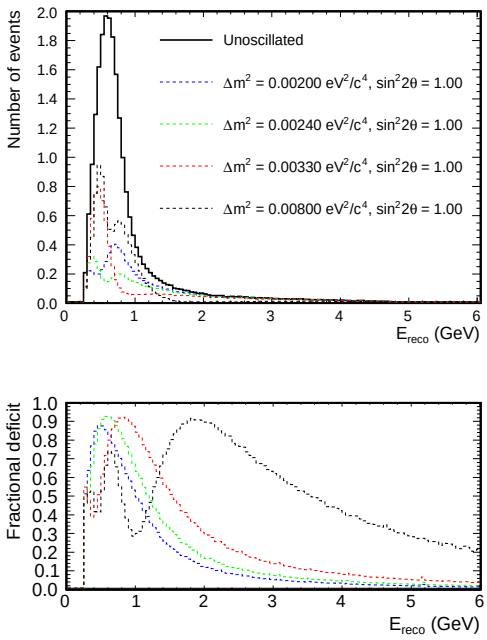


Figure 11: Predicted reconstructed-energy spectrum of 1-ring μ-like events (after extrapolation) for $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23}= 2.0×10^{-3} eV^2/c^4, 2.4×10^{-3} eV^2/c^4, 3.3×10^{-3} eV^2/c^4 and 8.0×10^{-3} eV^2/c^4. The non-oscillated spectrum is also shown (NEUT).

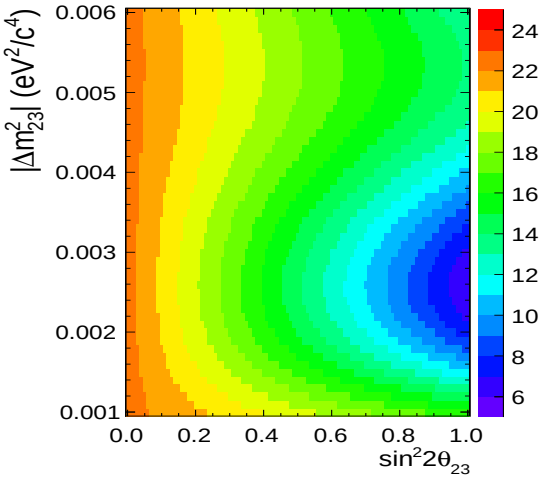


Figure 12: Expected number of 1-ring μ-like events (after extrapolation) in the 0-10 GeV reconstructed energy range (GENIE).

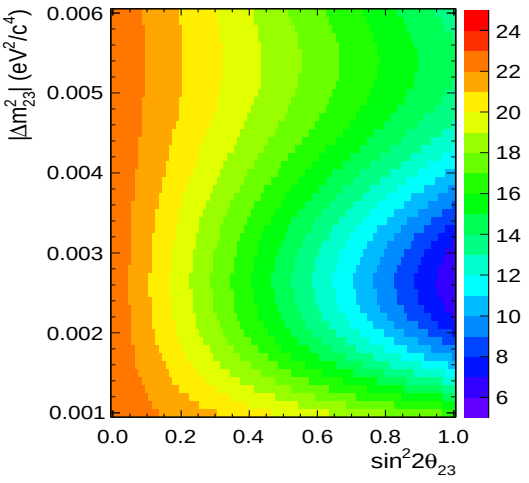


Figure 13: Expected number of 1-ring μ-like events (after extrapolation) in the 0-10 GeV reconstructed energy range (NEUT).

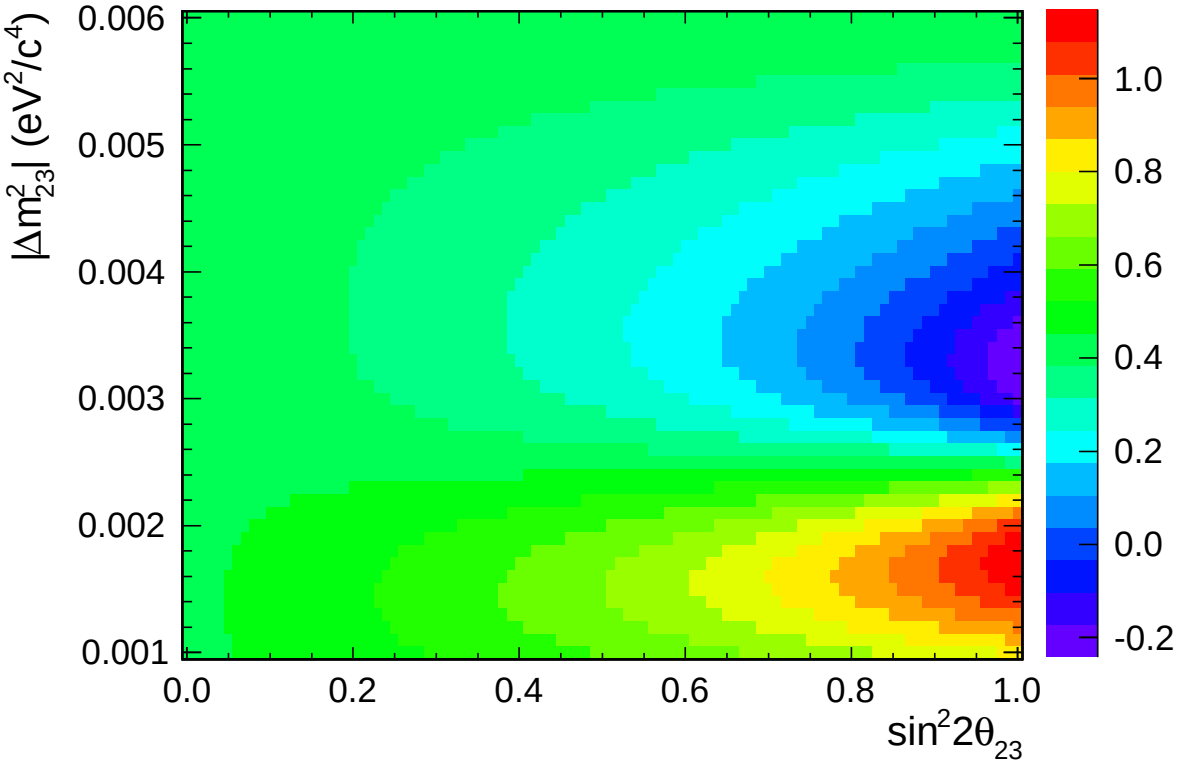


Figure 14: Percentage difference $100 \cdot (N_{SK}^{NEUT} - N_{SK}^{GENIE})/(N_{SK}^{NEUT} + N_{SK}^{GENIE})/2$ between the GENIE and NEUT expected numbers of 1-ring μ-like events (after extrapolation) in the 0-10 GeV reconstructed energy range. The percentage difference in the -0.2% to +1.2% range (Compare with Fig. 18).

3. Oscillation fitting method

3.1. Outline

Measurements of the oscillation parameters $sin^2 2\theta_{23}$, Δm^2_{23} are obtained by comparing the observed and predicted SuperK reconstructed energy spectra for 1-ring μ-like events. Let N be the number of reconstructed energy bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the SuperK reconstructed energy spectrum prediction (e.g parameters that control the beam and neutrino interaction simulations and the detector response). Best-fit values are obtained by minimizing:

$$\chi^2(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a}) = 2 \cdot \sum_{i=0}^{N-1} \left(n_i^{obs} \cdot ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp} - n_i^{obs}) \right) + (\mathbf{a} - \mathbf{a_0})^T \cdot \mathbf{C}^{-1} \cdot (\mathbf{a} - \mathbf{a_0}) \quad (18)$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of considered systematics parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, n_i^{obs} is the observed number of events in the ith bin, and $n_i^{exp} = n_i^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$ is the corresponding expected number of events.

Due to the limited statistics in SuperK and the limited amount of information available to this analysis from the near detectors, we do not fit for systematics. Instead, we evaluate oscillation systematics as described in section 4.

3.2. Construction of the likelihood function and comparison of the 'MINOS' and 'K2K' methods

In this analys, we use a fitting method previously employed by MINOS [5]. The construction of the likelihood function is described in this section in detail. We use a likelihood ratio method, whereas J.Kameda's analysis uses a maximum likelihood method previously employed by K2K [6]. These two methods give identical results for the best-fit oscillation parameters (for same input p.d.f's).

Let $\mathbf{n^{obs}}$ be a set of N measurements (the number of events in each of the N bins [6]). The probability density function (p.d.f.) for the total number of events $n_{tot}^{obs} = \sum_{i=0}^{N-1} n_i^{obs}$ observed in SuperK has a Poisson distribution, and the mean of that distribution is equal to the predicted total number of events $n_{tot}^{exp} = n_{tot}^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a}) = \sum_{i=0}^{N-1} n_i^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$. The p.d.f. is

$$f_{poisson} = f_{poisson}(\mathbf{n^{obs}}; sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a}) = \frac{[n_{tot}^{exp}]^{n_{tot}^{obs}} \, e^{-n_{tot}^{exp}}}{n_{tot}^{obs}!} \quad (19)$$

The reconstructed energies of the observed events are distributed over N bins. Since there are N possible outcomes for the process of placing an event in a bin, the p.d.f. of the distribution of those outcomes is multinomial:

$$f_{multinomial} = f_{multinomial}(\mathbf{n^{obs}}; sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a}) = n_{tot}^{obs}! \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} P_i^{n_i^{obs}} \quad (20)$$

where $P_i = P_i(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$ is the probability that an event is placed in the ith bin. This probability can be calculated as the ratio of the number of events predicted for that bin to the predicted total number of events:

$$P_i = \frac{n_i^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})}{n_{tot}^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})} \quad (21)$$

Therefore, $f_{multinomial}$ can be written as

$$f_{multinomial} = n_{tot}^{obs}! \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} \left[\frac{n_i^{exp}}{n_{tot}^{exp}} \right]^{n_i^{obs}} \quad (22)$$

The joint p.d.f. $f_{joint} = f(\mathbf{n^{obs}}; sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$ for the outcome of the reconstructed energy distribution is

[6] In this analysis we use 200 50-MeV bins in the 0-10 GeV reconstructed neutrino energy range.

the product of the Poisson and multinomial distributions shown in Eq. 19 and Eq. 22 respectively:

$$f_{joint} = e^{-n_{tot}^{exp}} \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} [n_i^{exp}]^{n_i^{obs}} \tag{23}$$

In our analysis, we use the p.d.f. in Eq. 23, and this is identical to the likelihood function used by K2K in [6]: L_{NORM} corresponds to the Poisson term and L_{SHAPE} to the multinomial term. However there is also a difference between the two analyses. K2K used the "maximum likelihood" method, i.e. the best-fit values of the oscillation parameters were obtained by taking the natural logarithm of Eq. 23, and maximising it with respect to those parameters. We use the "likelihood ratio" method, which involves dividing the p.d.f. in Eq. 23 by a factor that does not depend on the oscillation parameters. The best-fit values of the oscillation parameters are unchanged by such a division (see [7] for further details).

In our analysis, the p.d.f. in Eq. 23 is divided by

$$f_0 = f_0(\mathbf{n^{obs}}) = \frac{[n_{tot}^{obs}]^{n_{tot}^{obs}} e^{-n_{tot}^{obs}}}{n_{tot}^{obs}!} n_{tot}^{obs}! \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} \left[\frac{n_i^{obs}}{n_{tot}^{obs}}\right]^{n_i^{obs}} \tag{24}$$

Cancelling the factors of $n_{tot}^{obs}!$ and using $n_{tot}^{obs} = \sum_{i=0}^{N-1} n_i^{obs}$, Eq. 24 simplifies to

$$f_0 = e^{-n_{tot}^{obs}} \prod_{i=0}^{N-1} \frac{1}{n_i^{obs}!} [n_i^{obs}]^{n_i^{obs}} \tag{25}$$

Dividing Eq. 23 by Eq. 25 gives the likelihood ratio λ:

$$\lambda = e^{n_{tot}^{obs} - n_{tot}^{exp}} \prod_{i=0}^{N-1} \left(\frac{n_i^{exp}}{n_i^{obs}}\right)^{n_i^{obs}} \tag{26}$$

In the large-sample limit, the quantity

$$-2\ln\lambda = 2 \sum_{i=0}^{N-1} \left[n_i^{obs} \ln\left(\frac{n_i^{obs}}{n_i^{exp}}\right) + n_i^{exp} - n_i^{obs} \right] \tag{27}$$

has a χ^2 distribution. Best-fit values of the oscillation parameters are obtained by varying their values in the input to n_i^{exp} until the minimum of -2lnλ is found. The reader should compare Eq. 27 with the statistical term of Eq. 18 shown in previous section.

3.3. Oscillation parameter range

The search for the minimum value of $\chi^2(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$ (see Eq. 18) is performed in the range $0 \leq sin^2 2\theta_{23} \leq 1$ and $1 \times 10^{-3} eV^2/c^4 \leq |\Delta m^2_{23}| \leq 6 \times 10^{-3} eV^2/c^4$. Values of $sin^2 2\theta_{23} > 1$ are not permitted; although it is numerically allowed and, occasionally, it yields better fits to data, it has no physical interpretation within the context in which we attempt to interpret the data (2-flavour neutrino oscillations) and causes p.d.f's to have pathological behaviour. The maximum allowed value of $|\Delta m^2_{23}|$ is $6 \times 10^{-3} eV^2/c^4$. Due to the sinusoidal nature of the oscillation survival probability function, statistical fluctuations in the present small-statistics dataset can, very rarely, cause a global minimum in large $|\Delta m^2_{23}|$ ($> 6 \times 10^{-3} eV^2/c^4$). As a $|\Delta m^2_{23}|$ of $6 \times 10^{-3} eV^2/c^4$) is already more than 10 σ away from the MINOS central value we, a priori, prevent $\chi^2(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$ minima in the wrong trough of the survival probability function.

3.4. Calculation of confidence contours

In this work we have used two different methods for calculating confidence regions: (1) The PDG recommendation on quoting uncertainties on parameters measured jointly (see Sec. 3.4.1) and (2) The Feldman-Cousins method (see Sec. 3.4.2). The PDG method is used for illustration purposes only because of its simplicity. Due to concerns regarding confidence region coverage, the Feldman-Cousins method is used for all physics results. Both methods are discussed briefly.

3.4.1. Method 1: PDG recommendation

The PDG recommendation on quoting uncertainties on parameters measured jointly ($sin^2 2\theta_{23}$, Δm^2_{23} in our analysis) is described in [9]. It is a fully frequentist treatment using the Gaussian approximation. One finds the best-fit point and calculates the confidence contours as lines of constant $\Delta\chi^2$ from this point. Therefore, a confidence region is defined by:

$$\chi^2(sin^2 2\theta_{23}, \Delta m^2_{23}) < \chi^2_{min} + \Delta\chi^2 \tag{28}$$

Values of $\Delta\chi^2$ for obtaining confidence regions of given statistical significance are given in Tab. 5 for 1, 2 and 3 simultaneously-determined parameters. In future discussions we will be referring to the confidence regions obtained using this method as 'plain'.

		Confidence level			
		68%	**90%**	**95%**	**99%**
N_{params}	**1**	1.00	2.71	3.84	6.63
	2	2.30	4.61	5.99	9.21
	3	3.53	6.25	7.82	11.34

Table 5: Values of $\Delta\chi^2$ used for calculating confidence regions in the ν_μ-disappearance parameter space, for a given confidence level and a given number of fit parameters.

3.4.2. Method 2: The Feldman-Cousins method

The work of Feldman and Cousins [10] uses the freedom which is inherent in the Neyman construction of confidence regions, to address cases (in particular, cases with Poisson processes with background and Gaussian errors near a physical boundary) where classical construction of confidence regions produces overcoverage, undercoverage, or in extreme cases, an empty region. Using the Feldman-Cousins method, construction of the confidence region proceeds as follows:

The $\sin^2 2\theta_{23}$ - Δm^2_{23} plane is divided into a square grid of size $\sin^2 2\theta_{23}$ = 0.01, Δm^2_{23} = 0.1 $\times$ 10^{-3} eV^2/c^4. Grid points are taken to be the values of $\sin^2 2\theta_{23}$ and Δm^2_{23} at the centre of each grid square. For each grid point, sets of toy MC experiments were generated with the oscillation parameters of that grid point. Typically, 10000 toy MC experiments were made for each grid point including systematic variations, and 3000 datasets were made for each grid point without these systematic variations.

The best-fit values of the oscillation parameters are found for each MC experiment by minimising -2 lnλ in Eq. 27. It must be emphasised that, for the Feldman-Cousins method, the best-fit values are restricted to the physical region ($\sin^2 2\theta_{23} < 1$). Using these best-fit parameters, the value of $\Delta\chi^2_{MC}$ is calculated as follows:

$$\Delta\chi^2_{MC} = \chi^2(\text{true}) - \chi^2(\text{MC, best fit}) \tag{29}$$

where χ^2 is as defined in Eq. 18, "true" means the true values of the oscillation parameters at that grid point, and "best fit" means the best-fit values for that MC experiment. The quantity $\Delta\chi^2_{MC}$ in Eq. 29 is equivalent to that given in Eq. 5.6 in [10]; it is always >=0 since the best fit is at the minimum value of χ^2.

For each grid point, the critical value of $\Delta\chi^2$ ($\Delta\chi^2_{CRIT}$) is found such that α of the fake experiments have $\Delta\chi^2_{MC}$ $< \Delta\chi^2_{CRIT}$, where α is the desired confidence level. The principal difference between the Feldman-Cousins and PDG methods is that the former determines the value of $\Delta\chi^2_{CRIT}$ individually for each grid point, whereas the latter uses the same value for all grid points (see Tab. 5). It is possible to draw a Feldman-Cousins $\Delta\chi^2_{CRIT}$ surface in the $\sin^2(2\theta_{23}$ - Δm^2_{23} plane, and examples of such surfaces are shown in Fig. 15. From this figure, it can be seen that grid points near the physical boundary of $\sin^2(2\theta_{23}$ = 1 have lower values of $\Delta\chi^2_{CRIT}$ than points in the middle of the grid; this is due to the requirement that best-fit points be within the physical region. In an unconstrained fit, some best-fit points would lie outside the physical region; requiring them to be in the physical region causes them to pile up at the physical boundary, where they have lower values of $\Delta\chi^2_{MC}$ than they would have in an unconstrained fit.

Using the T2K 2010a data, Feldman-Cousins confidence contours are drawn such that grid points with $\Delta\chi^2_{DATA}$

$< \Delta\chi^2_{CRIT}$ are included within the contour, where $\Delta\chi^2_{DATA}$ is given by

$$\Delta\chi^2_{DATA} = \chi^2(\text{true}) - \chi^2(\text{data, best fit}) \tag{30}$$

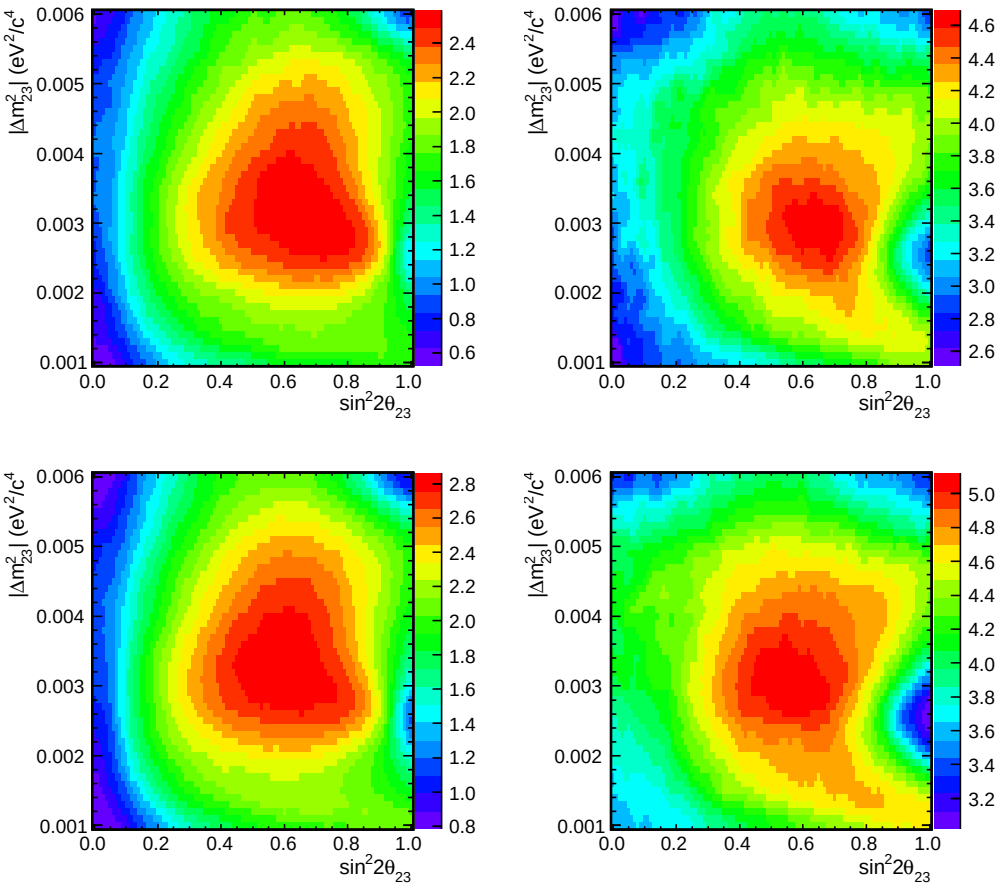


Figure 15: Monte Carlo estimation, according to the Feldman-Cousins method, of $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right) as functions of $sin^2 2\theta_{23}$ and Δm^2_{23}, for calculation of the 68% CL and 90% CL, respectively, confidence regions for the 3.23×10^{19}-POT T2K ν_μ-disappearance analysis. The top row shows $\Delta\chi^2$ surfaces calculated using statistical errors only whereas the bottom row shows $\Delta\chi^2$ surfaces calculated using both statistical and systematic erros. We observe that values of $\Delta\chi^2$ decrease as one moves closer to the physical boundaries at $sin^2 2\theta_{23}$ = 0 and $sin^2 2\theta_{23}$ = 1 and the fit ('search'/'sensitivity') Δm^2_{23} boundaries.

4. Oscillation systematics

4.1. Inputs uncertainties for systematic error evaluation

The uncertainties considered in this analysis are summarized in Tab. 6. In the presence of these systematic uncertainties, Eq. 16 may be re-written as

$$N^{exp}_{SK;r} = f_{norm} \cdot f_{FSI;r} \cdot (1 + \Delta N_{SK;r}(f^{SK}_E)) \frac{N^{obs}_{ND280;CC-like}}{f^{ND280}_\epsilon \sum_{m\prime} N^{MC}_{ND280;m\prime} \cdot \left(\prod_{m_s} f_{R_\sigma;m_s} \delta_{m\prime;m_s} \right)} \cdot$$
$$\sum_{m,t} P^{surv}_{m;t} \cdot N^{MC}_{SK;m;r;t} \cdot \left(\prod_{m_s} f_{R_\sigma;m_s} \delta_{m;m_s} \right) \cdot \left(f_{\sigma_{CCQE};t} \delta_{m;CCQE} \right) \cdot w^{Ch.}_{m;t}(\rho^{\mathbf{\Phi_{SK}}}) \tag{31}$$

As explained in Tab. 6, the energy scale uncertainty is taken into account by scaling the reconstructed energy distribution bin edges and, assuming uniform distribution of events within the 50 MeV bins, calculating the number of

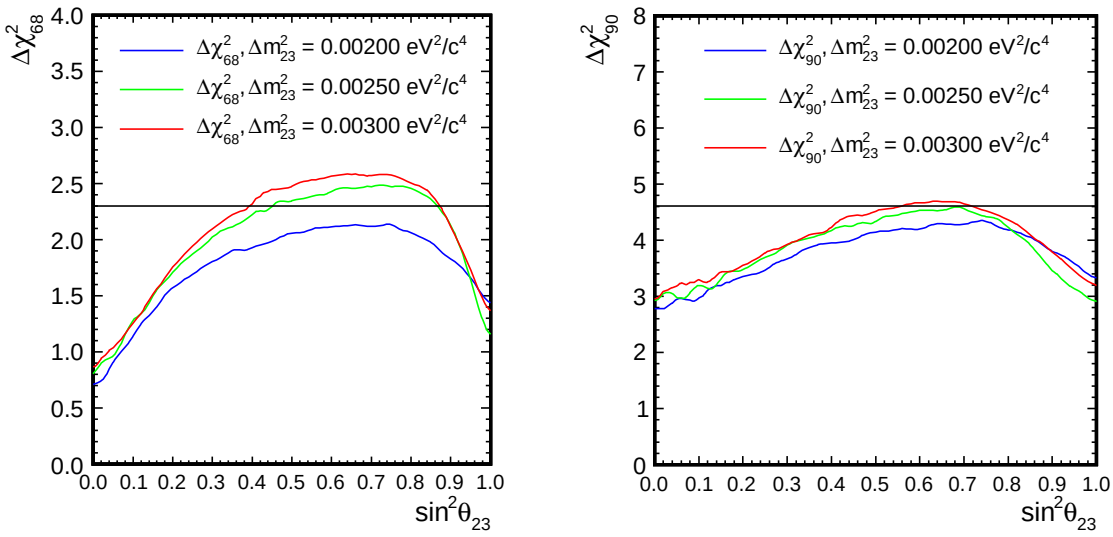


Figure 16: Monte Carlo estimation, according to the Feldman-Cousins method (statistical errors only), of $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right), as functions of $sin^2 2\theta_{23}$ for 3 different values of Δm^2_{23} ($2.0 \times 10^{-3} eV^2/c^4$, $2.5 \times 10^{-3} eV^2/c^4$ and $3.0 \times 10^{-3} eV^2/c^4$). For comparison, the canonical value of $\Delta\chi^2$ used in the PDG (see Tab. 5) is also shown as a horizontal line.

events gained from / lost to neighbouring bins. The term $\Delta N_{SK;r}$ in above equation expresses the net gain of events at the reconstructed energy bin r as a result of SuperK energy scale variations. The term $f_{R_\sigma;m_s}$, where the index m_s runs over CC1π, CCoth and NC, denotes the MC template modification factor due to the uncertainty on any of the cross section ratios (relative to CCQE) described in Tab. 6, while the term $\delta_{m\prime;m}$ is 1 for $m\prime = m$ and 0 otherwise. Finally, $w^{Ch.}_{m;t}(\rho^{\mathbf{\Phi_{SK}}})$ denotes flux weights generated from the SuperK ν_μ flux true energy bin-to-bin correlation matrix $\rho^{\mathbf{\Phi_{SK}}}$ using the Cholesky method (see Sec. 4.5). These weights (ratios of tweaked over nominal 2010d(v2) ν_μ neutrino fluxes) are applied only to the ν_μ SuperK MC templates and depend on the true energy bin t. The weights are constructed so that they have no effect on the integrated number of SuperK 1-ring μ-like events. All other factors appearing in Eq. 31 are explained in Tab. 6.

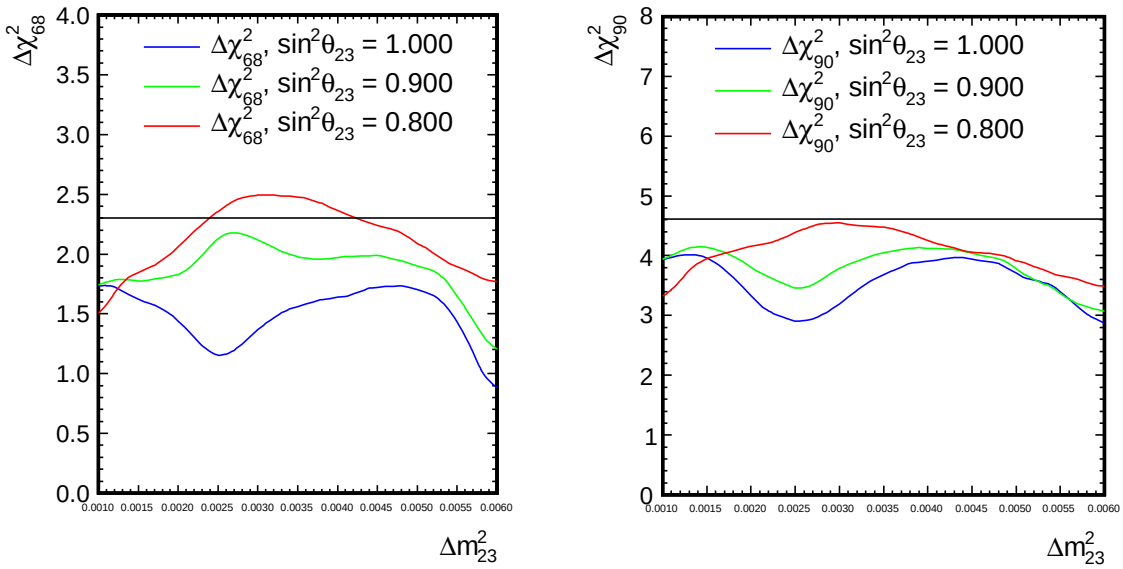


Figure 17: Monte Carlo estimation, according to the Feldman-Cousins method (statistical errors only), of $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right), as functions of Δm^2_{23} for 3 different values of $sin^2 2\theta_{23}$ (0.8, 0.9 and 1.0). For comparison, the canonical value of $\Delta\chi^2$ used in the PDG (see Tab. 5) is also shown as a horizontal line.

SuperK detector uncertainties	
f_{ϵ}^{SK}	Combined efficiency uncertainty due to fiducial volume, reduction chain, Outer Detector cut, ring counting and PID cut uncertainties. (CCQE: 7.8%, CCnonQE: 25.5%, NC: 115.1%, ν_e CC: 100% [1]). Applied to corresponding SuperK MC templates.
f_{E}^{SK}	Reconstructed energy scale error (1% [1]). Applied to SuperK MC templates by scaling bin edges and, assuming uniform distribution of events within the 50 MeV bins, calculating the number of events gained from / lost to neighbouring bins.
ND280 detector uncertainties	
f_{ϵ}^{ND280}	Efficiency error (5% [14]). Applied to ND280 MC templates.
Flux uncertainties	
f_{norm}	Flux normalization uncertainty. Uncertainty, expressed relatively to the N_{SK}/N_{ND280} ratio predicted by the MC, is calculated against oscillation and other systematic parameters using code provided by the Beam WG [20]. Calculated uncertainty for nominal MC and oscillations with parameters $sin^2 2\theta_{23}$ = 1.00 and Δm^2_{23} = 2.4$\times 10^{-3}\ eV^2/c^4$ is around 6.6%.
$\rho^{\Phi_{\mathrm{SK}}}$	Flux shape uncertainty. Expressed in terms of true energy bin-to-bin correlation matrix for SuperK ν_μ flux provided by the Beam WG [19]. Uncertainty is propagated using the Choleskly method (see Sec. 4.5).
Neutrino Generator (GENIE/NEUT) model uncertainties	
$f_{\sigma_{CCQE};t}$	CCQE cross section uncertainty. It respresents only "residual" CCQE uncertainty not cancelled-out in the near to far extrapolation due to the fact that the near and far detectors sample different flux energy ranges and they are made of different nuclear targets. Therefore the uncertainty is applied to SuperK MC templates only. The uncertainty has true neutrino energy dependence [24] which is denoted here by the true neutrino energy bin number t.
$f_{\sigma_{CC1\pi}/\sigma_{CCQE}}$	Fractional uncertainty on CC1π/CCQE cross section ratio (30% for true energy below 2 GeV and 20% for energy above 2 GeV [24]). Applied both to corresponding SuperK and ND280 MC templates.
$f_{\sigma_{CCoth}/\sigma_{CCQE}}$	Fractional uncertainty on CCother/CCQE cross section ratio (30% for true energy below 2 GeV and 25% for energy above 2 GeV [24]). Applied both to corresponding SuperK and ND280 MC templates.
$f_{\sigma_{NC}/\sigma_{CCQE}}$	Fractional uncertainty on NC/CCQE cross section ratio (36% [24]). Applied both to corresponding SuperK and ND280 MC templates.
$f_{FSI;r}$	Uncertainty on effect of FSI (final state interaction / intranuclear hadron transport). The uncertainty affects mainly the SuperK exclusive CCQE-like (1 μ-like ring) measurement (eg. effect of pion absorption in oxygen) and not the ND280 inclusive CC measurement. Therefore, it is applied only to SuperK MC templates. The error has a reconstructed energy dependence [24] which is denoted here by the reconstructed neutrino energy bin number r.

Table 6: Inputs for ν_μ disappearance systematic error evaluation. Each parameter f_P is a MC template modification factor accounting for the uncertainty on quantity P (P may be a detector efficiency, energy scale, absolute cross section, cross section ratio etc.) If the relative uncerianty on P is expressed by $\frac{\delta P}{P}$, then $f_P = 1 + g(\mu = 0; \sigma = 1) * \frac{\delta P}{P}$ where $g(\mu = 0; \sigma = 1)$ is a gaussian error with mean 0 and standard deviation of 1.

4.2. Effect of systematic uncertainties on SuperK predictions

The effect of input systematic uncertainties on predicted number of 1 μ-like ring events (after extrapolation) is shown for both GENIE and NEUT in Tab. 7 (no oscillations) and in Tab. 8 (for oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}\, eV^2/c^4$). The effect of systematic parameters on various SuperK and ND280 components of the calculation as well as in the predicted 1 μ-like ring event reconstructed energy spectrum is presented in detail in Appendix B.

Error source	GENIE $\delta N^{exp}_{SK}/N^{exp}_{SK}$	NEUT $\delta N^{exp}_{SK}/N^{exp}_{SK}$
SuperK CCQE efficiency	$\pm$6.4%	$\pm$6.4%
SuperK CCnonQE efficiency	$\pm$3.9%	$\pm$4.3%
SuperK NC efficiency	$\pm$2.6%	$\pm$2.1%
SuperK ν_e CC efficiency	$\pm$0.0%	$\pm$0.0%
ND280 efficiency	+5.3% -4.8%	+5.3% -4.8%
Flux normalization	$\pm$9.6%	$\pm$9.7%
CCQE cross section	$\pm$4.3%	$\pm$4.1%
CC1pi/CCQE cross section ratio	+2.7% -2.4%	+2.2% -2.0%
CCother/CCQE cross section ratio	+5.9% -5.2%	+5.4% -4.8%
NC/CCQE cross section ratio	$\pm$0.7%	$\pm$0.8%
FSI	$\pm$3.3%	$\pm$3.3%
Total	+16.0% -15.5%	+15.7% -15.3%

Table 7: Effect of 1σ systematic parameter variation on the non-oscillated number of 1 μ-like ring events predicted by GENIE and NEUT (after extarpolation).

4.3. Evaluating systematics at oscillation best-fit point

Systematics, at a given point of the oscillation parameter space, are calculated by fitting a large number (typically 10,000) of toy experiments, generated with the given oscillation parameters and with randomly varied systematic parameters. The systematic error on $sin^2 2\theta_{23}$ and Δm^2_{23} are taken to be the one standard deviation of the corresponding best-fit parameter distribution. In order to decouple statistical and systematic effects, no statistical fluctuations are included in the toy experiments. The effect of each systematic parameter is calculated separately and then they are combined by adding them in quadratures. An evaluation of the 3.23$\times 10^{19}$-POT T2K disapearance analysis systematics for oscillation parameters near the MINOS best fit point ($sin^2 2\theta_{23}$ = 0.9 and Δm^2_{23} = 2.4$\times 10^{-3}\, eV^2/c^4$) is shown in Tab. 9.

4.4. Including systematics in calculation of confidence regions

In the context of the Feldman-Cousins analysis, systematic errors are naturally included in the confidence regions by including systematic variations in the toy experiments used for calculating the $\Delta\chi^2_a$ surfaces (where a denotes the specified confidence level) as a function of $sin^2 2\theta_{23}$ and Δm^2_{23}. The procedure was outlined in Sec. 3.4.2.

Error source	GENIE $\delta N_{SK}^{exp}/N_{SK}^{exp}$	NEUT $\delta N_{SK}^{exp}/N_{SK}^{exp}$
SuperK CCQE efficiency	$\pm$4.8%	$\pm$4.7%
SuperK CCnonQE efficiency	$\pm$7.7%	$\pm$8.4%
SuperK NC efficiency	$\pm$9.5%	$\pm$7.5%
SuperK ν_e CC efficiency	$\pm$0.1%	$\pm$0.1%
ND280 efficiency	+5.3% -4.8%	+5.3% -4.8%
Flux normalization	$\pm$6.6%	$\pm$6.3%
CCQE cross section	$\pm$3.0%	$\pm$2.5%
CC1pi/CCQE cross section ratio	+0.1% -0.2%	+0.4% -0.5%
CCother/CCQE cross section ratio	+5.1% -4.5%	+4.1% -3.7%
NC/CCQE cross section ratio	$\pm$1.5%	$\pm$0.9%
FSI	$\pm$5.8%	$\pm$6.0%
Total	+17.8% -17.5%	+16.6% -16.3%

Table 8: Effect of 1σ systematic parameter variation on the number of 1 μ-like ring events predicted by GENIE and NEUT (after extarpolation) for oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4.

4.5. Handling correlated systematics: Cholesky decomposition method

The LU decomposition is a well-known factorization of a matrix $\mathbf{C}$ as a product of a lower triangular matrix $\mathbf{L}$ and an upper triangular matrix $\mathbf{U}$. The LU decomposition has several applications in scientific computing [8]. Let $\mathbf{C}$ be a square, symmetric, positive-definite[7] matrix. Under the stated conditions, there exists a special factorization of $\mathbf{C}$, the Cholesky decomposition, which is significantly *more efficient* than the generic LU decomposition. Rather than factorizing $\mathbf{C}$ as the product of two independent matrices $\mathbf{L}$ and $\mathbf{U}$, the Cholesky method factorizes $\mathbf{C}$ as the product of a lower triangular matrix $\mathbf{L}$ and its transpose matrix:

$$\mathbf{C} = \mathbf{L} \cdot \mathbf{L}^T \tag{32}$$

The Cholesky decomposition algorithm can be found in [8].

The Cholesky decomposition is commonly used for Monte Carlo simulation of systems with correlated variables. Let S_N be such a system whose dimensionality is N and let $\mathbf{C}$ be the $N \times N$ matrix which describes the covariance of the N simulated variables. The covariance matrix $\mathbf{C}$ is decomposed to obtain the lower triangular matrix $\mathbf{L}$. Let $\mathbf{u}$ be a N-dimensional vector of uncorrelated variables; each vector element u_i can be randomly generated from a gaussian distribution with a mean of 0 and a standard deviation of 1. The vector

$$\mathbf{x} = \mathbf{L} \cdot \mathbf{u} \tag{33}$$

has the covariance properties of S_N.

[7] $\mathbf{v} \cdot \mathbf{C} \cdot \mathbf{v} > 0$ for every vector $\mathbf{v}$.

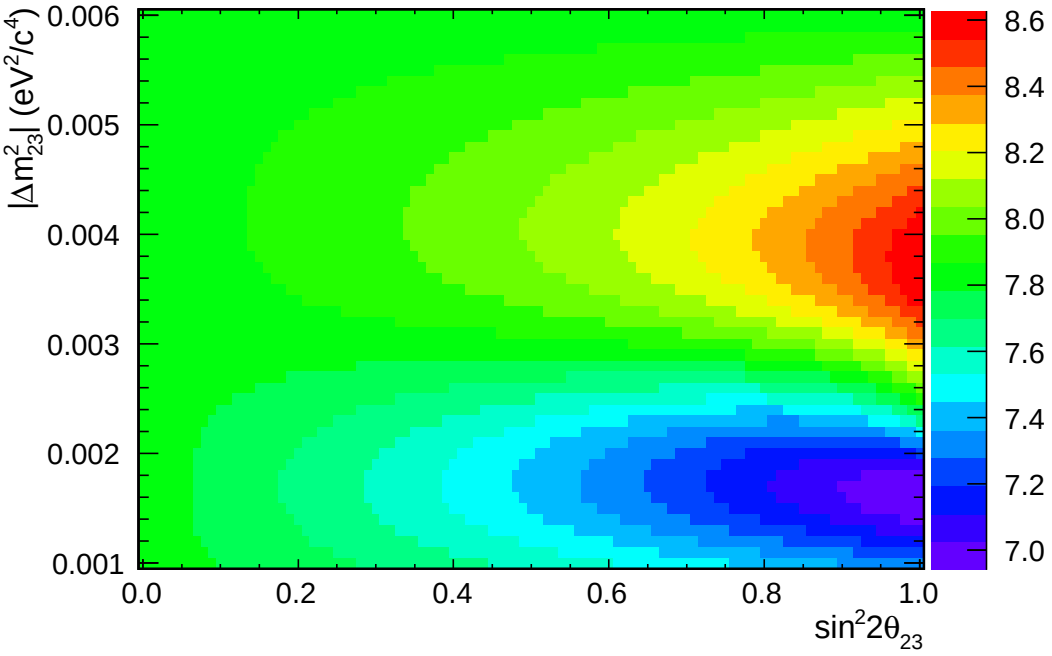


Figure 18: Percentage uncertainty on expected numbers of 1-ring μ-like events (after extrapolation) in the 0-10 GeV reconstructed energy range due to quoted cross uncertainties only. Over the oscillation parameter space considered by this analysis, the uncertainty in the 7% to 8.6% range. This plot can be directtly compared with the GENIE-NEUT difference shown in Fig. 14.

5. Oscillation fit to the 3.23×10^{19} POT dataset

Using the method outlined in the previous sections, an oscillation fit was performed to the 2010a 3.23×10^{19} POT dataset. For GENIE, this analysis finds $|\Delta m^2_{23}|$ = 2.8×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.81 as the best-fit values. For NEUT, the corresponding values are $|\Delta m^2_{23}|$ = 2.9×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.83. The reconstructed neutrino energy distribution for the 8 1-ring μ-like events of the 2010a dataset, along with the best-fit predictions from GENIE and NEUT are shown in Figs. 20 and 22.

The systematic errors at the best-fit point were evaluated as described in Sec. 4.3. The systematic errors at the GENIE best-fit point are given, in Tab. 10. For NEUT, the systematic errors at the best-fit point are given in Tab. 11. We list the effect on $|\Delta m^2_{23}|$ and $sin^2 2\theta_{23}$ of all systematic uncertainties considered in this analysis. The total systematic error is computed by adding in quadrature the errors due to individual systematic uncertainties. This analysis finds that the combined systematic uncertainty at the GENIE (NEUT) best-fit point is 4.3% (3.9%) for $|\Delta m^2_{23}|$ and 6.7% (6.0%) for $sin^2 2\theta_{23}$. The SuperK NC efficiency, the CCQE cross section and the flux shape uncertainties are the ones with the largest impact on $|\Delta m^2_{23}|$. The dominant systematics on $sin^2 2\theta_{23}$ are the SuperK CCQE efficiency, the flux normalization, the ND280 efficiency and the CCother/CCQE cross section ratio.

Confidence regions were calculated using both methods described in Sec. 3.4. A comparison of the Feldman-Cousins and plain confidence regions (including statistical errors only) for the 2010a dataset is shown in Fig. 19. As discussed in Sec. 3.4 the Feldman-Cousins confidence regions have correct coverage by construction. As it can be seen in Fig. 19, the Feldman-Cousins and plain confidence regions differ significantly and, therefore, the coverage of the latter is spurious. The plain confidence regions tend to overcover near a physical boundary, and either undercover or overcover away from that boundary. Consequently, this analysis considers only the Feldman-Cousins method for quoting the final results. The 68%, 90% and 95% Feldman-Cousins confidence regions, with and without systematics, for the 2010a ν_μ disappearance analysis are shown in Fig. 21 for GENIE and in Fig. 23 for NEUT. The estimated systematic error at the best-fit point (listed in Tabs. 10 and 11) is also shown.

Upon seeing the reconstructed energy distribution of the 2010a dataset, it was clear to the authors of this note that large values of $|\Delta m^2_{23}|$, in the wrong trough of the oscillation survival probability function, may provide a better fit to the data. The oscillation parameter limits (discussed in Sec. 3.3) were relaxed and a new fit was performed. The result is shown in Fig. 24. Indeed, a better fit to the data exists at maximal mixing angle and large $|\Delta m^2_{23}|$ (of the order of

Error source	GENIE		NEUT	
	$\left(\frac{\delta\Delta m^2_{23}}{\Delta m^2_{23}}\right)$	$\left(\frac{\delta \sin^2 2\theta_{23}}{sin^2 2\theta_{23}}\right)$	$\left(\frac{\delta\Delta m^2_{23}}{\Delta m^2_{23}}\right)$	$\left(\frac{\delta \sin^2 2\theta_{23}}{sin^2 2\theta_{23}}\right)$
SuperK CCQE efficiency	0.3%	1.7%	0.5%	1.6%
SuperK CCnonQE efficiency	0.5%	1.1%	0.5%	1.3%
SuperK NC efficiency	3.9%	3.1%	3.5%	3.0%
SuperK ν_e CC efficiency	0.0%	0.0%	0.0%	0.0%
SuperK energy scale	0.3%	0.2%	0.5%	0.2%
ND280 efficiency	0.1%	1.5%	0.3%	1.4%
Flux normalization	0.1%	1.6%	0.3%	1.5%
Flux shape	2.3%	0.5%	2.2%	0.6%
CCQE cross section	1.2%	1.5%	1.0%	1.5%
CC1pi/CCQE cross section ratio	0.2%	0.6%	0.1%	0.3%
CCother/CCQE cross section ratio	0.1%	1.7%	0.1%	1.5%
NC/CCQE cross section ratio	1.2%	0.5%	1.2%	0.5%
FSI	0.1%	0.7%	0.3%	0.6%
Total	5.1%	5.3%	4.7%	5.0%

Table 9: Oscillation systematics evaluated at $sin^2 2\theta_{23}$ = 0.9 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4.

10^{-2} eV^2/c^4, some $\sim 20\sigma$ away from the MINOS best-fit value). As it can be seen in Fig. 24, much of the χ^2 valley at large $|\Delta m^2_{23}|$, is lower than the best-fit point in the $|\Delta m^2_{23}|$ window from 10^{-3} eV^2/c^4 to 6×10^{-3} eV^2/c^4. Fig. 25 shows the marginalised χ^2 distributions from the fit shown in Fig. 24.

Although the high $|\Delta m^2_{23}|$ solution was pointed-out by the authors of this analysis, the majority of the authors feel uneasy modifying the pre-defined analysis procedure after seeing the actual data (in response to a statistical fluctuation in the observed dataset). The pre-defined $|\Delta m^2_{23}|$ limits of 1×10^{-3} eV^2/c^4 and 6×10^{-3} eV^2/c^4 correspond to the T2K sensitivity window, as it can be seen clearly in Fig. 26. The limits were set (before looking at the data) exactly so as to avoid fluctuations such as the one observed in the data. This entire experiment (and analysis) was built on the premise that the first oscillation minimum is around the T2K peak energy, as was determined by MINOS and K2K. In addition, including this prior knowledge (Bayesian prior) into the oscillation result does not invalidate the frequentist interpretation of the confidence regions as it has been incorrectly indicated [25]. The SuperK expectation already employs a host of other Bayesian priors; they are physics parameters, such as the quasi-elastic axial mass, which are selected based on belief from (sometimes) conflicting experimental results. Still, one can certainly assert that the 68% confidence regions shown in this analysis is the region where 68% of the measurements would fall if the same measurement could be repeated N times. The statement implies analysing N independent 3.23E+19 POT datasets using the exact same input assumptions (Bayesian priors). The confidence regions shown here have correct coverage and proper frequestist interpretation in the $|\Delta m^2_{23}|$ window from 10^{-3} eV^2/c^4 to 6×10^{-3} eV^2/c^4. The arguments against showing confidence regions from a fit to an enlarged $|\Delta m^2_{23}|$ range were made convincingly in [26] and will not be repeated here. For the actual publication it is proposed that a) we show the best-fit value and confidence regions in the restricted $|\Delta m^2_{23}|$ range, and b) we indicate the existence of a global minimum outside the restricted $|\Delta m^2_{23}|$ range and we quote the probability for such a fluctuation. The probability for such a $|\Delta m^2_{23}|$ fluctuation with a 3.23×10^{19} POT exposure is about 5.6% (see Fig. 27). With 1×10^{20} POT (dataset already accumulated at the time

of revisiting this note) the probability for such a fluctuation has already fallen to about 0.2% (see Fig. 28).

Error source	$\left(\frac{\delta\Delta m^2_{23}}{\Delta m^2_{23}}\right)$	$\left(\frac{\delta \sin^2 2\theta_{23}}{sin^2 2\theta_{23}}\right)$
SuperK CCQE efficiency	0.5%	2.9%
SuperK CCnonQE efficiency	1.0%	1.6%
SuperK NC efficiency	2.0%	1.8%
SuperK ν_e CC efficiency	0.0%	0.0%
SuperK energy scale	0.5%	0.3%
ND280 efficiency	0.2%	2.2%
Flux normalization	0.3%	2.8%
Flux shape	2.6%	1.0%
CCQE cross section	1.8%	1.8%
CC1pi/CCQE cross section ratio	1.1%	1.1%
CCother/CCQE cross section ratio	0.4%	2.6%
NC/CCQE cross section ratio	0.5%	0.1%
FSI	0.2%	1.1%
Total	4.3%	6.7%

Table 10: Oscillation systematics computed using GENIE, at the GENIE best-fit point ($sin^2 2\theta_{23}$ = 0.81 and Δm^2_{23} = 2.8$\times 10^{-3}$ eV^2/c^4).

6. Future directions of the Rutherford/Oxford analysis

Over the next few months, in preparation for the ν_μ disappearance analysis using the extended dataset from the November 2010 - June 2011 physics run, this analysis will be incrementaly updated using additional inputs from the Rutherford/Oxford/Lancs ND280 analyses (CC inclusive reconstructed energy spectrum, measurements of CCQE / CC inclusive and CC1π^+ / CC inclusive, measurements of π^+ momentum spectrum in CC1π^+ etc). The full oscillation analysis will be used to benchmark the utility of each additional ND280 input.

Informed by these studies, we will prepare a document, well in advance of the next global T2K analysis effort, detailing our views on the optimum methodology for the next ν_μ disappearance analysis.

7. Summary

To be written.

Error source	$\left(\frac{\delta\Delta m^2_{23}}{\Delta m^2_{23}}\right)$	$\left(\frac{\delta \sin^2 2\theta_{23}}{sin^2 2\theta_{23}}\right)$
SuperK CCQE efficiency	0.6%	2.6%
SuperK CCnonQE efficiency	1.0%	1.7%
SuperK NC efficiency	1.6%	1.5%
SuperK ν_e CC efficiency	0.0%	0.0%
SuperK energy scale	0.5%	0.4%
ND280 efficiency	0.2%	2.1%
Flux normalization	0.3%	2.6%
Flux shape	2.3%	1.1%
CCQE cross section	1.7%	1.4%
CC1pi/CCQE cross section ratio	1.0%	0.8%
CCother/CCQE cross section ratio	0.5%	2.1%
NC/CCQE cross section ratio	0.4%	0.1%
FSI	0.4%	1.1%
Total	3.9%	6.0%

Table 11: Oscillation systematics computed using NEUT, at the NEUT best-fit point ($sin^2 2\theta_{23}$ = 0.83 and Δm^2_{23} = 2.9$\times 10^{-3}$ eV^2/c^4).

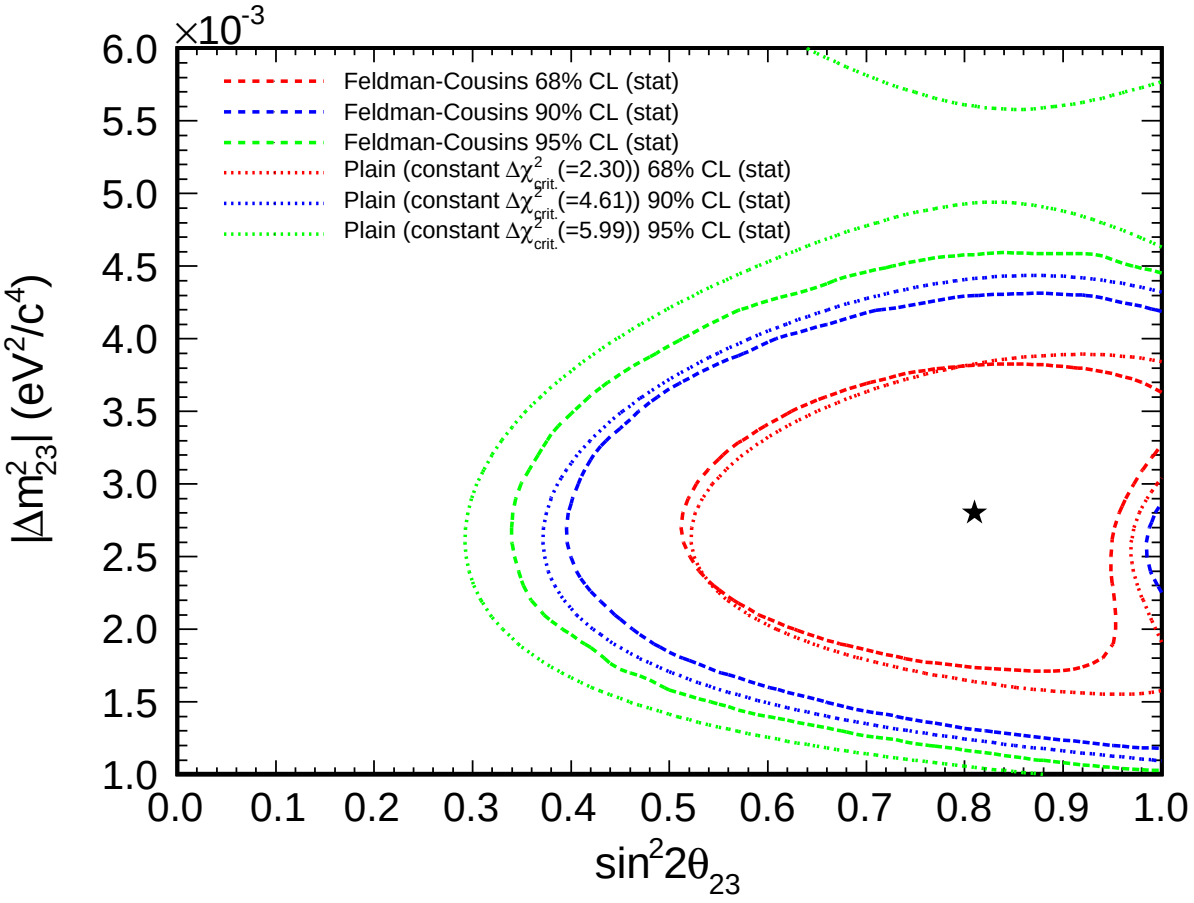


Figure 19: Comparison of the Feldman-Cousins and plain confidence regions (including statistical errors only) for the 2010a dataset. As discussed in Sec. 3.4 the Feldman-Cousins confidence regions have correct coverage by construction. The Feldman-Cousins and plain confidence regions differ significantly and, therefore, coverage of the plain contours is spurious. The latter seem to overcover significantly near the physical boundary and undercover away from it. In addition, the non plain contour appears to ignore the existence of Δm^2_{23} boundaries (fitter limits). The contours are shown for GENIE. Identical conclusions are drawn by the comparison of the Feldman-Cousins and plain confidence regions generated by NEUT.

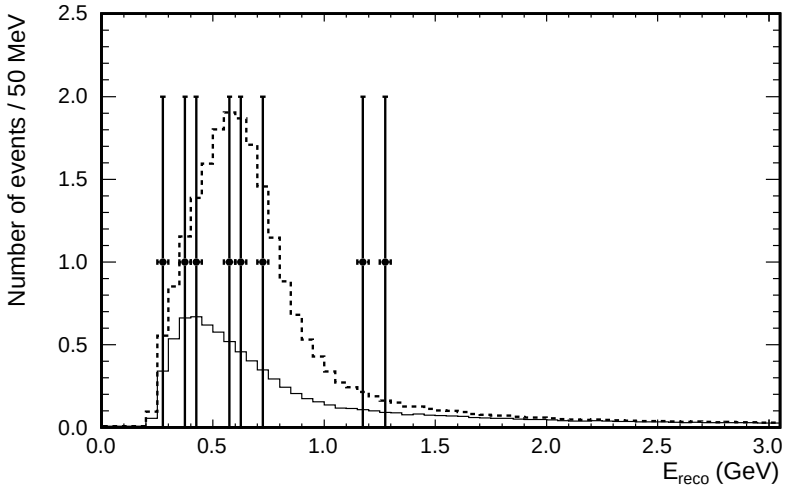


Figure 20: Reconstructed neutrino energy distribution for the 8 1-ring μ-like events of the 2010a 3.23×10^{19} POT dataset (points) and GENIE best-fit SuperK prediction (solid line). The best-fit oscillation parameters (in the search window defined in Sec. 3.3) are $|\Delta m^2_{23}|$ = 2.8×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.81. The dotted line shows the GENIE prediction without oscillations.

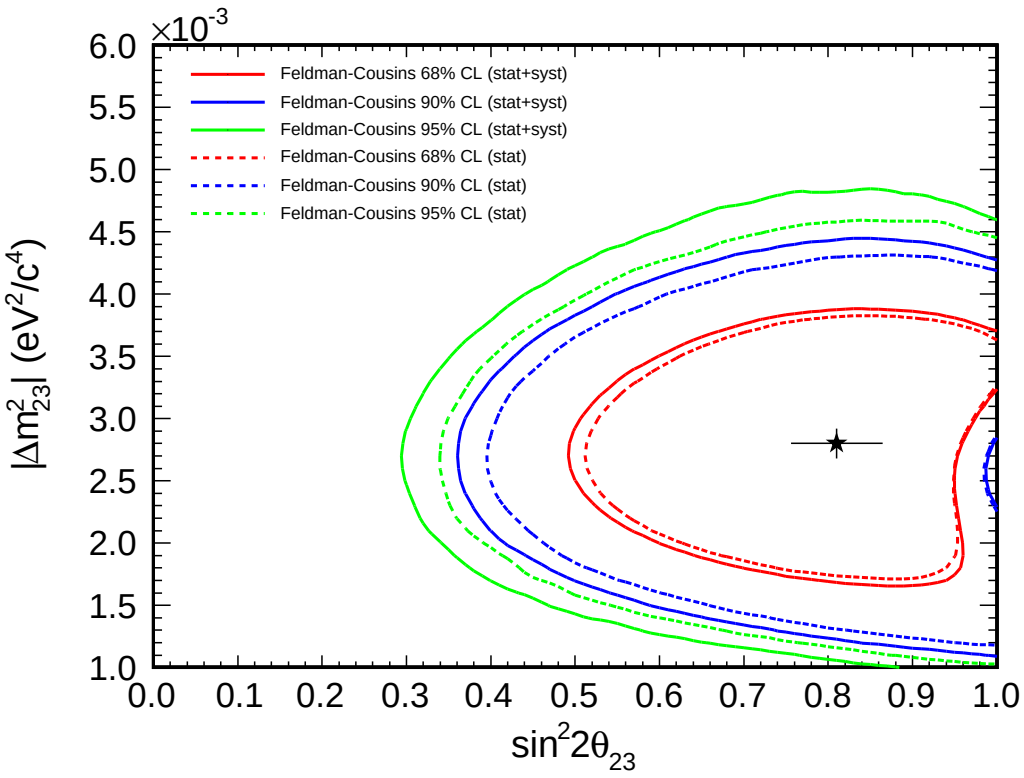


Figure 21: 68%, 90% and 95% Feldman-Cousins confidence regions, with and without systematics, for the GENIE disappearance fit to the 2010a dataset. The estimated systematic error at the best-fit point is also shown with a cross.

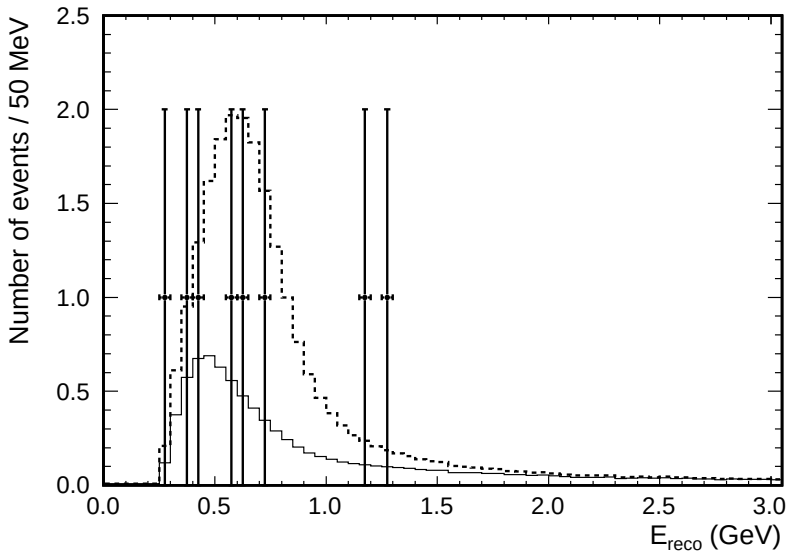


Figure 22: Reconstructed neutrino energy distribution for the 8 1-ring μ-like events of the 2010a 3.23$\times$10^{19} POT dataset (points) and NEUT best-fit SuperK prediction (solid line). The best-fit oscillation parameters (in the search window defined in Sec. 3.3) are $|\Delta m^2_{23}|$ = 2.9$\times$10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.83. The dotted line shows the NEUT prediction without oscillations.

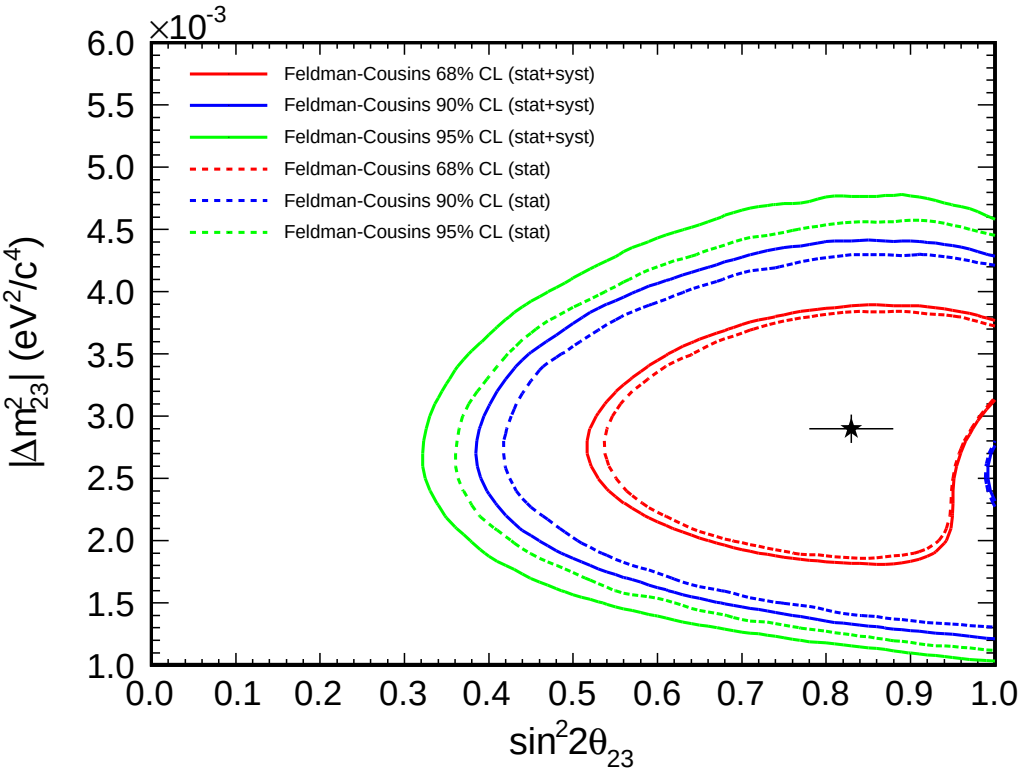


Figure 23: 68%, 90% and 95% Feldman-Cousins confidence regions, with and without systematics, for the NEUT disappearance fit to the 2010a dataset. The estimated systematic error at the best-fit point is also shown with a cross.

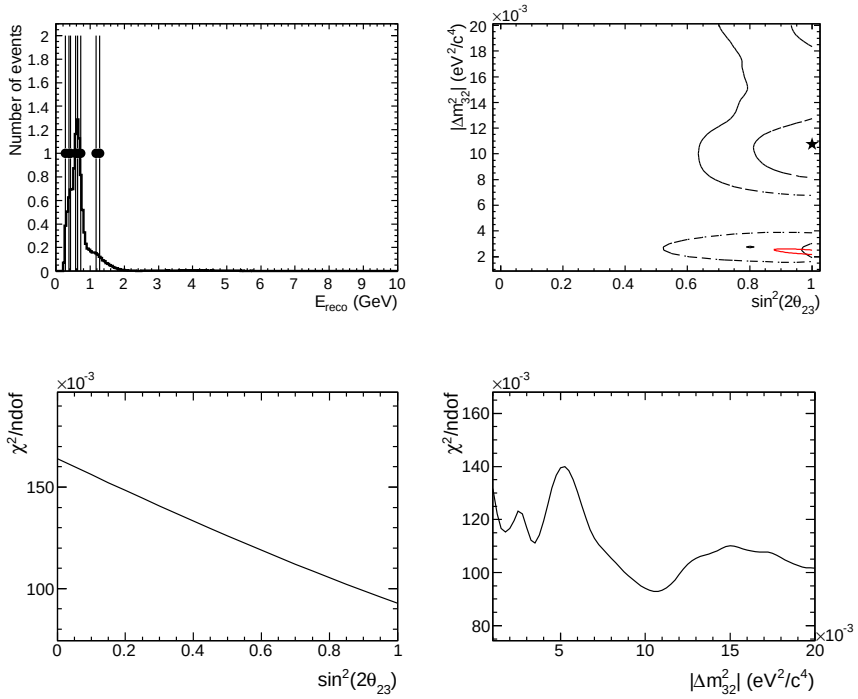


Figure 24: Fit to the 2010a set with relaxed upper limit of $|\Delta m^2_{23}|$. The top-left plot shows the reconstructed energies of the 2010a dataset and the best-fit prediction. The confidence regions shown on the top-right are the 68% and 90% CL regions (statistical errors only) calculated, for simplicity, using the PDG method. For reference, the MINOS 90% CL region is also shown (red line). The best-fit point is shown with a black-star. The bottom-left and bottom-right plots show the χ^2/ndof along the $sin^2 2\theta_{23}$ and $|\Delta m^2_{23}|$ lines that pass through the best-fit point.

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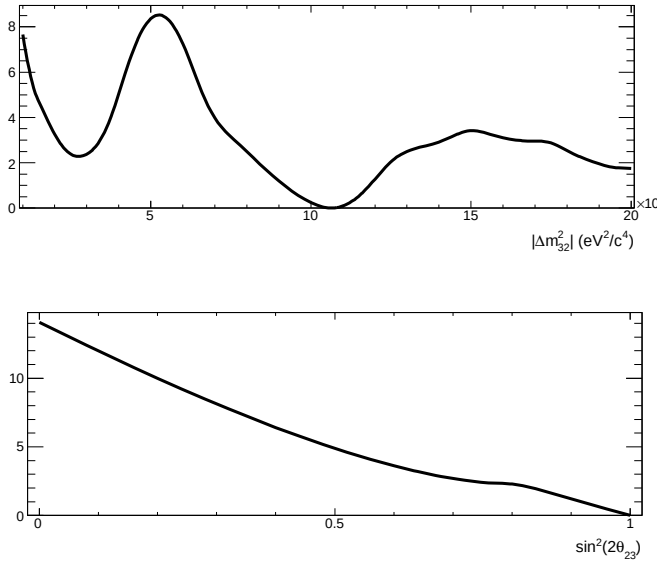


Figure 25: Marginalised χ^2 distributions for the fit shown in Fig. 24.

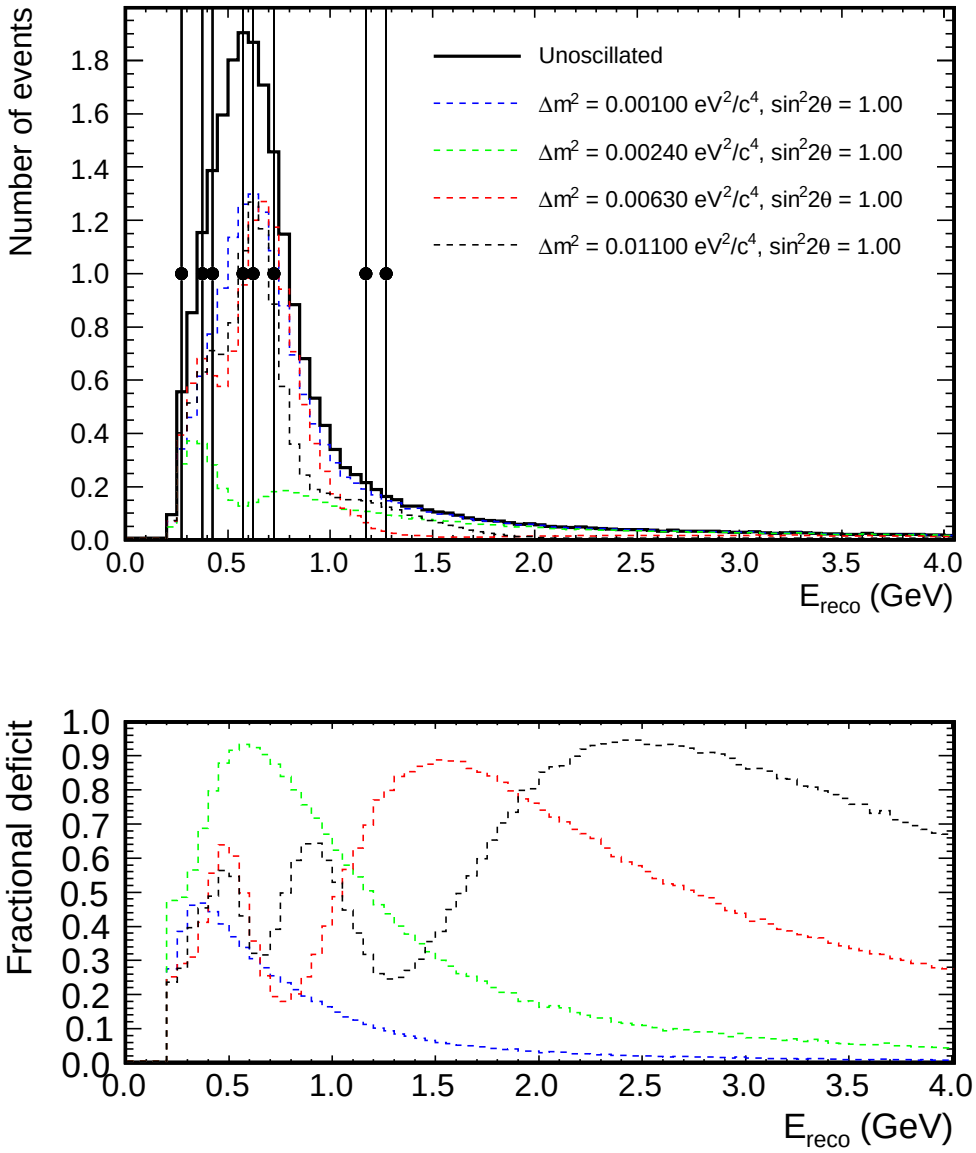


Figure 26: Predicted reconstructed-energy spectrum of 1-ring μ-like events (after extrapolation) for $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 1.0×10^{-3} eV^2/c^4, 2.4×10^{-3} eV^2/c^4, 6.0×10^{-3} eV^2/c^4 and 1.1×10^{-2} eV^2/c^4. The non-oscillated spectrum (GENIE) and the 2010a dataset are also shown. With $|\Delta m^2_{23}|$ = 1×10^{-3} eV^2/c^4 the first oscillation minimum is in the region dominated by NC and non-CCQE interactions. On the other hand, with $|\Delta m^2_{23}|$ = 6×10^{-3} eV^2/c^4 the (first) oscillation minimum is already well above the T2K peak energy. In both of these cases, our $|\Delta m^2_{23}|$ messurement capabilities are much deteriorated. With Δm^2_{23}=1.1×10^{-2} eV^2/c^4, the first oscillation minimum appears at 2.5 GeV! The high-Δm^2_{23} solution provides a better fit to the small 2010a dataset only because the second and third oscillation minina happen to be accomodated between gaps (pseudo-dips) left by the 8 observed events at around 0.5 GeV and 1 GeV. The probability for such a Δm^2_{23} fluctuation is given in the text.

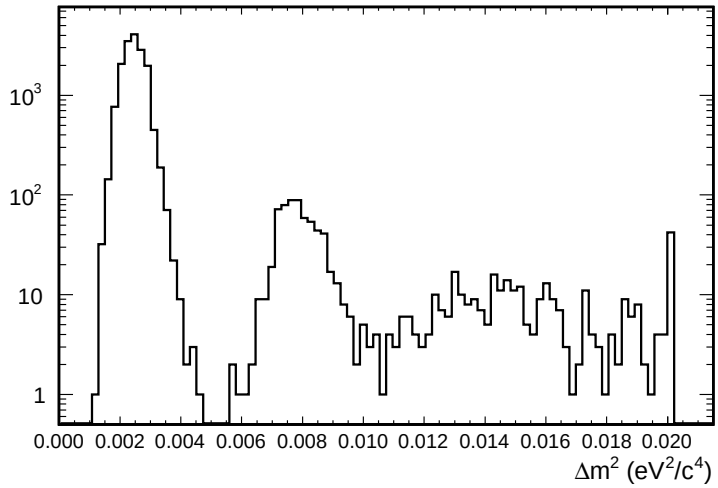


Figure 27: Distribution of best-fit Δm^2_{23} values for a set of toy MC experiments generated with the MINOS best-fit values (Δm^2_{23} = 2.35$\times 10^{-3}$ eV^2/c^4 and maximal mixing). The toy MC experiments were generated for an integrated neutrino beam exposure of 3.23×10^{19} POT and systematic variations were included. 5.6% of fits gave a Δm^2_{23} value above 6×10^{-3} eV^2/c^4.

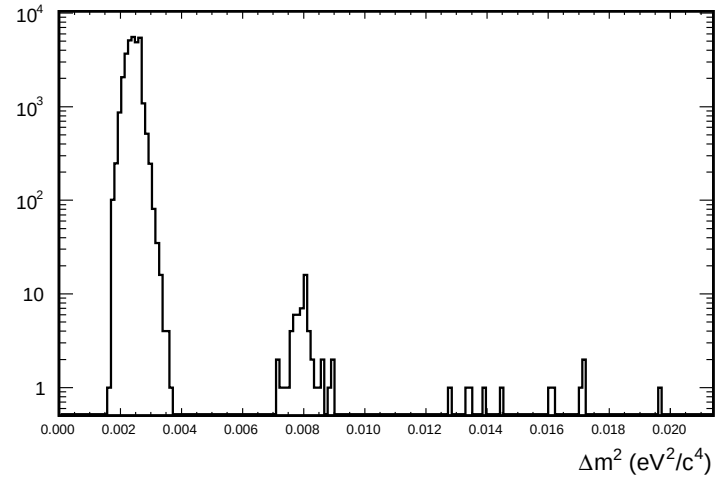


Figure 28: Distribution of best-fit Δm^2_{23} values for a set of toy MC experiments generated with the MINOS best-fit values (Δm^2_{23} = 2.35$\times 10^{-3}$ eV^2/c^4 and maximal mixing). The toy MC experiments were generated for an integrated neutrino beam exposure of 1×10^{20} POT ($3 \times$ 2010a) and systematic variations were included. 0.2% of fits gave a Δm^2_{23} value above 6×10^{-3} eV^2/c^4.

A. Sample mock data fits

In order for the reader to gain an understanding of the difficulties involved in fitting the reconstructed energy spectrum from a 3.23×10^{19} POT dataset (just over 6 1-ring μ-like events are expected by both GENIE and NEUT for oscillations with $sin^2 2\theta_{23}$ = 1.00 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4), we present a series of mock-dataset fits (Figs. 29 - 36). The mock-dataset were both generated (with true oscillation parameters $sin^2 2\theta_{23}$ = 1.00 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4) and fitted using the GENIE p.d.f's. In each figure, the top-left plot shows the reconstructed energies of the mock-dataset and the best-fit prediction. The confidence regions shown on the top-right are the 68% and 90% CL regions (statistical errors only) calculated using the PDG method (Sec. 3.4.1). For reference, the MINOS 90% CL region is also shown (red line). The best-fit point is shown with a black-star while the input point is indicated by a red circle. The bottom-left and bottom-right plots show the χ^2/ndof along the $sin^2 2\theta_{23}$ and Δm^2_{23} lines that pass through the best-fit point[8]. Finally, in Fig. 37 we show the best-fit point distribution from a set of 5,000 mock-dataset fits.

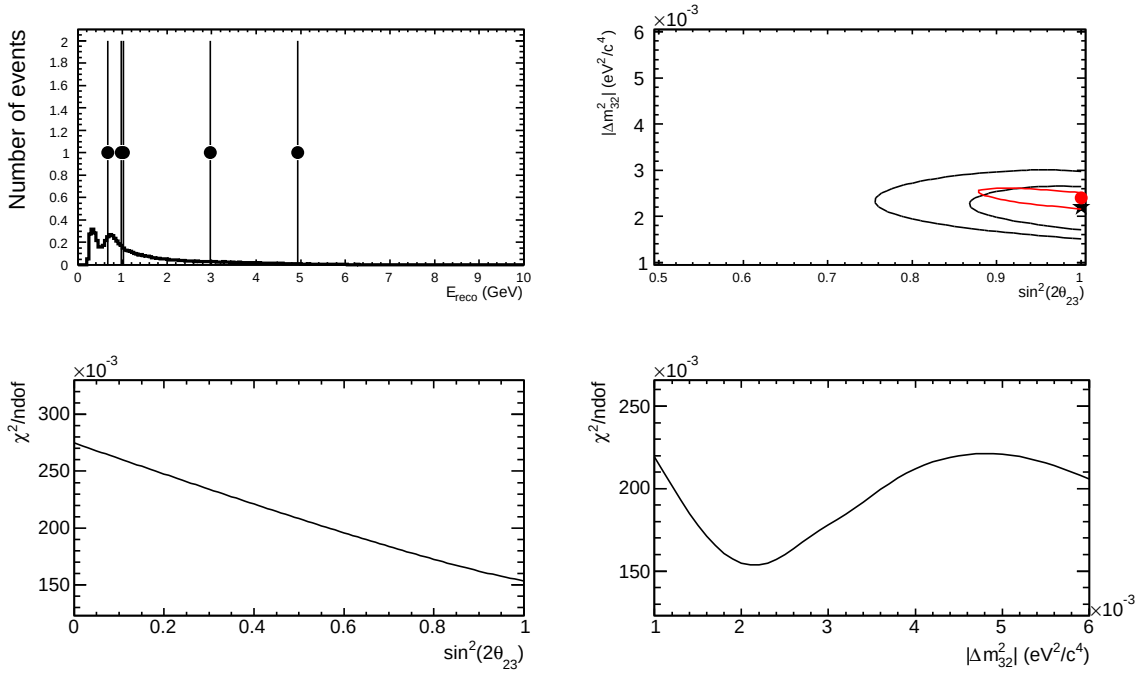


Figure 29: Mock-dataset fit 1: The reconstructed neutrino energies of the mock 1-ring μ-like data sample are {0.652999, 0.972174, 1.0032, 2.98126, 4.91308} GeV. The best-fit oscillation parameters are $sin^2 2\theta_{23}$ = 1.00 and Δm^2_{23} = 2.2×10^{-3} eV^2/c^4.

[8]Note that they *are not* marginalized χ^2/ndof distributions as functions of $sin^2 2\theta_{23}$ and Δm^2_{23}.

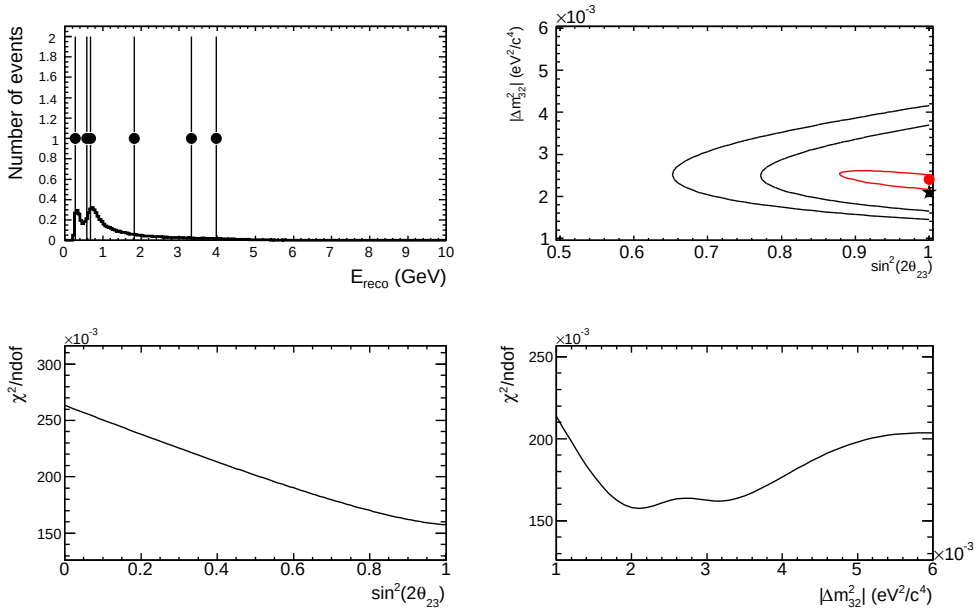


Figure 30: Mock-dataset fit 2: The reconstructed neutrino energies of the mock 1-ring μ-like data sample are {0.253796, 0.574917, 0.654082, 1.82992, 3.33207, 3.95891} GeV. The best-fit oscillation parameters are $sin^2 2\theta_{23}$ = 1.00 and Δm^2_{23} = 2.1×10^{-3} eV^2/c^4.

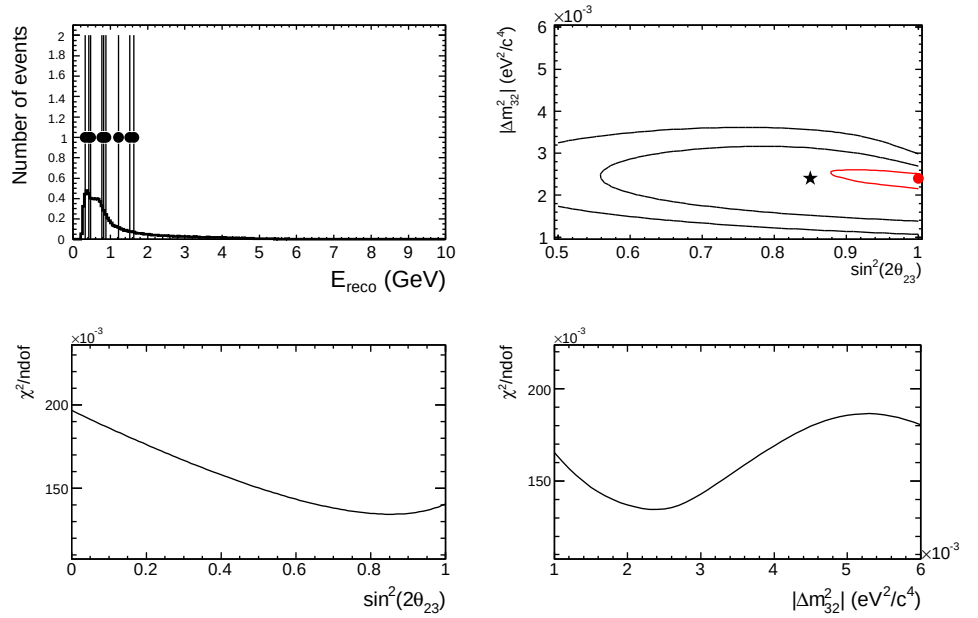


Figure 31: Mock-dataset fit 3: The reconstructed neutrino energies of the mock 1-ring μ-like data sample are {0.32655, 0.427434, 0.480255, 0.774868, 0.841949, 0.85971, 1.24938, 1.5061, 1.63616} GeV. The best-fit oscillation parameters are $sin^2 2\theta_{23}$ = 0.85 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4.

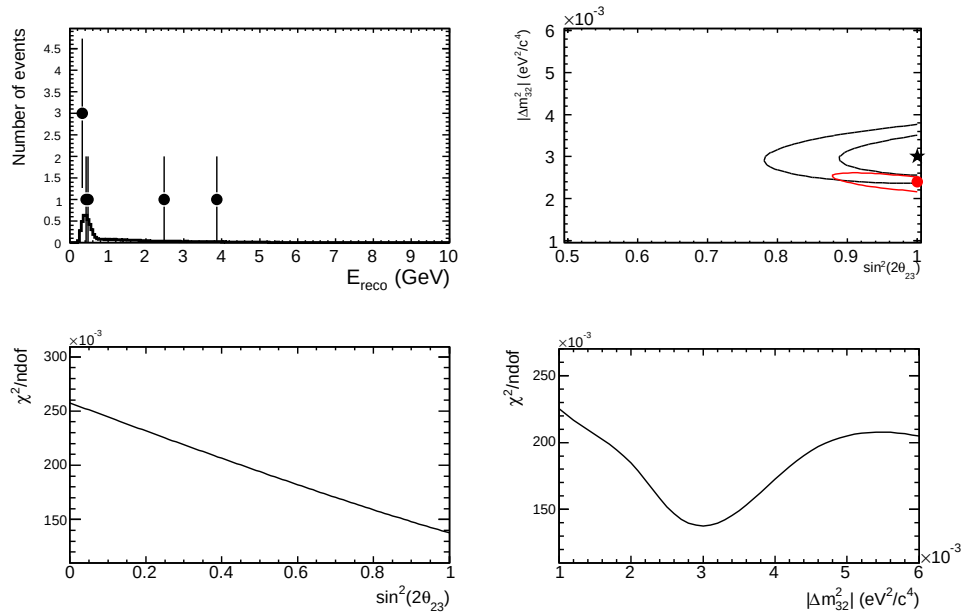


Figure 32: Mock-dataset fit 4: The reconstructed neutrino energies of the mock 1-ring μ-like data sample are {0.310258, 0.343666, 0.346476, 0.429402, 0.455564, 2.48224, 3.87468} GeV. The best-fit oscillation parameters are $sin^2 2\theta_{23}$ = 1.00 and Δm^2_{23} = 3.0×10^{-3} eV^2/c^4.

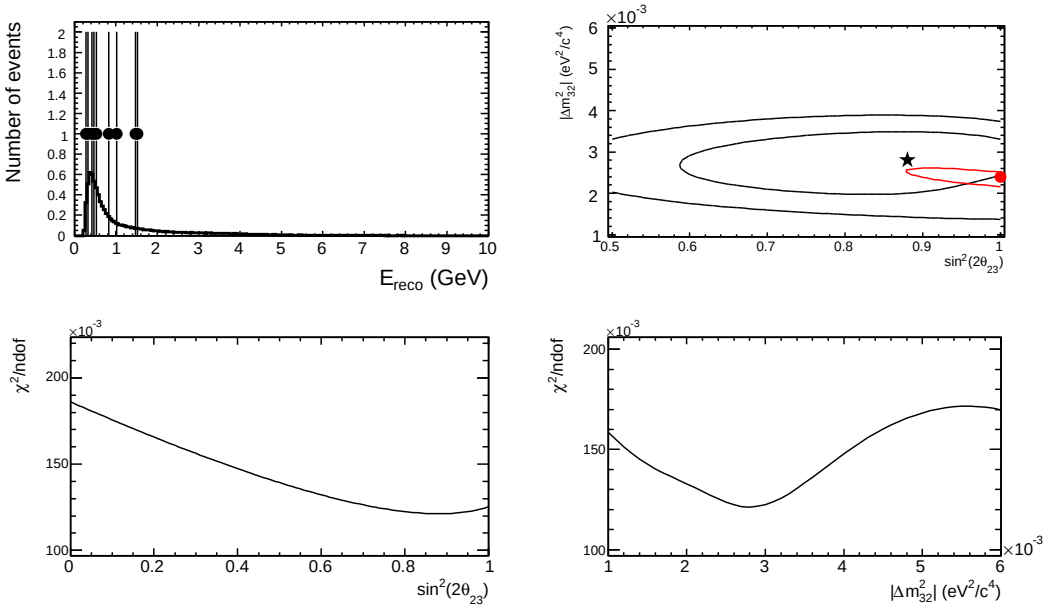


Figure 33: Mock-dataset fit 5: The reconstructed neutrino energies of the mock 1-ring μ-like data sample are {0.28082, 0.328775, 0.44747, 0.45361, 0.543383, 0.812863, 1.03911, 1.45479, 1.53207} GeV. The best-fit oscillation parameters are $sin^2 2\theta_{23}$ = 0.88 and Δm^2_{23} = 2.8×10^{-3} eV^2/c^4.

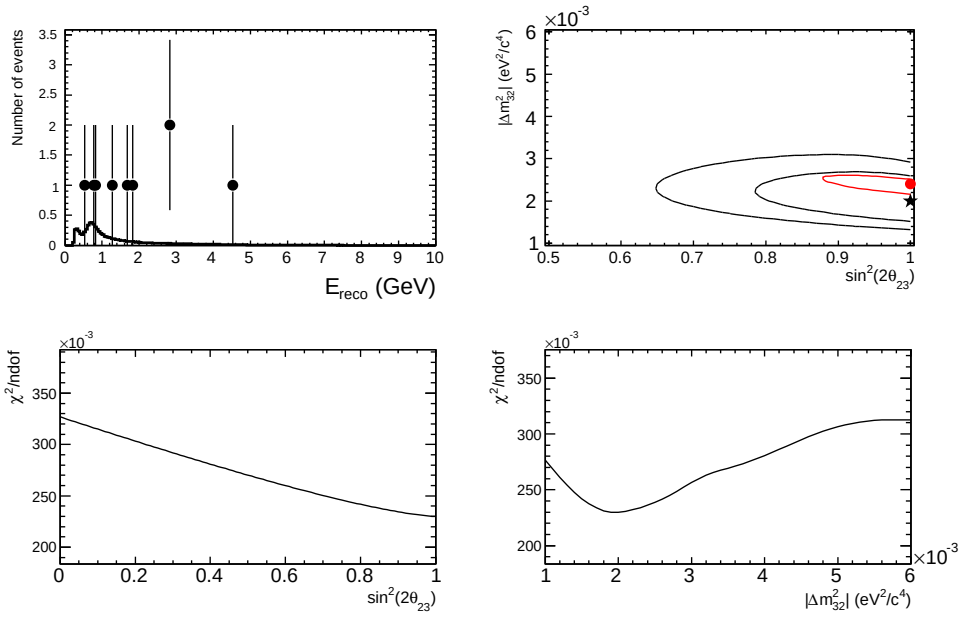


Figure 34: Mock-dataset fit 6: The reconstructed neutrino energies of the mock 1-ring μ-like data sample are {0.540056, 0.771411, 0.815833, 1.27384, 1.69788, 1.8422, 2.84393, 2.8057, 4.5466} GeV. The best-fit oscillation parameters are $sin^2 2\theta_{23}$ = 1.00 and Δm^2_{23} = 2.0×10^{-3} eV^2/c^4.

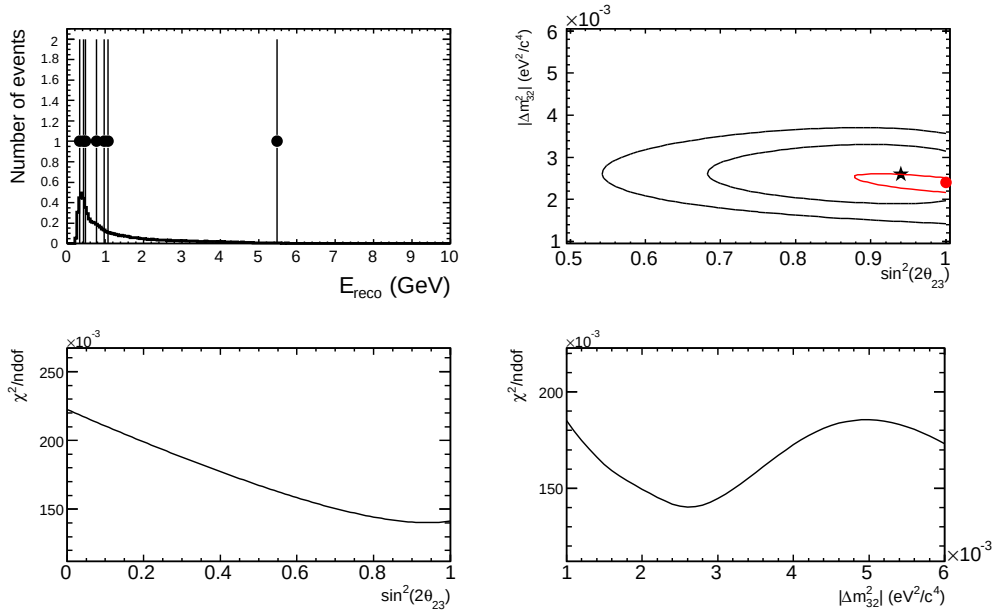


Figure 35: Mock-dataset fit 7: The reconstructed neutrino energies of the mock 1-ring μ-like data sample are {0.319674, 0.407844, 0.462414, 0.759538, 0.968712, 1.06938, 5.4639} GeV. The best-fit oscillation parameters are $sin^2 2\theta_{23}$ = 0.94 and Δm^2_{23} = 2.6×10^{-3} eV^2/c^4.

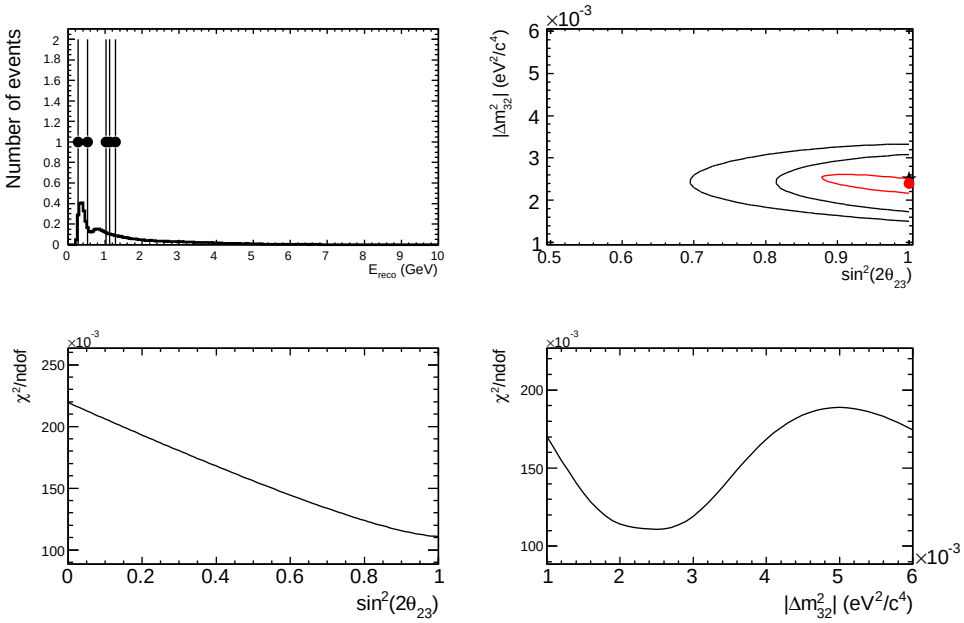


Figure 36: Mock-dataset fit 8: The reconstructed neutrino energies of the mock 1-ring μ-like data sample are {0.264395, 0.530563, 1.01775, 1.12194, 1.28155} GeV. The best-fit oscillation parameters are $sin^2 2\theta_{23}$ = 1.00 and Δm^2_{23} = 2.5×10^{-3} eV^2/c^4.

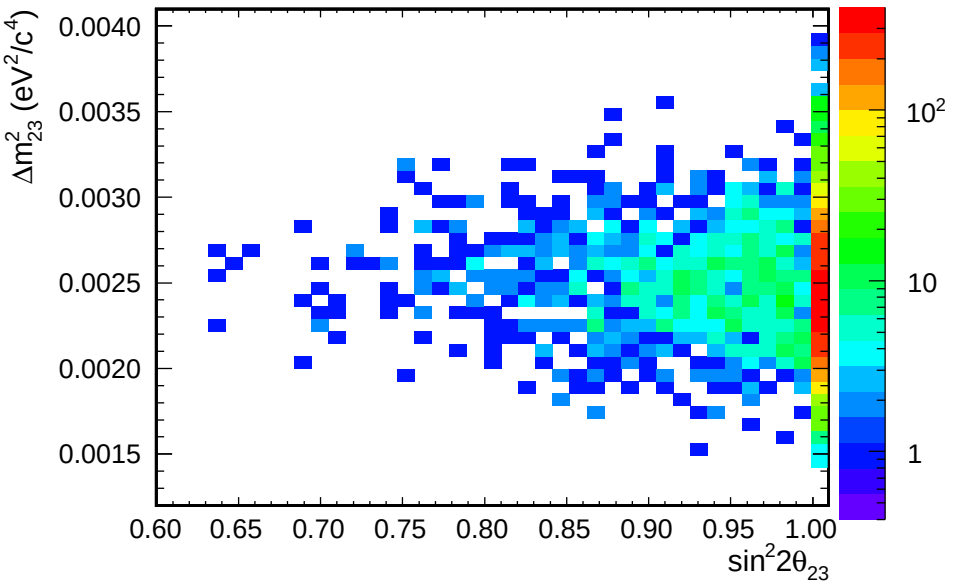


Figure 37: Distribution of best-fit oscillation parameters for 5,000 mock-dataset fits.

B. Effect of systematic parameters on predicted SuperK energy spectrum

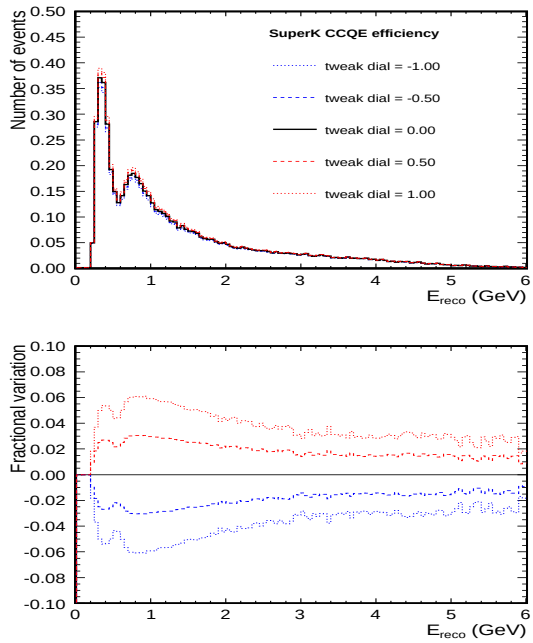


Figure 38: Effect of SuperK CCQE efficiency variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

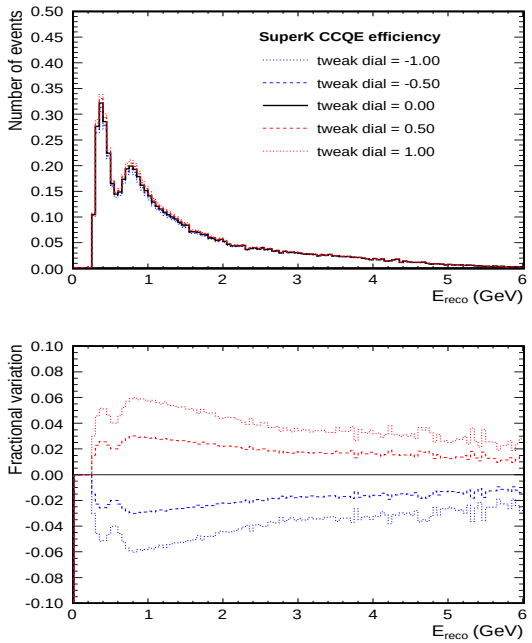


Figure 39: Effect of SuperK CCQE efficiency variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

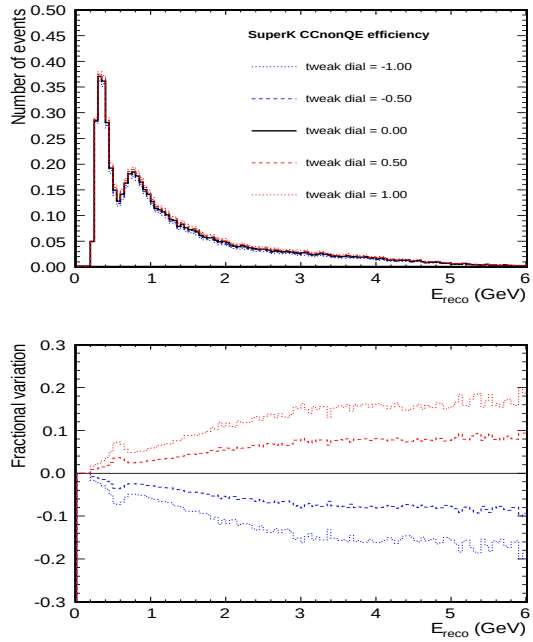


Figure 40: Effect of SuperK CCnonQE efficiency variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

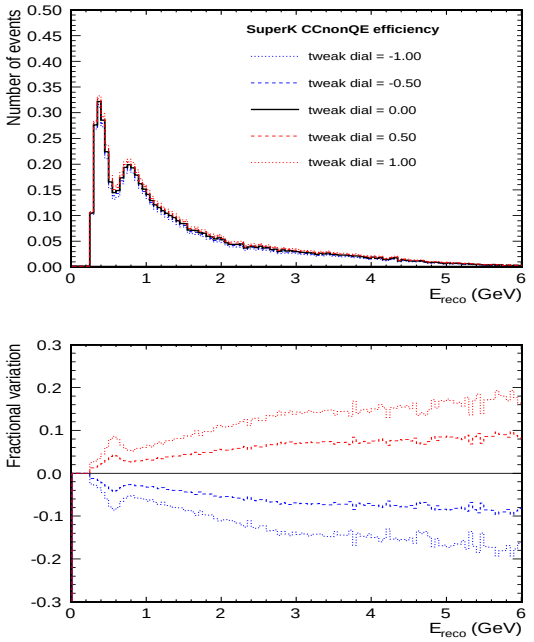


Figure 41: Effect of SuperK CCnonQE efficiency variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

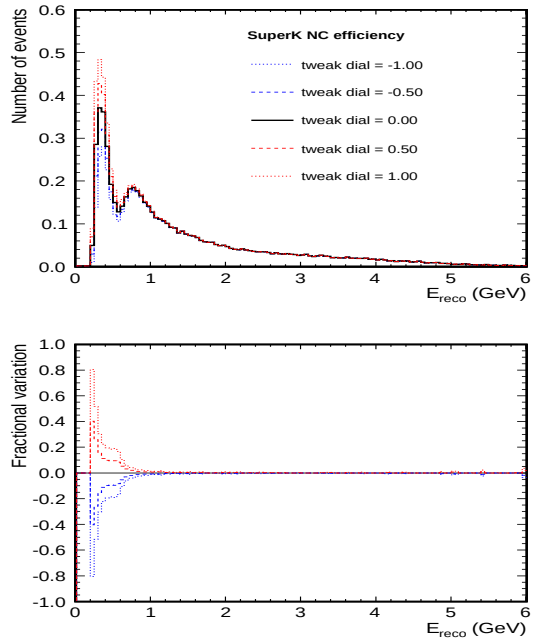


Figure 42: Effect of SuperK NC efficiency variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

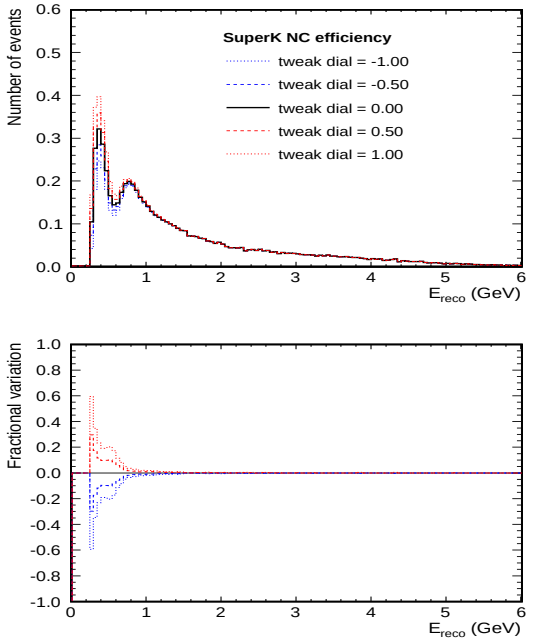


Figure 43: Effect of SuperK NC efficiency variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

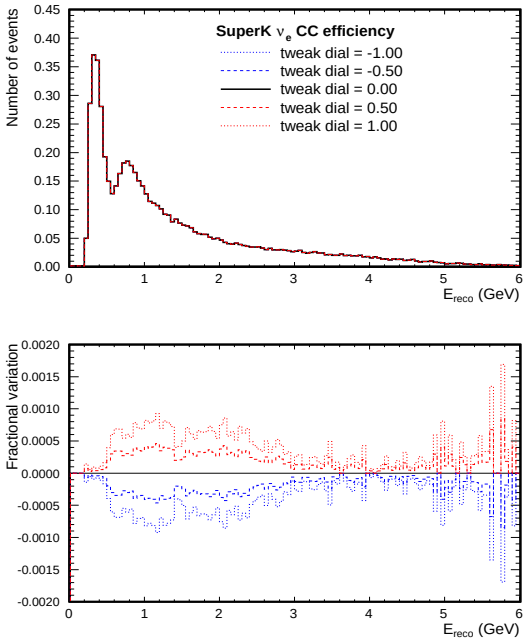


Figure 44: Effect of SuperK ν_e CC efficiency variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

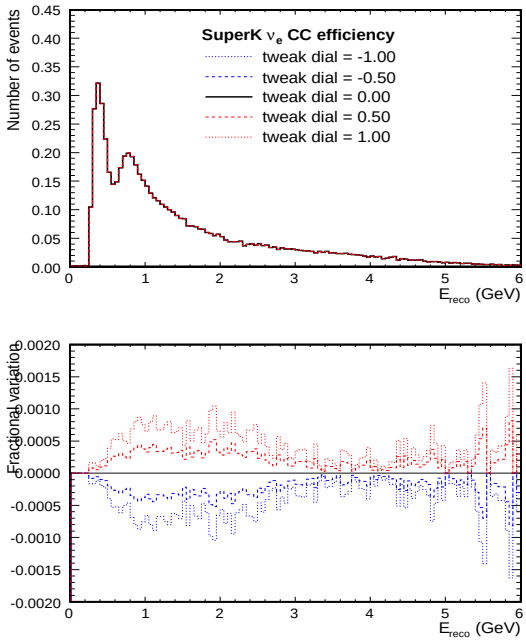


Figure 45: Effect of SuperK ν_e CC efficiency variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

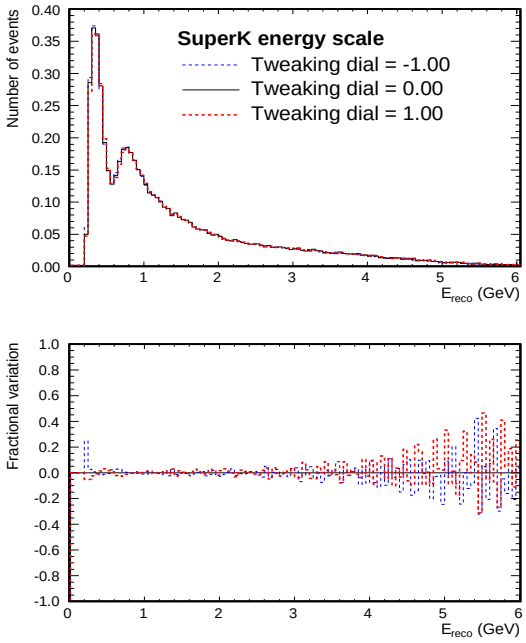


Figure 46: Effect of SuperK energy scale variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

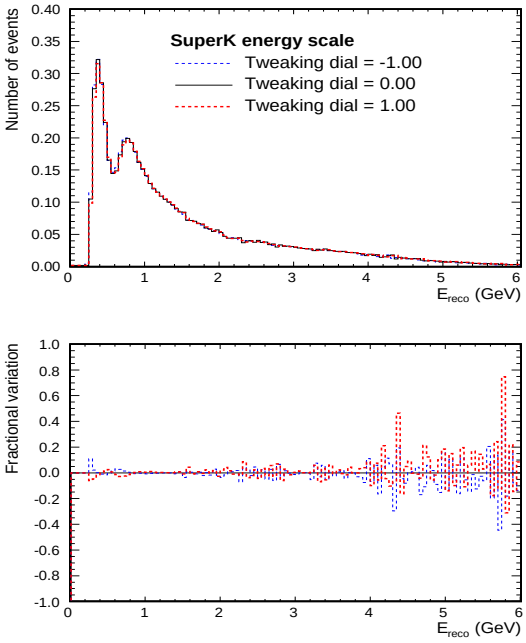


Figure 47: Effect of SuperK energy scale variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

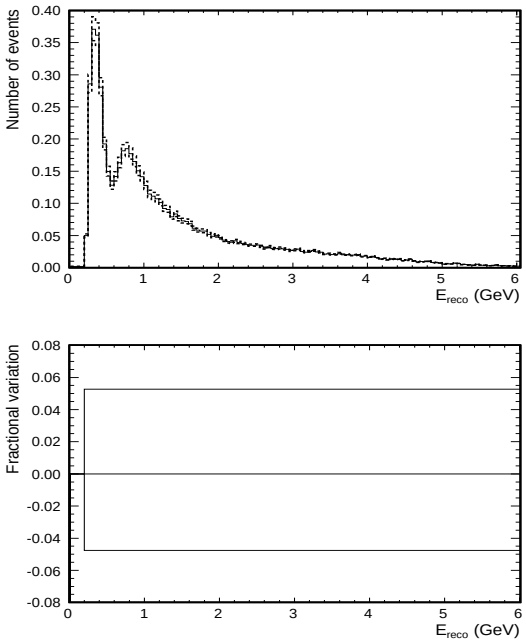


Figure 48: Effect of ND280 efficiency variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

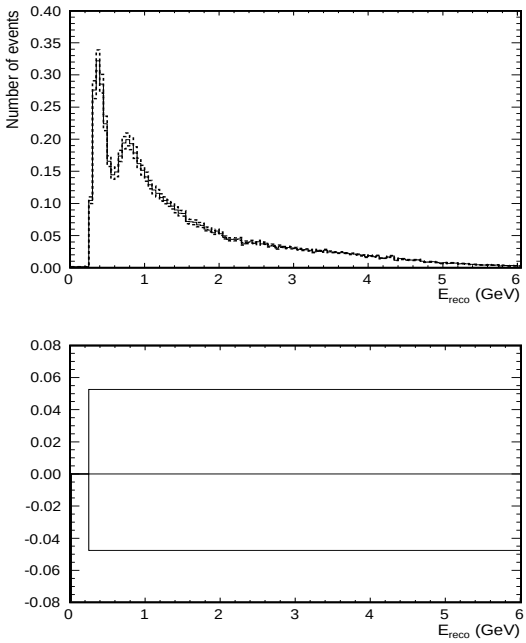


Figure 49: Effect of ND280 efficiency variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

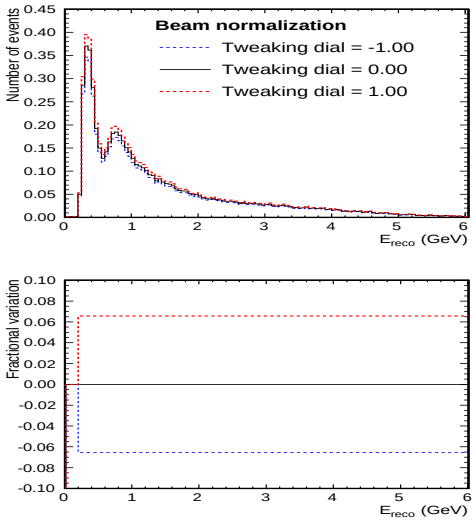


Figure 50: Effect of beam normalization variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

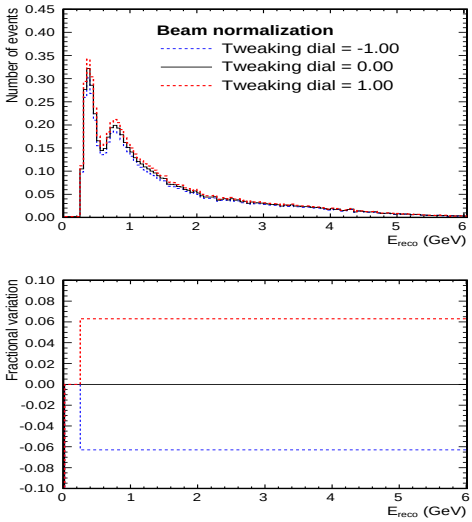


Figure 51: Effect of beam normalization variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

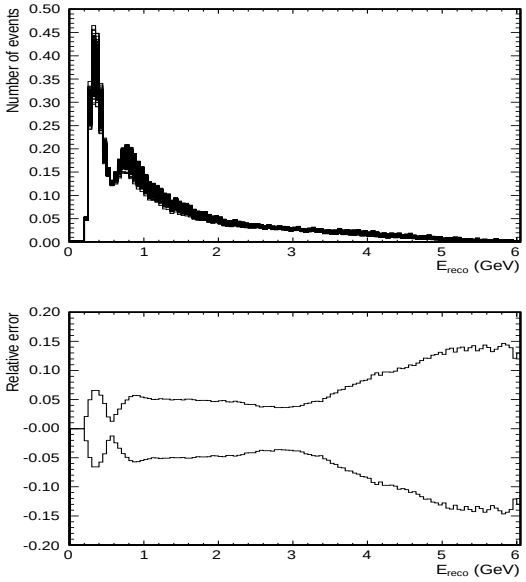


Figure 52: Effect of beam shape systematic variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). The top plot shows 2000 reconstructed-energy spectra corresponding to 2000 fluxes generated using the Cholesky method (see Sec. 4.5). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied and they are normalized to the number of events predicted using the nominal flux. The relative error shown in the bottom plot is calculated from the standard deviation of the randomly generated spectra.

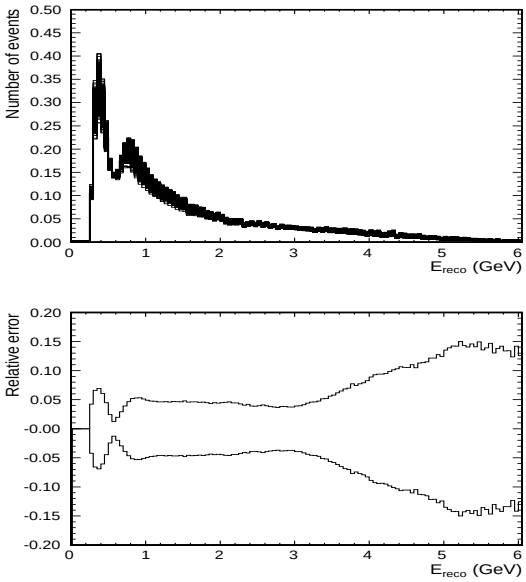


Figure 53: Effect of beam shape variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). The plots were generated as described in Fig. 52.

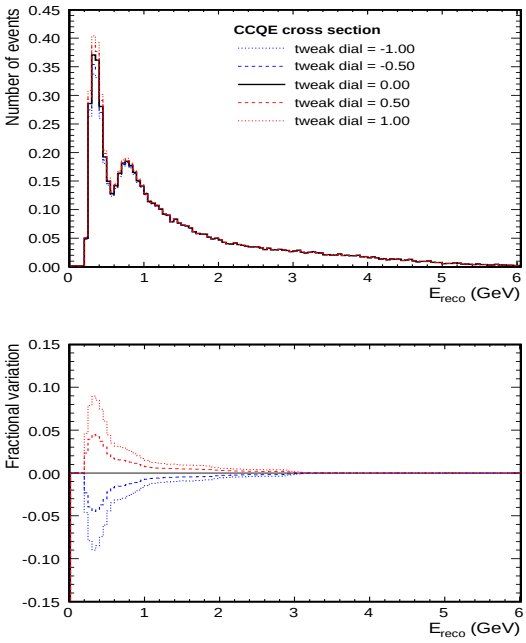


Figure 54: Effect of CCQE cross section variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

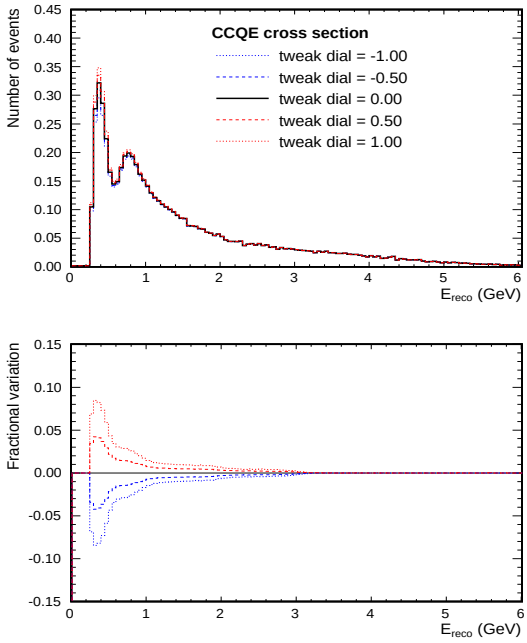


Figure 55: Effect of CCQE cross section variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

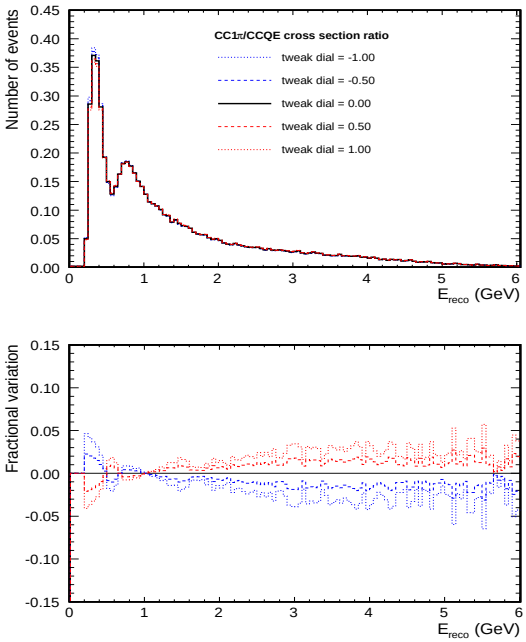


Figure 56: Effect of CC1π/CCQE cross section ratio variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

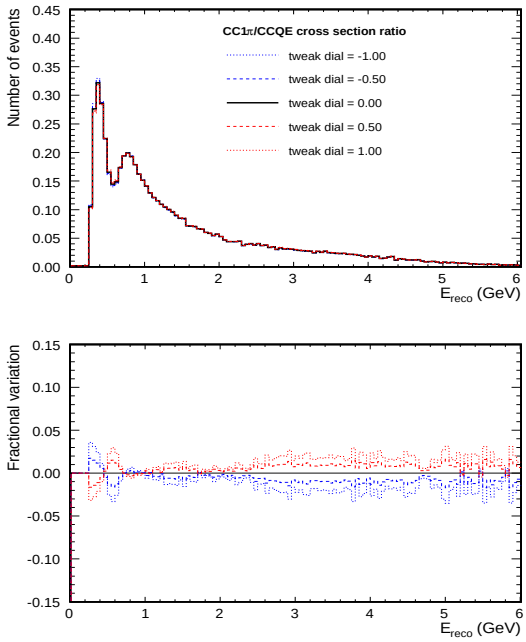


Figure 57: Effect of CC1π/CCQE cross section ratio variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

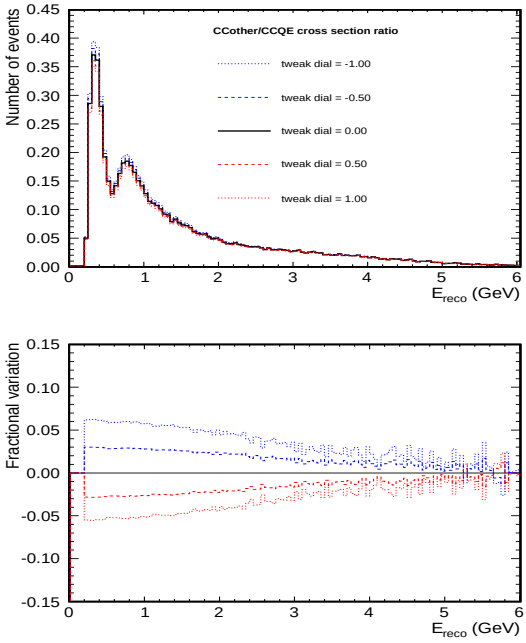


Figure 58: Effect of CCother/CCQE cross section ratio variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

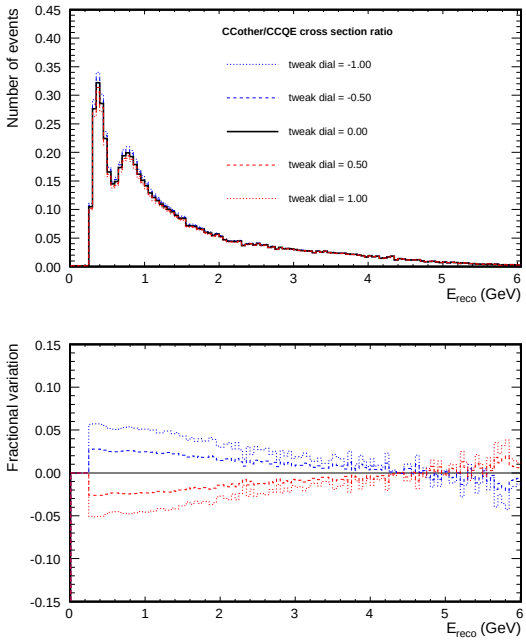


Figure 59: Effect of CCother/CCQE cross section ratio variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

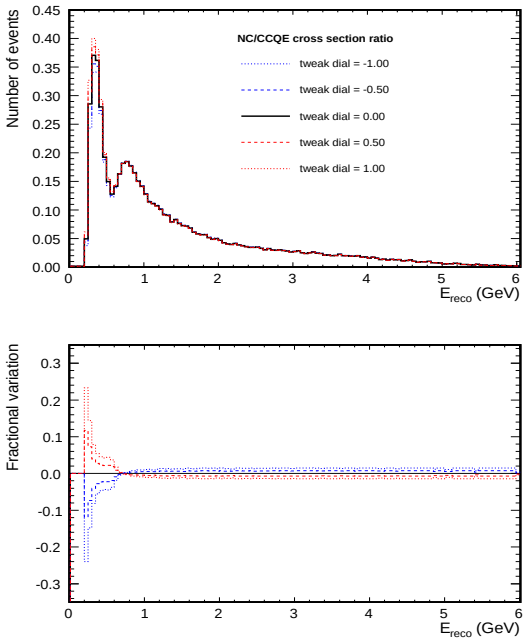


Figure 60: Effect of NC/CCQE cross section ratio variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

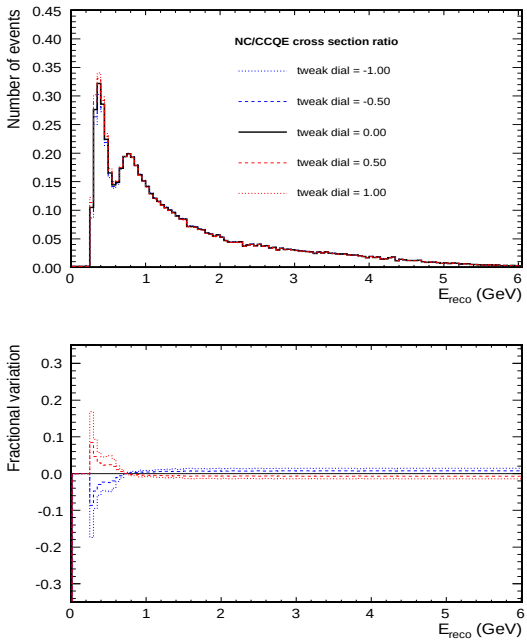


Figure 61: Effect of NC/CCQE cross section ratio variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

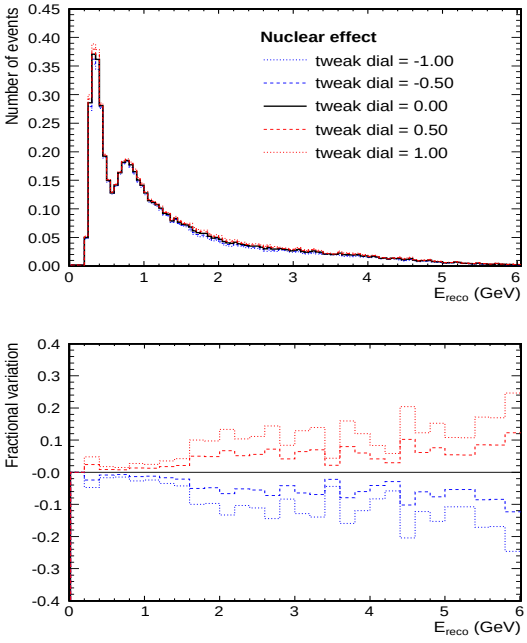


Figure 62: Effect of FSI variation to the GENIE reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

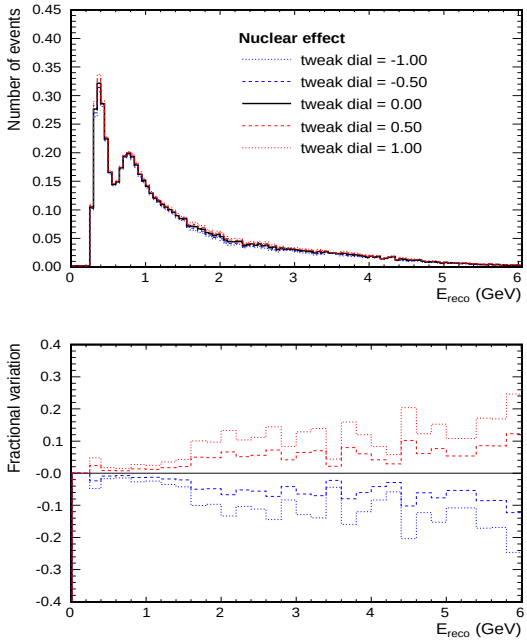


Figure 63: Effect of FSI variation to the NEUT reconstructed-energy spectrum of 1-ring muon-like events (after extrapolation). All spectra have oscillations with $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	6.5019	4.80 %	5.9066	-4.80 %
SK MC (1 μ-like ring)	5.3721	5.6299	4.80 %	5.1144	-4.80 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.3402	7.80 %	2.8568	-7.80 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.2545	0.00 %	1.2545	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2177	0.00 %	0.2177	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.8124	5.29 %	4.3290	-5.29 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.4087	0.00 %	0.4087	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2217	7.80 %	0.1896	-7.80 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1271	0.00 %	0.1271	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0215	0.00 %	0.0215	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3702	4.53 %	0.3382	-4.53 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0236	0.00 %	0.0236	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0014	0.00 %	0.0014	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0008	0.00 %	0.0008	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0002	0.00 %	0.0002	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0024	0.00 %	0.0024	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0125	0.00 %	0.0125	0.00 %
ND280 CC-like Observed/MC ratio	1.1549	1.1549	0.00 %	1.1549	0.00 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1251.1924	0.00 %	1251.1924	0.00 %
ND280 MC: Total CC	1199.6960	1199.6960	0.00 %	1199.6960	0.00 %
ND280 MC: Total NC background	51.4964	51.4964	0.00 %	51.4964	0.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	291.9388	0.00 %	291.9388	0.00 %
ND280 MC: ν_μ CC other	280.0197	280.0197	0.00 %	280.0197	0.00 %
ND280 MC: ν_μ CC all	1189.5883	1189.5883	0.00 %	1189.5883	0.00 %
ND280 MC: ν_μ NC all	48.8820	48.8820	0.00 %	48.8820	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.0898	0.00 %	3.0898	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	4.9767	0.00 %	4.9767	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	8.4480	0.00 %	8.4480	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.8276	0.00 %	1.8276	0.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.4312	0.00 %	0.4312	0.00 %
ND280 MC: ν_e CC other	1.0397	1.0397	0.00 %	1.0397	0.00 %
ND280 MC: ν_e CC all	1.6598	1.6598	0.00 %	1.6598	0.00 %
ND280 MC: ν_e NC all	0.7868	0.7868	0.00 %	0.7868	0.00 %

Table 12: Effect of SuperK CCQE efficiency variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	6.6816	7.69 %	5.7270	-7.69 %
SK MC (1 μ-like ring)	5.3721	5.7854	7.69 %	4.9588	-7.69 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.0985	0.00 %	3.0985	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.5744	25.50 %	0.9346	-25.50 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2732	25.50 %	0.1622	-25.50 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.9461	8.21 %	4.1953	-8.21 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.4087	0.00 %	0.4087	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2056	0.00 %	0.2056	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1595	25.50 %	0.0947	-25.50 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0270	25.50 %	0.0160	-25.50 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3921	10.69 %	0.3163	-10.69 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0236	0.00 %	0.0236	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0014	0.00 %	0.0014	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0008	0.00 %	0.0008	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0002	0.00 %	0.0002	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0024	0.00 %	0.0024	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0125	0.00 %	0.0125	0.00 %
ND280 CC-like Observed/MC ratio	1.1549	1.1549	0.00 %	1.1549	0.00 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1251.1924	0.00 %	1251.1924	0.00 %
ND280 MC: Total CC	1199.6960	1199.6960	0.00 %	1199.6960	0.00 %
ND280 MC: Total NC background	51.4964	51.4964	0.00 %	51.4964	0.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	291.9388	0.00 %	291.9388	0.00 %
ND280 MC: ν_μ CC other	280.0197	280.0197	0.00 %	280.0197	0.00 %
ND280 MC: ν_μ CC all	1189.5883	1189.5883	0.00 %	1189.5883	0.00 %
ND280 MC: ν_μ NC all	48.8820	48.8820	0.00 %	48.8820	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.0898	0.00 %	3.0898	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	4.9767	0.00 %	4.9767	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	8.4480	0.00 %	8.4480	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.8276	0.00 %	1.8276	0.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.4312	0.00 %	0.4312	0.00 %
ND280 MC: ν_e CC other	1.0397	1.0397	0.00 %	1.0397	0.00 %
ND280 MC: ν_e CC all	1.6598	1.6598	0.00 %	1.6598	0.00 %
ND280 MC: ν_e NC all	0.7868	0.7868	0.00 %	0.7868	0.00 %

Table 13: Effect of SuperK CCnonQE efficiency variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	6.7956	9.53 %	5.6130	-9.53 %
SK MC (1 μ-like ring)	5.3721	5.8841	9.53 %	4.8601	-9.53 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.0985	0.00 %	3.0985	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.2545	0.00 %	1.2545	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2177	0.00 %	0.2177	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.5707	0.00 %	4.5707	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.8792	115.10 %	-0.0617	-115.10 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2056	0.00 %	0.2056	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1271	0.00 %	0.1271	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0215	0.00 %	0.0215	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3542	0.00 %	0.3542	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0509	115.10 %	-0.0036	-115.10 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0014	0.00 %	0.0014	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0008	0.00 %	0.0008	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0002	0.00 %	0.0002	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0024	0.00 %	0.0024	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0268	115.10 %	-0.0019	-115.10 %
ND280 CC-like Observed/MC ratio	1.1549	1.1549	0.00 %	1.1549	0.00 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1251.1924	0.00 %	1251.1924	0.00 %
ND280 MC: Total CC	1199.6960	1199.6960	0.00 %	1199.6960	0.00 %
ND280 MC: Total NC background	51.4964	51.4964	0.00 %	51.4964	0.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	291.9388	0.00 %	291.9388	0.00 %
ND280 MC: ν_μ CC other	280.0197	280.0197	0.00 %	280.0197	0.00 %
ND280 MC: ν_μ CC all	1189.5883	1189.5883	0.00 %	1189.5883	0.00 %
ND280 MC: ν_μ NC all	48.8820	48.8820	0.00 %	48.8820	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.0898	0.00 %	3.0898	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	4.9767	0.00 %	4.9767	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	8.4480	0.00 %	8.4480	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.8276	0.00 %	1.8276	0.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.4312	0.00 %	0.4312	0.00 %
ND280 MC: ν_e CC other	1.0397	1.0397	0.00 %	1.0397	0.00 %
ND280 MC: ν_e CC all	1.6598	1.6598	0.00 %	1.6598	0.00 %
ND280 MC: ν_e NC all	0.7868	0.7868	0.00 %	0.7868	0.00 %

Table 14: Effect of SuperK NC efficiency variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	6.2071	0.05 %	6.2015	-0.05 %
SK MC (1 μ-like ring)	5.3721	5.3745	0.05 %	5.3697	-0.05 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.0985	0.00 %	3.0985	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.2545	0.00 %	1.2545	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2177	0.00 %	0.2177	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.5707	0.00 %	4.5707	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.4087	0.00 %	0.4087	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2056	0.00 %	0.2056	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1271	0.00 %	0.1271	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0215	0.00 %	0.0215	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3542	0.00 %	0.3542	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0236	0.00 %	0.0236	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0027	100.00 %	0.0000	-100.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0016	100.00 %	0.0000	-100.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0005	100.00 %	0.0000	-100.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0048	100.00 %	0.0000	-100.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0125	0.00 %	0.0125	0.00 %
ND280 CC-like Observed/MC ratio	1.1549	1.1549	0.00 %	1.1549	0.00 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1251.1924	0.00 %	1251.1924	0.00 %
ND280 MC: Total CC	1199.6960	1199.6960	0.00 %	1199.6960	0.00 %
ND280 MC: Total NC background	51.4964	51.4964	0.00 %	51.4964	0.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	291.9388	0.00 %	291.9388	0.00 %
ND280 MC: ν_μ CC other	280.0197	280.0197	0.00 %	280.0197	0.00 %
ND280 MC: ν_μ CC all	1189.5883	1189.5883	0.00 %	1189.5883	0.00 %
ND280 MC: ν_μ NC all	48.8820	48.8820	0.00 %	48.8820	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.0898	0.00 %	3.0898	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	4.9767	0.00 %	4.9767	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	8.4480	0.00 %	8.4480	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.8276	0.00 %	1.8276	0.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.4312	0.00 %	0.4312	0.00 %
ND280 MC: ν_e CC other	1.0397	1.0397	0.00 %	1.0397	0.00 %
ND280 MC: ν_e CC all	1.6598	1.6598	0.00 %	1.6598	0.00 %
ND280 MC: ν_e NC all	0.7868	0.7868	0.00 %	0.7868	0.00 %

Table 15: Effect of SuperK $\nu_e CC$ efficiency variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	5.9088	-4.76 %	6.5308	5.26 %
SK MC (1 μ-like ring)	5.3721	5.3721	0.00 %	5.3721	0.00 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.0985	0.00 %	3.0985	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.2545	0.00 %	1.2545	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2177	0.00 %	0.2177	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.5707	0.00 %	4.5707	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.4087	0.00 %	0.4087	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2056	0.00 %	0.2056	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1271	0.00 %	0.1271	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0215	0.00 %	0.0215	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3542	0.00 %	0.3542	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0236	0.00 %	0.0236	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0014	0.00 %	0.0014	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0008	0.00 %	0.0008	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0002	0.00 %	0.0002	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0024	0.00 %	0.0024	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0125	0.00 %	0.0125	0.00 %
ND280 CC-like Observed/MC ratio	1.1549	1.0999	-4.76 %	1.2157	5.26 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1313.7520	5.00 %	1188.6327	-5.00 %
ND280 MC: Total CC	1199.6960	1259.6808	5.00 %	1139.7112	-5.00 %
ND280 MC: Total NC background	51.4964	54.0712	5.00 %	48.9215	-5.00 %
ND280 MC: ν_μ CCQE	617.6298	648.5112	5.00 %	586.7483	-5.00 %
ND280 MC: ν_μ CC1π	291.9388	306.5358	5.00 %	277.3419	-5.00 %
ND280 MC: ν_μ CC other	280.0197	294.0206	5.00 %	266.0187	-5.00 %
ND280 MC: ν_μ CC all	1189.5883	1249.0677	5.00 %	1130.1088	-5.00 %
ND280 MC: ν_μ NC all	48.8820	51.3261	5.00 %	46.4379	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.4006	5.00 %	0.3624	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.2443	5.00 %	2.9353	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	5.2255	5.00 %	4.7278	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	8.8704	5.00 %	8.0256	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.9190	5.00 %	1.7362	-5.00 %
ND280 MC: ν_e CCQE	0.1889	0.1983	5.00 %	0.1794	-5.00 %
ND280 MC: ν_e CC1π	0.4312	0.4528	5.00 %	0.4097	-5.00 %
ND280 MC: ν_e CC other	1.0397	1.0917	5.00 %	0.9877	-5.00 %
ND280 MC: ν_e CC all	1.6598	1.7427	5.00 %	1.5768	-5.00 %
ND280 MC: ν_e NC all	0.7868	0.8262	5.00 %	0.7475	-5.00 %

Table 16: Effect of ND280 efficiency variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	6.6112	6.56 %	5.7973	-6.56 %
SK MC (1 μ-like ring)	5.3721	5.3721	0.00 %	5.3721	0.00 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.0985	0.00 %	3.0985	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.2545	0.00 %	1.2545	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2177	0.00 %	0.2177	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.5707	0.00 %	4.5707	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.4087	0.00 %	0.4087	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2056	0.00 %	0.2056	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1271	0.00 %	0.1271	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0215	0.00 %	0.0215	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3542	0.00 %	0.3542	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0236	0.00 %	0.0236	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0014	0.00 %	0.0014	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0008	0.00 %	0.0008	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0002	0.00 %	0.0002	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0024	0.00 %	0.0024	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0125	0.00 %	0.0125	0.00 %
ND280 CC-like Observed/MC ratio	1.1549	1.1549	0.00 %	1.1549	0.00 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1251.1924	0.00 %	1251.1924	0.00 %
ND280 MC: Total CC	1199.6960	1199.6960	0.00 %	1199.6960	0.00 %
ND280 MC: Total NC background	51.4964	51.4964	0.00 %	51.4964	0.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	291.9388	0.00 %	291.9388	0.00 %
ND280 MC: ν_μ CC other	280.0197	280.0197	0.00 %	280.0197	0.00 %
ND280 MC: ν_μ CC all	1189.5883	1189.5883	0.00 %	1189.5883	0.00 %
ND280 MC: ν_μ NC all	48.8820	48.8820	0.00 %	48.8820	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.0898	0.00 %	3.0898	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	4.9767	0.00 %	4.9767	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	8.4480	0.00 %	8.4480	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.8276	0.00 %	1.8276	0.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.4312	0.00 %	0.4312	0.00 %
ND280 MC: ν_e CC other	1.0397	1.0397	0.00 %	1.0397	0.00 %
ND280 MC: ν_e CC all	1.6598	1.6598	0.00 %	1.6598	0.00 %
ND280 MC: ν_e NC all	0.7868	0.7868	0.00 %	0.7868	0.00 %

Table 17: Effect of flux normalization variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	6.3896	2.99 %	6.0189	-2.99 %
SK MC (1 μ-like ring)	5.3721	5.5326	2.99 %	5.2117	-2.99 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.2545	5.04 %	2.9425	-5.04 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.2545	0.00 %	1.2545	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2177	0.00 %	0.2177	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.7267	3.41 %	4.4147	-3.41 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.4087	0.00 %	0.4087	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2100	2.11 %	0.2013	-2.11 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1271	0.00 %	0.1271	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0215	0.00 %	0.0215	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3585	1.23 %	0.3499	-1.23 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0236	0.00 %	0.0236	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0015	6.79 %	0.0013	-6.79 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0008	0.00 %	0.0008	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0002	0.00 %	0.0002	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0025	3.85 %	0.0023	-3.85 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0125	0.00 %	0.0125	0.00 %
ND280 CC-like Observed/MC ratio	1.1549	1.1549	0.00 %	1.1549	0.00 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1251.1924	0.00 %	1251.1924	0.00 %
ND280 MC: Total CC	1199.6960	1199.6960	0.00 %	1199.6960	0.00 %
ND280 MC: Total NC background	51.4964	51.4964	0.00 %	51.4964	0.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	291.9388	0.00 %	291.9388	0.00 %
ND280 MC: ν_μ CC other	280.0197	280.0197	0.00 %	280.0197	0.00 %
ND280 MC: ν_μ CC all	1189.5883	1189.5883	0.00 %	1189.5883	0.00 %
ND280 MC: ν_μ NC all	48.8820	48.8820	0.00 %	48.8820	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.0898	0.00 %	3.0898	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	4.9767	0.00 %	4.9767	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	8.4480	0.00 %	8.4480	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.8276	0.00 %	1.8276	0.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.4312	0.00 %	0.4312	0.00 %
ND280 MC: ν_e CC other	1.0397	1.0397	0.00 %	1.0397	0.00 %
ND280 MC: ν_e CC all	1.6598	1.6598	0.00 %	1.6598	0.00 %
ND280 MC: ν_e NC all	0.7868	0.7868	0.00 %	0.7868	0.00 %

Table 18: Effect of CCQE cross section variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	6.2128	0.14 %	6.1946	-0.16 %
SK MC (1 μ-like ring)	5.3721	5.7121	6.33 %	5.0322	-6.33 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.0985	0.00 %	3.0985	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.5634	24.63 %	0.9455	-24.63 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2177	0.00 %	0.2177	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.8796	6.76 %	4.2618	-6.76 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.4087	0.00 %	0.4087	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2056	0.00 %	0.2056	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1579	24.23 %	0.0963	-24.23 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0215	0.00 %	0.0215	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3850	8.69 %	0.3234	-8.69 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0236	0.00 %	0.0236	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0014	0.00 %	0.0014	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0010	27.33 %	0.0006	-27.33 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0002	0.00 %	0.0002	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0026	9.27 %	0.0022	-9.27 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0125	0.00 %	0.0125	0.00 %
ND280 CC-like Observed/MC ratio	1.1549	1.0877	-5.82 %	1.2310	6.59 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1328.5368	6.18 %	1173.8479	-6.18 %
ND280 MC: Total CC	1199.6960	1277.0405	6.45 %	1122.3515	-6.45 %
ND280 MC: Total NC background	51.4964	51.4964	0.00 %	51.4964	0.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	364.5956	24.89 %	219.2821	-24.89 %
ND280 MC: ν_μ CC other	280.0197	283.8437	1.37 %	276.1956	-1.37 %
ND280 MC: ν_μ CC all	1189.5883	1266.0691	6.43 %	1113.1075	-6.43 %
ND280 MC: ν_μ NC all	48.8820	48.8820	0.00 %	48.8820	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.8106	23.33 %	2.3690	-23.33 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	5.0146	0.76 %	4.9387	-0.76 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	9.2067	8.98 %	7.6892	-8.98 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.8276	0.00 %	1.8276	0.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.5309	23.11 %	0.3316	-23.11 %
ND280 MC: ν_e CC other	1.0397	1.0449	0.50 %	1.0344	-0.50 %
ND280 MC: ν_e CC all	1.6598	1.7647	6.32 %	1.5548	-6.32 %
ND280 MC: ν_e NC all	0.7868	0.7868	0.00 %	0.7868	0.00 %

Table 19: Effect of CC1π/CCQE cross section ratio variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	5.9230	-4.53 %	6.5210	5.11 %
SK MC (1 μ-like ring)	5.3721	5.4332	1.14 %	5.3111	-1.14 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.0985	0.00 %	3.0985	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.2545	0.00 %	1.2545	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2731	25.46 %	0.1623	-25.46 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.6261	1.21 %	4.5153	-1.21 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.4087	0.00 %	0.4087	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2056	0.00 %	0.2056	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1271	0.00 %	0.1271	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0271	26.02 %	0.0159	-26.02 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3598	1.58 %	0.3486	-1.58 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0236	0.00 %	0.0236	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0014	0.00 %	0.0014	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0008	0.00 %	0.0008	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0003	26.33 %	0.0002	-26.33 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0025	2.46 %	0.0024	-2.46 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0125	0.00 %	0.0125	0.00 %
ND280 CC-like Observed/MC ratio	1.1549	1.0902	-5.61 %	1.2278	6.31 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1325.5003	5.94 %	1176.8844	-5.94 %
ND280 MC: Total CC	1199.6960	1274.0039	6.19 %	1125.3881	-6.19 %
ND280 MC: Total NC background	51.4964	51.4964	0.00 %	51.4964	0.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	296.0404	1.40 %	287.8373	-1.40 %
ND280 MC: ν_μ CC other	280.0197	348.6666	24.52 %	211.3727	-24.52 %
ND280 MC: ν_μ CC all	1189.5883	1262.3367	6.12 %	1116.8398	-6.12 %
ND280 MC: ν_μ NC all	48.8820	48.8820	0.00 %	48.8820	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.1319	1.36 %	3.0477	-1.36 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	6.2304	25.19 %	3.7229	-25.19 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	9.7439	15.34 %	7.1521	-15.34 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.8276	0.00 %	1.8276	0.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.4371	1.36 %	0.4254	-1.36 %
ND280 MC: ν_e CC other	1.0397	1.2974	24.79 %	0.7820	-24.79 %
ND280 MC: ν_e CC all	1.6598	1.9233	15.88 %	1.3962	-15.88 %
ND280 MC: ν_e NC all	0.7868	0.7868	0.00 %	0.7868	0.00 %

Table 20: Effect of CCother/CCQE cross section ratio variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	6.2949	1.46 %	6.1109	-1.50 %
SK MC (1 μ-like ring)	5.3721	5.5314	2.96 %	5.2129	-2.96 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.0985	0.00 %	3.0985	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.2545	0.00 %	1.2545	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2177	0.00 %	0.2177	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.5707	0.00 %	4.5707	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.5551	35.80 %	0.2624	-35.80 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2056	0.00 %	0.2056	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1271	0.00 %	0.1271	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0215	0.00 %	0.0215	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3542	0.00 %	0.3542	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0321	35.75 %	0.0152	-35.75 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0014	0.00 %	0.0014	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0008	0.00 %	0.0008	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0002	0.00 %	0.0002	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0024	0.00 %	0.0024	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0169	35.89 %	0.0080	-35.89 %
ND280 CC-like Observed/MC ratio	1.1549	1.1380	-1.46 %	1.1723	1.50 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1269.7310	1.48 %	1232.6537	-1.48 %
ND280 MC: Total CC	1199.6960	1199.6960	0.00 %	1199.6960	0.00 %
ND280 MC: Total NC background	51.4964	70.0351	36.00 %	32.9577	-36.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	291.9388	0.00 %	291.9388	0.00 %
ND280 MC: ν_μ CC other	280.0197	280.0197	0.00 %	280.0197	0.00 %
ND280 MC: ν_μ CC all	1189.5883	1189.5883	0.00 %	1189.5883	0.00 %
ND280 MC: ν_μ NC all	48.8820	66.4795	36.00 %	31.2845	-36.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.0898	0.00 %	3.0898	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	4.9767	0.00 %	4.9767	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	8.4480	0.00 %	8.4480	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	2.4855	36.00 %	1.1696	-36.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.4312	0.00 %	0.4312	0.00 %
ND280 MC: ν_e CC other	1.0397	1.0397	0.00 %	1.0397	0.00 %
ND280 MC: ν_e CC all	1.6598	1.6598	0.00 %	1.6598	0.00 %
ND280 MC: ν_e NC all	0.7868	1.0701	36.00 %	0.5036	-36.00 %

Table 21: Effect of NC/CCQE cross section ratio variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.2043	6.5664	5.84 %	5.8421	-5.84 %
SK MC (1 μ-like ring)	5.3721	5.6857	5.84 %	5.0585	-5.84 %
SK MC (1 μ-like ring): ν_μ CCQE	3.0985	3.2495	4.87 %	2.9476	-4.87 %
SK MC (1 μ-like ring): ν_μ CC1π	1.2545	1.3479	7.45 %	1.1610	-7.45 %
SK MC (1 μ-like ring): ν_μ CC other	0.2177	0.2431	11.65 %	0.1923	-11.65 %
SK MC (1 μ-like ring): ν_μ CC all	4.5707	4.8404	5.90 %	4.3009	-5.90 %
SK MC (1 μ-like ring): ν_μ NC all	0.4087	0.4244	3.84 %	0.3930	-3.84 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2056	0.2204	7.16 %	0.1909	-7.16 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1271	0.1369	7.71 %	0.1173	-7.71 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0215	0.0237	10.13 %	0.0193	-10.13 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3542	0.3809	7.54 %	0.3275	-7.54 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0236	0.0245	3.63 %	0.0228	-3.63 %
SK MC (1 μ-like ring): ν_e CCQE	0.0014	0.0014	5.46 %	0.0013	-5.46 %
SK MC (1 μ-like ring): ν_e CC1π	0.0008	0.0009	6.26 %	0.0008	-6.26 %
SK MC (1 μ-like ring): ν_e CC other	0.0002	0.0002	8.69 %	0.0002	-8.69 %
SK MC (1 μ-like ring): ν_e CC all	0.0024	0.0026	6.03 %	0.0023	-6.03 %
SK MC (1 μ-like ring): ν_e NC all	0.0125	0.0129	3.59 %	0.0120	-3.59 %
ND280 CC-like Observed/MC ratio	1.1549	1.1549	0.00 %	1.1549	0.00 %
ND280 Observed CC-like	1445.0000	1445.0000	0.00 %	1445.0000	0.00 %
ND280 MC: Total CC-like	1251.1924	1251.1924	0.00 %	1251.1924	0.00 %
ND280 MC: Total CC	1199.6960	1199.6960	0.00 %	1199.6960	0.00 %
ND280 MC: Total NC background	51.4964	51.4964	0.00 %	51.4964	0.00 %
ND280 MC: ν_μ CCQE	617.6298	617.6298	0.00 %	617.6298	0.00 %
ND280 MC: ν_μ CC1π	291.9388	291.9388	0.00 %	291.9388	0.00 %
ND280 MC: ν_μ CC other	280.0197	280.0197	0.00 %	280.0197	0.00 %
ND280 MC: ν_μ CC all	1189.5883	1189.5883	0.00 %	1189.5883	0.00 %
ND280 MC: ν_μ NC all	48.8820	48.8820	0.00 %	48.8820	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.3815	0.3815	0.00 %	0.3815	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	3.0898	3.0898	0.00 %	3.0898	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	4.9767	4.9767	0.00 %	4.9767	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	8.4480	8.4480	0.00 %	8.4480	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.8276	1.8276	0.00 %	1.8276	0.00 %
ND280 MC: ν_e CCQE	0.1889	0.1889	0.00 %	0.1889	0.00 %
ND280 MC: ν_e CC1π	0.4312	0.4312	0.00 %	0.4312	0.00 %
ND280 MC: ν_e CC other	1.0397	1.0397	0.00 %	1.0397	0.00 %
ND280 MC: ν_e CC all	1.6598	1.6598	0.00 %	1.6598	0.00 %
ND280 MC: ν_e NC all	0.7868	0.7868	0.00 %	0.7868	0.00 %

Table 22: Effect of FSI variation on various components of the oscillation analysis (GENIE). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4×10^{-3} eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.6395	4.73 %	6.0394	-4.73 %
SK MC (1 μ-like ring)	5.9735	6.2562	4.73 %	5.6907	-4.73 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.6511	7.80 %	3.1227	-7.80 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.4288	0.00 %	1.4288	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.3822	0.00 %	0.3822	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.4620	5.08 %	4.9337	-5.08 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.3549	0.00 %	0.3549	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2566	7.80 %	0.2195	-7.80 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1110	0.00 %	0.1110	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0364	0.00 %	0.0364	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.4040	4.82 %	0.3669	-4.82 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0206	0.00 %	0.0206	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0016	0.00 %	0.0016	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0010	0.00 %	0.0010	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0004	0.00 %	0.0004	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0030	0.00 %	0.0030	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0116	0.00 %	0.0116	0.00 %
ND280 CC-like Observed/MC ratio	1.0613	1.0613	0.00 %	1.0613	0.00 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1371.9471	0.00 %	1371.9471	0.00 %
ND280 MC: Total CC	1316.7019	1316.7019	0.00 %	1316.7019	0.00 %
ND280 MC: Total NC background	55.2452	55.2452	0.00 %	55.2452	0.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	314.5124	0.00 %	314.5124	0.00 %
ND280 MC: ν_μ CC other	294.8727	294.8727	0.00 %	294.8727	0.00 %
ND280 MC: ν_μ CC all	1305.4926	1305.4926	0.00 %	1305.4926	0.00 %
ND280 MC: ν_μ NC all	52.4573	52.4573	0.00 %	52.4573	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.5675	0.00 %	2.5675	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	5.7370	0.00 %	5.7370	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.0117	0.00 %	9.0117	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.4559	0.00 %	1.4559	0.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.2581	0.00 %	0.2581	0.00 %
ND280 MC: ν_e CC other	1.7210	1.7210	0.00 %	1.7210	0.00 %
ND280 MC: ν_e CC all	2.1976	2.1976	0.00 %	2.1976	0.00 %
ND280 MC: ν_e NC all	1.3320	1.3320	0.00 %	1.3320	0.00 %

Table 23: Effect of SuperK CCQE efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.8694	8.36 %	5.8095	-8.36 %
SK MC (1 μ-like ring)	5.9735	6.4729	8.36 %	5.4741	-8.36 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.3869	0.00 %	3.3869	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.7931	25.50 %	1.0644	-25.50 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.4796	25.50 %	0.2847	-25.50 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.6597	8.88 %	4.7361	-8.88 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.3549	0.00 %	0.3549	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2380	0.00 %	0.2380	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1393	25.50 %	0.0827	-25.50 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0457	25.50 %	0.0271	-25.50 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.4230	9.75 %	0.3479	-9.75 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0206	0.00 %	0.0206	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0016	0.00 %	0.0016	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0010	0.00 %	0.0010	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0004	0.00 %	0.0004	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0030	0.00 %	0.0030	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0116	0.00 %	0.0116	0.00 %
ND280 CC-like Observed/MC ratio	1.0613	1.0613	0.00 %	1.0613	0.00 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1371.9471	0.00 %	1371.9471	0.00 %
ND280 MC: Total CC	1316.7019	1316.7019	0.00 %	1316.7019	0.00 %
ND280 MC: Total NC background	55.2452	55.2452	0.00 %	55.2452	0.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	314.5124	0.00 %	314.5124	0.00 %
ND280 MC: ν_μ CC other	294.8727	294.8727	0.00 %	294.8727	0.00 %
ND280 MC: ν_μ CC all	1305.4926	1305.4926	0.00 %	1305.4926	0.00 %
ND280 MC: ν_μ NC all	52.4573	52.4573	0.00 %	52.4573	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.5675	0.00 %	2.5675	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	5.7370	0.00 %	5.7370	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.0117	0.00 %	9.0117	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.4559	0.00 %	1.4559	0.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.2581	0.00 %	0.2581	0.00 %
ND280 MC: ν_e CC other	1.7210	1.7210	0.00 %	1.7210	0.00 %
ND280 MC: ν_e CC all	2.1976	2.1976	0.00 %	2.1976	0.00 %
ND280 MC: ν_e NC all	1.3320	1.3320	0.00 %	1.3320	0.00 %

Table 24: Effect of SuperK CCnonQE efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.8124	7.46 %	5.8665	-7.46 %
SK MC (1 μ-like ring)	5.9735	6.4191	7.46 %	5.5278	-7.46 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.3869	0.00 %	3.3869	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.4288	0.00 %	1.4288	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.3822	0.00 %	0.3822	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.1979	0.00 %	5.1979	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.7634	115.10 %	-0.0536	-115.10 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2380	0.00 %	0.2380	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1110	0.00 %	0.1110	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0364	0.00 %	0.0364	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.3854	0.00 %	0.3854	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0443	115.10 %	-0.0031	-115.10 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0016	0.00 %	0.0016	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0010	0.00 %	0.0010	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0004	0.00 %	0.0004	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0030	0.00 %	0.0030	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0250	115.10 %	-0.0018	-115.10 %
ND280 CC-like Observed/MC ratio	1.0613	1.0613	0.00 %	1.0613	0.00 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1371.9471	0.00 %	1371.9471	0.00 %
ND280 MC: Total CC	1316.7019	1316.7019	0.00 %	1316.7019	0.00 %
ND280 MC: Total NC background	55.2452	55.2452	0.00 %	55.2452	0.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	314.5124	0.00 %	314.5124	0.00 %
ND280 MC: ν_μ CC other	294.8727	294.8727	0.00 %	294.8727	0.00 %
ND280 MC: ν_μ CC all	1305.4926	1305.4926	0.00 %	1305.4926	0.00 %
ND280 MC: ν_μ NC all	52.4573	52.4573	0.00 %	52.4573	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.5675	0.00 %	2.5675	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	5.7370	0.00 %	5.7370	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.0117	0.00 %	9.0117	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.4559	0.00 %	1.4559	0.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.2581	0.00 %	0.2581	0.00 %
ND280 MC: ν_e CC other	1.7210	1.7210	0.00 %	1.7210	0.00 %
ND280 MC: ν_e CC all	2.1976	2.1976	0.00 %	2.1976	0.00 %
ND280 MC: ν_e NC all	1.3320	1.3320	0.00 %	1.3320	0.00 %

Table 25: Effect of SuperK NC efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.3426	0.05 %	6.3363	-0.05 %
SK MC (1 μ-like ring)	5.9735	5.9765	0.05 %	5.9705	-0.05 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.3869	0.00 %	3.3869	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.4288	0.00 %	1.4288	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.3822	0.00 %	0.3822	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.1979	0.00 %	5.1979	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.3549	0.00 %	0.3549	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2380	0.00 %	0.2380	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1110	0.00 %	0.1110	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0364	0.00 %	0.0364	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.3854	0.00 %	0.3854	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0206	0.00 %	0.0206	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0032	100.00 %	0.0000	-100.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0020	100.00 %	0.0000	-100.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0008	100.00 %	0.0000	-100.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0060	100.00 %	0.0000	-100.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0116	0.00 %	0.0116	0.00 %
ND280 CC-like Observed/MC ratio	1.0613	1.0613	0.00 %	1.0613	0.00 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1371.9471	0.00 %	1371.9471	0.00 %
ND280 MC: Total CC	1316.7019	1316.7019	0.00 %	1316.7019	0.00 %
ND280 MC: Total NC background	55.2452	55.2452	0.00 %	55.2452	0.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	314.5124	0.00 %	314.5124	0.00 %
ND280 MC: ν_μ CC other	294.8727	294.8727	0.00 %	294.8727	0.00 %
ND280 MC: ν_μ CC all	1305.4926	1305.4926	0.00 %	1305.4926	0.00 %
ND280 MC: ν_μ NC all	52.4573	52.4573	0.00 %	52.4573	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.5675	0.00 %	2.5675	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	5.7370	0.00 %	5.7370	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.0117	0.00 %	9.0117	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.4559	0.00 %	1.4559	0.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.2581	0.00 %	0.2581	0.00 %
ND280 MC: ν_e CC other	1.7210	1.7210	0.00 %	1.7210	0.00 %
ND280 MC: ν_e CC all	2.1976	2.1976	0.00 %	2.1976	0.00 %
ND280 MC: ν_e NC all	1.3320	1.3320	0.00 %	1.3320	0.00 %

Table 26: Effect of SuperK ν_e CC efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.0376	-4.76 %	6.6731	5.26 %
SK MC (1 μ-like ring)	5.9735	5.9735	0.00 %	5.9735	0.00 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.3869	0.00 %	3.3869	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.4288	0.00 %	1.4288	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.3822	0.00 %	0.3822	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.1979	0.00 %	5.1979	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.3549	0.00 %	0.3549	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2380	0.00 %	0.2380	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1110	0.00 %	0.1110	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0364	0.00 %	0.0364	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.3854	0.00 %	0.3854	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0206	0.00 %	0.0206	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0016	0.00 %	0.0016	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0010	0.00 %	0.0010	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0004	0.00 %	0.0004	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0030	0.00 %	0.0030	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0116	0.00 %	0.0116	0.00 %
ND280 CC-like Observed/MC ratio	1.0613	1.0107	-4.76 %	1.1171	5.26 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1440.5444	5.00 %	1303.3497	-5.00 %
ND280 MC: Total CC	1316.7019	1382.5370	5.00 %	1250.8668	-5.00 %
ND280 MC: Total NC background	55.2452	58.0075	5.00 %	52.4829	-5.00 %
ND280 MC: ν_μ CCQE	696.1075	730.9129	5.00 %	661.3021	-5.00 %
ND280 MC: ν_μ CC1π	314.5124	330.2381	5.00 %	298.7868	-5.00 %
ND280 MC: ν_μ CC other	294.8727	309.6163	5.00 %	280.1290	-5.00 %
ND280 MC: ν_μ CC all	1305.4926	1370.7672	5.00 %	1240.2179	-5.00 %
ND280 MC: ν_μ NC all	52.4573	55.0802	5.00 %	49.8344	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7426	5.00 %	0.6719	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.6958	5.00 %	2.4391	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	6.0239	5.00 %	5.4502	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.4623	5.00 %	8.5611	-5.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.5286	5.00 %	1.3831	-5.00 %
ND280 MC: ν_e CCQE	0.2186	0.2295	5.00 %	0.2076	-5.00 %
ND280 MC: ν_e CC1π	0.2581	0.2710	5.00 %	0.2452	-5.00 %
ND280 MC: ν_e CC other	1.7210	1.8070	5.00 %	1.6349	-5.00 %
ND280 MC: ν_e CC all	2.1976	2.3075	5.00 %	2.0877	-5.00 %
ND280 MC: ν_e NC all	1.3320	1.3986	5.00 %	1.2654	-5.00 %

Table 27: Effect of ND280 efficiency variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.7384	6.29 %	5.9405	-6.29 %
SK MC (1 μ-like ring)	5.9735	5.9735	0.00 %	5.9735	0.00 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.3869	0.00 %	3.3869	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.4288	0.00 %	1.4288	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.3822	0.00 %	0.3822	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.1979	0.00 %	5.1979	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.3549	0.00 %	0.3549	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2380	0.00 %	0.2380	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1110	0.00 %	0.1110	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0364	0.00 %	0.0364	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.3854	0.00 %	0.3854	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0206	0.00 %	0.0206	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0016	0.00 %	0.0016	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0010	0.00 %	0.0010	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0004	0.00 %	0.0004	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0030	0.00 %	0.0030	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0116	0.00 %	0.0116	0.00 %
ND280 CC-like Observed/MC ratio	1.0613	1.0613	0.00 %	1.0613	0.00 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1371.9471	0.00 %	1371.9471	0.00 %
ND280 MC: Total CC	1316.7019	1316.7019	0.00 %	1316.7019	0.00 %
ND280 MC: Total NC background	55.2452	55.2452	0.00 %	55.2452	0.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	314.5124	0.00 %	314.5124	0.00 %
ND280 MC: ν_μ CC other	294.8727	294.8727	0.00 %	294.8727	0.00 %
ND280 MC: ν_μ CC all	1305.4926	1305.4926	0.00 %	1305.4926	0.00 %
ND280 MC: ν_μ NC all	52.4573	52.4573	0.00 %	52.4573	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.5675	0.00 %	2.5675	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	5.7370	0.00 %	5.7370	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.0117	0.00 %	9.0117	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.4559	0.00 %	1.4559	0.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.2581	0.00 %	0.2581	0.00 %
ND280 MC: ν_e CC other	1.7210	1.7210	0.00 %	1.7210	0.00 %
ND280 MC: ν_e CC all	2.1976	2.1976	0.00 %	2.1976	0.00 %
ND280 MC: ν_e NC all	1.3320	1.3320	0.00 %	1.3320	0.00 %

Table 28: Effect of beam normalization variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.4998	2.53 %	6.1791	-2.53 %
SK MC (1 μ-like ring)	5.9735	6.1245	2.53 %	5.8224	-2.53 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.5330	4.31 %	3.2408	-4.31 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.4288	0.00 %	1.4288	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.3822	0.00 %	0.3822	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.3440	2.81 %	5.0518	-2.81 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.3549	0.00 %	0.3549	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2429	2.05 %	0.2332	-2.05 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1110	0.00 %	0.1110	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0364	0.00 %	0.0364	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.3903	1.26 %	0.3806	-1.26 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0206	0.00 %	0.0206	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0017	5.97 %	0.0015	-5.97 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0010	0.00 %	0.0010	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0004	0.00 %	0.0004	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0031	3.22 %	0.0029	-3.22 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0116	0.00 %	0.0116	0.00 %
ND280 CC-like Observed/MC ratio	1.0613	1.0613	0.00 %	1.0613	0.00 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1371.9471	0.00 %	1371.9471	0.00 %
ND280 MC: Total CC	1316.7019	1316.7019	0.00 %	1316.7019	0.00 %
ND280 MC: Total NC background	55.2452	55.2452	0.00 %	55.2452	0.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	314.5124	0.00 %	314.5124	0.00 %
ND280 MC: ν_μ CC other	294.8727	294.8727	0.00 %	294.8727	0.00 %
ND280 MC: ν_μ CC all	1305.4926	1305.4926	0.00 %	1305.4926	0.00 %
ND280 MC: ν_μ NC all	52.4573	52.4573	0.00 %	52.4573	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.5675	0.00 %	2.5675	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	5.7370	0.00 %	5.7370	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.0117	0.00 %	9.0117	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.4559	0.00 %	1.4559	0.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.2581	0.00 %	0.2581	0.00 %
ND280 MC: ν_e CC other	1.7210	1.7210	0.00 %	1.7210	0.00 %
ND280 MC: ν_e CC all	2.1976	2.1976	0.00 %	2.1976	0.00 %
ND280 MC: ν_e NC all	1.3320	1.3320	0.00 %	1.3320	0.00 %

Table 29: Effect of CCQE cross section variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.3673	0.44 %	6.3081	-0.49 %
SK MC (1 μ-like ring)	5.9735	6.3556	6.40 %	5.5914	-6.40 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.3869	0.00 %	3.3869	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.7839	24.86 %	1.0736	-24.86 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.3822	0.00 %	0.3822	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.5531	6.83 %	4.8427	-6.83 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.3549	0.00 %	0.3549	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2380	0.00 %	0.2380	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1376	24.00 %	0.0843	-24.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0364	0.00 %	0.0364	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.4121	6.91 %	0.3588	-6.91 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0206	0.00 %	0.0206	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0016	0.00 %	0.0016	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0013	27.60 %	0.0007	-27.60 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0004	0.00 %	0.0004	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0033	9.18 %	0.0027	-9.18 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0116	0.00 %	0.0116	0.00 %
ND280 CC-like Observed/MC ratio	1.0613	1.0018	-5.60 %	1.1282	6.31 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1453.3314	5.93 %	1290.5627	-5.93 %
ND280 MC: Total CC	1316.7019	1398.0862	6.18 %	1235.3175	-6.18 %
ND280 MC: Total NC background	55.2452	55.2452	0.00 %	55.2452	0.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	395.2631	25.67 %	233.7618	-25.67 %
ND280 MC: ν_μ CC other	294.8727	294.8727	0.00 %	294.8727	0.00 %
ND280 MC: ν_μ CC all	1305.4926	1386.2432	6.19 %	1224.7419	-6.19 %
ND280 MC: ν_μ NC all	52.4573	52.4573	0.00 %	52.4573	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	3.1437	22.44 %	1.9912	-22.44 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	5.7370	0.00 %	5.7370	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.5879	6.39 %	8.4355	-6.39 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.4559	0.00 %	1.4559	0.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.3156	22.28 %	0.2006	-22.28 %
ND280 MC: ν_e CC other	1.7210	1.7210	0.00 %	1.7210	0.00 %
ND280 MC: ν_e CC all	2.1976	2.2551	2.62 %	2.1401	-2.62 %
ND280 MC: ν_e NC all	1.3320	1.3320	0.00 %	1.3320	0.00 %

Table 30: Effect of CC1π/CCQE cross section variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.1059	-3.68 %	6.6013	4.13 %
SK MC (1 μ-like ring)	5.9735	6.0821	1.82 %	5.8649	-1.82 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.3869	0.00 %	3.3869	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.4288	0.00 %	1.4288	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.4811	25.88 %	0.2833	-25.88 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.2968	1.90 %	5.0990	-1.90 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.3549	0.00 %	0.3549	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2380	0.00 %	0.2380	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1110	0.00 %	0.1110	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0461	26.44 %	0.0268	-26.44 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.3951	2.50 %	0.3758	-2.50 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0206	0.00 %	0.0206	0.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0016	0.00 %	0.0016	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0010	0.00 %	0.0010	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0005	26.90 %	0.0003	-26.90 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0031	3.47 %	0.0029	-3.47 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0116	0.00 %	0.0116	0.00 %
ND280 CC-like Observed/MC ratio	1.0613	1.0039	-5.40 %	1.1256	6.06 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1450.3320	5.71 %	1293.5621	-5.71 %
ND280 MC: Total CC	1316.7019	1395.0868	5.95 %	1238.3169	-5.95 %
ND280 MC: Total NC background	55.2452	55.2452	0.00 %	55.2452	0.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	314.5124	0.00 %	314.5124	0.00 %
ND280 MC: ν_μ CC other	294.8727	371.3097	25.92 %	218.4356	-25.92 %
ND280 MC: ν_μ CC all	1305.4926	1381.9297	5.86 %	1229.0555	-5.86 %
ND280 MC: ν_μ NC all	52.4573	52.4573	0.00 %	52.4573	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.5675	0.00 %	2.5675	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	7.2515	26.40 %	4.2225	-26.40 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	10.5262	16.81 %	7.4972	-16.81 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.4559	0.00 %	1.4559	0.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.2581	0.00 %	0.2581	0.00 %
ND280 MC: ν_e CC other	1.7210	2.1544	25.18 %	1.2876	-25.18 %
ND280 MC: ν_e CC all	2.1976	2.6310	19.72 %	1.7642	-19.72 %
ND280 MC: ν_e NC all	1.3320	1.3320	0.00 %	1.3320	0.00 %

Table 31: Effect of CCother/CCQE cross section ratio variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.3947	0.87 %	6.2826	-0.90 %
SK MC (1 μ-like ring)	5.9735	6.1129	2.33 %	5.8341	-2.33 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.3869	0.00 %	3.3869	0.00 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.4288	0.00 %	1.4288	0.00 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.3822	0.00 %	0.3822	0.00 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.1979	0.00 %	5.1979	0.00 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.4827	36.00 %	0.2271	-36.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2380	0.00 %	0.2380	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1110	0.00 %	0.1110	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0364	0.00 %	0.0364	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.3854	0.00 %	0.3854	0.00 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0280	36.00 %	0.0132	-36.00 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0016	0.00 %	0.0016	0.00 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0010	0.00 %	0.0010	0.00 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0004	0.00 %	0.0004	0.00 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0030	0.00 %	0.0030	0.00 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0158	36.00 %	0.0075	-36.00 %
ND280 CC-like Observed/MC ratio	1.0613	1.0461	-1.43 %	1.0769	1.47 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1391.8353	1.45 %	1352.0588	-1.45 %
ND280 MC: Total CC	1316.7019	1316.7019	0.00 %	1316.7019	0.00 %
ND280 MC: Total NC background	55.2452	75.1335	36.00 %	35.3569	-36.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	314.5124	0.00 %	314.5124	0.00 %
ND280 MC: ν_μ CC other	294.8727	294.8727	0.00 %	294.8727	0.00 %
ND280 MC: ν_μ CC all	1305.4926	1305.4926	0.00 %	1305.4926	0.00 %
ND280 MC: ν_μ NC all	52.4573	71.3419	36.00 %	33.5727	-36.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.5675	0.00 %	2.5675	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	5.7370	0.00 %	5.7370	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.0117	0.00 %	9.0117	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.9800	36.00 %	0.9317	-36.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.2581	0.00 %	0.2581	0.00 %
ND280 MC: ν_e CC other	1.7210	1.7210	0.00 %	1.7210	0.00 %
ND280 MC: ν_e CC all	2.1976	2.1976	0.00 %	2.1976	0.00 %
ND280 MC: ν_e NC all	1.3320	1.8116	36.00 %	0.8525	-36.00 %

Table 32: Effect of NC/CCQE cross section ratio variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

Component	**Nominal**	**+1σ**		**-1σ**	
N_{SK} (1 μ-like ring)	6.3394	6.7185	5.98 %	5.9604	-5.98 %
SK MC (1 μ-like ring)	5.9735	6.3307	5.98 %	5.6163	-5.98 %
SK MC (1 μ-like ring): ν_μ CCQE	3.3869	3.5572	5.03 %	3.2166	-5.03 %
SK MC (1 μ-like ring): ν_μ CC1π	1.4288	1.5302	7.10 %	1.3273	-7.10 %
SK MC (1 μ-like ring): ν_μ CC other	0.3822	0.4244	11.03 %	0.3400	-11.03 %
SK MC (1 μ-like ring): ν_μ CC all	5.1979	5.5118	6.04 %	4.8839	-6.04 %
SK MC (1 μ-like ring): ν_μ NC all	0.3549	0.3669	3.37 %	0.3429	-3.37 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CCQE	0.2380	0.2560	7.52 %	0.2201	-7.52 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC1π	0.1110	0.1197	7.87 %	0.1022	-7.87 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC other	0.0364	0.0398	9.37 %	0.0330	-9.37 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ CC all	0.3854	0.4155	7.80 %	0.3554	-7.80 %
SK MC (1 μ-like ring): $\bar{\nu_\mu}$ NC all	0.0206	0.0213	3.14 %	0.0200	-3.14 %
SK MC (1 μ-like ring): ν_e CCQE	0.0016	0.0017	7.80 %	0.0015	-7.80 %
SK MC (1 μ-like ring): ν_e CC1π	0.0010	0.0011	5.35 %	0.0009	-5.35 %
SK MC (1 μ-like ring): ν_e CC other	0.0004	0.0004	7.72 %	0.0004	-7.72 %
SK MC (1 μ-like ring): ν_e CC all	0.0030	0.0032	6.97 %	0.0028	-6.97 %
SK MC (1 μ-like ring): ν_e NC all	0.0116	0.0120	3.23 %	0.0113	-3.23 %
ND280 CC-like Observed/MC ratio	1.0613	1.0613	0.00 %	1.0613	0.00 %
ND280 Observed CC-like	1456.0000	1456.0000	0.00 %	1456.0000	0.00 %
ND280 MC: Total CC-like	1371.9471	1371.9471	0.00 %	1371.9471	0.00 %
ND280 MC: Total CC	1316.7019	1316.7019	0.00 %	1316.7019	0.00 %
ND280 MC: Total NC background	55.2452	55.2452	0.00 %	55.2452	0.00 %
ND280 MC: ν_μ CCQE	696.1075	696.1075	0.00 %	696.1075	0.00 %
ND280 MC: ν_μ CC1π	314.5124	314.5124	0.00 %	314.5124	0.00 %
ND280 MC: ν_μ CC other	294.8727	294.8727	0.00 %	294.8727	0.00 %
ND280 MC: ν_μ CC all	1305.4926	1305.4926	0.00 %	1305.4926	0.00 %
ND280 MC: ν_μ NC all	52.4573	52.4573	0.00 %	52.4573	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CCQE	0.7072	0.7072	0.00 %	0.7072	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC1π	2.5675	2.5675	0.00 %	2.5675	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC other	5.7370	5.7370	0.00 %	5.7370	0.00 %
ND280 MC: $\bar{\nu_\mu}$ CC all	9.0117	9.0117	0.00 %	9.0117	0.00 %
ND280 MC: $\bar{\nu_\mu}$ NC all	1.4559	1.4559	0.00 %	1.4559	0.00 %
ND280 MC: ν_e CCQE	0.2186	0.2186	0.00 %	0.2186	0.00 %
ND280 MC: ν_e CC1π	0.2581	0.2581	0.00 %	0.2581	0.00 %
ND280 MC: ν_e CC other	1.7210	1.7210	0.00 %	1.7210	0.00 %
ND280 MC: ν_e CC all	2.1976	2.1976	0.00 %	2.1976	0.00 %
ND280 MC: ν_e NC all	1.3320	1.3320	0.00 %	1.3320	0.00 %

Table 33: Effect of FSI variation on various components of the oscillation analysis (NEUT). Oscillations with parameters $sin^2 2\theta_{23}$ = 1.0 and Δm^2_{23} = 2.4$\times 10^{-3}$ eV^2/c^4 were applied.

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